

Words That Rule

A Computational Linguistic Analysis of Executive Speeches between Presidential and Parliamentary Systems

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Data Science PhD

Dissertation



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Dedication

I dedicate this dissertation work to my father. I extend my heartfelt gratitude to my mother and brother for their unwavering support, to my wife Remya and my daughters Janaki and Parvathi for standing by me, and to all my family members and friends who have supported me in this journey.

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Abstract

Presidential and parliamentary systems are the two most popular forms of democratic governments worldwide. Understanding how institutional setups influence executive communication patterns in these democratic systems can reveal the role of governance structures in shaping executive speeches across different cultural and political contexts. However, multi-national comparative research on how governance structures influence executive communication patterns remains limited, with most studies confined to single-country analyses or qualitative observations. The main objective of this quantitative study is to use advances in computational linguistics to systematically compare and quantify pronoun usage, narrative frameworks, repetition patterns, linguistic complexity, and agenda-setting in executive speeches across sixteen nations representing both presidential and parliamentary systems. Analysis of over 20,000 speeches from sixteen nations revealed that presidential speeches employed significantly more first-person pronouns and narrative dramatization, whereas parliamentary speeches exhibited higher linguistic complexity and a greater focus on domestic policy. Machine learning analysis of linguistic features predicted electoral outcomes with 73% accuracy across both systems. These findings demonstrate that political practitioners must adopt system-specific rather than universal communication approaches. The study establishes data-driven computational linguistics as an effective tool for cross-institutional political analysis and electoral prediction. Results demonstrate that institutional design fundamentally shapes executive rhetoric.

1

Introduction

1.1 Overview

In today's rapidly changing world, the role of executive speeches in shaping public discourse and policy has become more critical than ever. Presidents and prime ministers are at the forefront of navigating complex domestic and international challenges (Broschek, 2018). Their language usage, political rhetoric, and policy emphasis influence public opinion and legislative outcomes differently based on their institutional frameworks. In presidential systems, independently elected executive heads can speak directly to the public, while executive heads in parliamentary systems must maintain legislative support through coalition-building rhetoric. This fundamental institutional difference shapes how executives communicate policies, build public support, and implement their agendas (Indriðason & Kam, 2020). The language used by executive heads in these systems sets the agenda for legislative deliberations and policymaking and projects national interests globally. Analyzing these linguistic patterns using data science methodologies and computational analysis can reveal how institutional structures shape executive communication, offering insights for academic understanding and practical applications across domains.

This dissertation compares executive speeches between presidential and parliamentary systems, examining pronoun usage, agenda setting, storytelling, linguistic complexity, repetition, and electoral outcomes. Using NLP and AI models, this study conducts a quantitative analysis of executive speeches across sixteen nations and thirteen different languages, including low-resource languages. The selected countries include the United States and Canada in North America; Brazil and Argentina in South America; the United Kingdom and Norway in Europe; Israel in the Middle East; Egypt, Kenya, Seychelles, and Mauritius in Africa; and India, South Korea, Japan, Australia, and Indonesia in the Asia-Pacific region. This study analyzes how leaders use executive speeches to frame policies, project leadership, establish authority and influence stakeholders across presidential and parliamentary systems while navigating the complexities and liabilities of governance communication. This research contributes to the field of data science by creating a novel multilingual, multi-national corpus of executive speeches from presidential and parliamentary systems, which enables us to conduct a comprehensive comparative analysis of executive speeches for this study and lays the foundation for reusing the data corpus for future innovative studies on multi-national political language.

1.2 Statement of Problem

Executive speeches influence domestic politics, policy formations, public discourse, and geopolitics, yet large systematic comparisons of executive speeches between presidential and parliamentary systems remain scarce. Existing research on executive speeches focuses mainly on single systems or geographical regions, creating gaps in our comparative understanding. This limitation is particularly acute due to the absence of multilingual, multi-region corpora spanning presidential and parliamentary systems and the limited application of computational text analysis in comparative analysis.

Presidents engage citizens through personalized rhetoric, while prime ministers do so through personalized yet party-coordinated leadership, shaped by the demands of parliamentary governance (Dowding, 2013). However, studies highlighting how these institutional differences reflect in executive speeches remain limited. This research addresses this gap by creating a novel corpus of executive speeches from sixteen nations across six regions in thirteen languages, including low-resource languages. Data science advances in computational text analysis are used for comparative analysis of pronoun usage, linguistic complexity, storytelling patterns, repetition structures, and agenda-setting approaches across presidential and parliamentary systems. This research contributes to data science by creating a novel multilingual, multicultural, geographically varied political executive speech corpus and utilizing advances in computational text analysis for comparative political analysis.

1.3 Definition of Terms

- Agenda Setting: The process by which political leaders transform issues into policy priorities through strategic communication (Cobb & Elder, 1971).
- Anaphora: The repetition of the same word or phrase at the beginning of successive clauses or sentences.
- BERT (Bidirectional Encoder Representations from Transformers): An advanced deep learning model that analyzes each word in both left and right contexts within a sentence, typically up to 512 tokens, allowing it to capture nuanced, bidirectional language patterns.
- Bigrams/Trigrams: Two-word or three-word sequences analyzed to identify patterns in speech.

- Bootstrapped Mean Differences: A statistical technique that uses random resampling to estimate group parameters and calculate differences between groups without assuming underlying data distributions.
- Coalition-building Rhetoric: The strategic use of rhetoric and language by political leaders in parliamentary systems to maintain legislative support while balancing competing demands of coalition partners and their respective constituencies (Martin & Vanberg, 2008).
- Corpus Creation: Gathering text to create a new data set for computational analysis.
- Cross-System Comparative Analysis: A research methodology that compares how things work in different political systems to find patterns shaped by their unique structures.
- Epistrophe: The repetition of the same word or phrase at the end of successive clauses or sentences.
- Epizeuxis: The immediate repetition of a word or phrase without any words in between.
- Executive Speeches: Formal addresses delivered by heads of government/state.
- Flesch-Kincaid Grade Level: A readability formula that measures how easy a text is to read based on sentence length and word difficulty. Scores range from U.S school grades 3.0 to 12.0 and above, where 3.0–6.0 indicates elementary level, 6.0–9.0 middle school, 9.0–12.0 high school, and 12.0+ college-level or advanced reading.
- Institutional Constraints: Limits placed on leaders by the rules and structure of their government system that affect how they communicate and present policies.
- Institutional Differences: Structural variations between different governance systems.
- Mesodiplosis: The repetition of a word or phrase in the middle of successive sentences or clauses.

- N-Grams: Recurring multi-word sequences that become standardized phrases in particular contexts.
- Linguistic Complexity: Measures of language sophistication in political discourse.
- Low-Resource Languages: Languages with limited available digital resources, computational tools, or training data for natural language processing.
- Moving-Average Type-Token Ratio (MATTR): A linguistic complexity measure that calculates vocabulary diversity while controlling text length.
- Multilingual Corpus: A collection of texts in multiple languages created for computational analysis and cross-lingual comparison.
- N-gram Diversity: Measure vocabulary variation based on the ratio of unique word sequences to total sequences.
- N-gram Entropy: Measure predictability and randomness in word sequences.
- Narrative Arc: The structural pattern of storytelling in speeches, including elements of staging, plot progression, and cognitive tension, as measured by the Narrative Arc function in LIWC-22 (Boyd et al., 2022).
- Parliamentary System: A governance system where the executive is selected from legislature and is accountable to the legislature.
- Political Communication: Exchange of messages between political actors and the public.
- Political Rhetoric: Strategic use of language to persuade and influence political contexts.
- Presidential System: A governance system where the president serves as head of state and government and is elected independently of the legislature.
- Pronoun Usage: How presidents and prime ministers employ pronouns like "I," "we," "our," "they," "them" in speech.

- Pronoun-verb Collocations: Common combinations of pronouns and verbs that reveal action frames in speech.
- Repetition: Strategic reiteration of key phrases or concepts for emphasis.
- Repetition Density: Measure how frequently the same phrases recur within defined text segments.
- Rhetorical Strategies: Communication strategies used by executive leaders to persuade audiences and frame political messages.
- Semantic Embedding: Vector representations of words that capture meaning and relationships between concepts in computational text analysis.
- Sentiment Analysis: The computational analysis that identifies and extracts subjective information from text to determine the writer's attitude toward a particular topic, product, or service, typically classifying sentiments as positive, negative, or neutral (Liu, 2022).
- Topic Modeling: Computational techniques to identify themes in text collections.
- Type-Token Ratio (TTR): The ratio of unique words (types) to total words (tokens) in a text, measuring vocabulary diversity.
- Zipf's Law: Linguistic principles describing how word frequency distributions follow a power law pattern.

1.4 Significance of Study

Understanding how institutional structures shape executive communication can help navigate today's complex political landscape. This research bridges gaps in comparative political communication studies and data science by applying computational text analysis to compare executive speeches in presidential and parliamentary systems, revealing patterns that traditional qualitative approaches cannot capture at scale.

Globalization, migration, changing world order, and the emergence of digital platforms have created a more interconnected world where executive leaders must engage beyond national boundaries to address international challenges (Reeves, Malone, & O'Driscoll, 2008). The institutional differences between presidential and parliamentary systems shape the rhetorical choices of how executive leaders approach diplomatic relationships and build consensus on contentious foreign policy issues. Widmaier (2007) shows how leaders' rhetorical choices in interpreting international crises reshape policy debates and redefine national interests globally. The outcome of international agreements often depends on an executive leader's strategic use of rhetoric to bridge diplomatic differences, as evidenced in numerous negotiations.

Political rhetoric in executive speeches in the presidential and parliamentary systems utilizes language and communication to sway public opinion, frame policies, craft debates, and bridge gaps. Political rhetoric also helps advance political leaders' and parties' influence and agenda (Krebs & Jackson, 2007). Presidents, as independently elected heads of state and government, speak directly to citizens by rising above party politics as their authority is derived from electoral mandate rather than legislative support, thus promoting rhetoric centered on individual leadership. In contrast, prime ministers govern through legislative confidence and employ rhetoric that balances party ideology with national interests while acknowledging parliamentary supremacy and maintaining coalition consensus. Regardless of the system, political rhetoric fosters public trust in political institutions to advance democratic functionality and civic engagement (Rottenberg, 2020).

Executive rhetoric plays a key role in shaping domestic and international policies. An executive leader's rhetoric can significantly influence public perception and unify the nation. Individual examples clearly illustrate this influence. President Kennedy's call for national service

in his famous inaugural speech, "Ask not what your country can do for you, but what you can do for your country," inspired Americans toward shared sacrifice and civic responsibility. In contrast, Prime Minister Churchill's wartime rhetoric—"His Majesty's Government will never enter into negotiations with Hitler"—demonstrated parliamentary systems' collective action. The influence of executive rhetoric often extends to complex, sensitive policy issues like immigration; executive speeches frame immigrants as economic threats ("job stealers") or valuable contributors to national prosperity. Presidents frame their rhetoric around personal values and national identity ("This is a nation of immigrants" or "We must secure our borders"). At the same time, prime ministers align themselves within party agendas and coalition agreements ("Our government believes" or "Our party's position is"), demonstrating how executive rhetorical choices can shape public discourse and unite or divide public opinion on critical issues.

On the global stage, executive rhetoric becomes even more critical in addressing global challenges. President Obama's speech at COP21 demonstrated how executive rhetoric promoting a shared sense of responsibility can bridge gaps, leading to the Paris Agreement. Similarly, executive speeches gained attention in diplomatic negotiations to end the Russia-Ukraine conflict, particularly at the SCO summit. Indian Prime Minister Modi's declaration, "This is not an era of war," demonstrated how carefully crafted rhetoric by executive leaders can influence global diplomacy and promote stability.

By creating a multilingual corpus of executive speeches, this research contributes to data science by developing methods for analyzing political texts across different languages and institutional contexts. Further, it provides empirical evidence of how governance systems influence communication patterns across diverse political systems, cultural contexts and

languages, contributing to more globally representative approaches in political communication and computational linguistics.

1.5 Framework

This study examines executive speeches across presidential and parliamentary systems to understand how differences in institutional structures shape political communication patterns. Building on communication theories, we analyze how presidents' direct mandates contrast with prime ministers' collective responsibilities in framing priorities, targeting audiences, and consensus-building strategies. Focusing on five key communication theories and using a data driven analytical approach, we try to understand how these dynamics vary between presidential and parliamentary systems. Agenda-setting theory (McCombs & Shaw, 1972) explains how presidents' direct mandates provide greater agenda-setting freedom compared to prime ministers constrained by legislative support; Two-Step Flow Theory (Katz & Lazarsfeld, 1955) reveals how presidential systems rely on direct public appeal while parliamentary systems operate through collective decision-making; Uses and Gratifications Theory (Katz et al., 1973) reveals how presidents target broader public audiences while prime ministers focus on legislative and coalition audiences; Rhetorical Presidency Theory (Tulis, 1987) frames modern executives' increased use of direct public communication; and Spiral of Silence Theory (Noelle-Neumann, 1993) demonstrates how both systems employ strategies to present agendas as enjoying majority support through executive rhetoric based on their institutional structures.

Analyzing executive speeches in presidential and parliamentary systems using NLP and deep learning techniques can provide valuable insights into rhetorical styles and governance models. Researchers increasingly use automatic text analysis techniques to study political decision-making, moving beyond traditional qualitative approaches (Van Atteveldt et al., 2019).

Chatsiou and Mikhaylov (2020) showcase how advances in machine learning and artificial intelligence are creating new opportunities for analyzing political datasets, while Mitrani et al. (2022) highlight computerized text analysis as a promising direction that brings both significant potential and methodological challenges to political research. These findings align with Warsaw's (2019) observation that improved analytical methods now enable studying political phenomena at an unprecedented scale, supporting Brandt and Wagemans' (2017) call for methodologies that develop more generalizable knowledge across political systems. While previous studies of executive speeches have typically focused on single political systems or limited geographical regions, our cross-system comparative approach spans multiple geographies and institutional setups. Our study builds on recent methodological advances by performing a comprehensive quantitative computational analysis of executive speeches, with each method grounded in a relevant communication theory. (1) Pronoun analysis (using spaCy) examines rhetorical positioning and audience engagement, reflecting the personalization of leadership emphasized in Rhetorical Presidency Theory. (2) Linguistic complexity assessment (via Type-Token Ratio and semantic embeddings) captures the accessibility and functional purpose of language, aligning with Uses and Gratifications Theory. (3) Storytelling analysis (through narrative structure and sentiment patterns) examines message dissemination through intermediary cues, drawing on Two-Step Flow Theory. (4) Repetition pattern analysis identifies how leaders reinforce dominant narratives, connecting to the Spiral of Silence Theory. (5) Agenda-setting analysis (via topic modeling) investigates issue prioritization in line with Agenda-Setting Theory. (6) Electoral prediction using machine learning and deep learning explores how rhetorical strategies may relate to political outcomes across institutional contexts. This quantitative analysis of executive speeches, using advances in data science and

computational text analysis, enables us to identify and quantify rhetorical patterns across diverse political contexts that previous qualitative studies might have overlooked. Our approach simultaneously contributes to data science by creating a novel multilingual corpus and developing cross-institutional text analytics methodologies.

1.6 Purpose and Research Questions

This research aims to apply advanced data science methodologies to examine how institutional differences between presidential and parliamentary systems shape executive rhetoric across diverse political and cultural contexts. By leveraging computational text analysis, machine learning, and cross-lingual NLP techniques, this study aims to identify systematic executive communication patterns reflecting institutional constraints and opportunities. This approach contributes to the theoretical understanding of political communication and the methodological advancement of data science by developing novel techniques for analyzing political text across languages and systems. The research bridges the gap between data science and political science through a rigorous quantitative analysis of the following research questions:

RQ1: How does the use of pronouns in executive speeches vary between presidential and parliamentary systems?

Pronoun usage varies between presidential and parliamentary systems due to their institutional differences. Presidents derive authority directly from electoral mandates, while prime ministers must maintain legislative support, leading to different rhetorical approaches. Addo-Danquah et al. (2023) and Romadlani (2021) demonstrate that presidential leaders strategically employ personal pronouns to craft identities and influence public opinion. Wageche and Chi (2016) further show how presidents use first-person pronouns differently to emphasize individual leadership and persuasive rhetoric. However, comparative research analyzing how

these institutional differences influence pronoun usage remains limited to specific systems and geographies, creating a significant gap in our understanding of executive rhetoric.

To address this gap, we analyze the frequency of five core pronoun types: first-person singular, first-person plural, second person, third-person singular, and third-person plural. We also track pronoun shifts from individual to collective and from collective to individual to capture rhetorical transitions in speaker identity. Additionally, we identify pronoun–verb constructions by pairing individual and collective pronouns with past, present, and future tense verbs. These variables collectively allow us to examine how executive leaders in presidential and parliamentary systems use language to frame leadership, consistent with the principles of the Rhetorical Presidency Theory.

RQ2. How does the use of storytelling in executive speeches vary between presidential and parliamentary systems?

Storytelling and emotional appeals manifest differently across presidential and parliamentary systems. Presidents employ personal narratives to connect directly with citizens and make complex policies relatable (Norrick, 2013), with emotional appeals prominently featured in their rhetoric to build public support (Mazlum & Afshin, 2016). In contrast, prime ministers operate within more complex institutional constraints, balancing the interests of party members, coalition partners, and legislators, potentially shaping their emotional appeals and storytelling techniques differently from their presidential counterparts (Kaarbo, 1997). This institutional influence on storytelling in executive speeches requires further investigation, with Prior and Leston-Bandeira (2022) noting growing interest in storytelling techniques within parliamentary systems. However, systematic comparative studies remain limited, necessitating

further studies on how institutional differences shape storytelling approaches in executive speeches in presidential and parliamentary systems.

To address this gap, we analyze three fundamental narrative processes: staging frequency (articles and prepositions), plot progression patterns (pronouns, auxiliary verbs, conjunctions, negations, and non-referential adverbs), and cognitive tension markers (cognitive process words like "think," "understand," "believe"). These variables collectively allow us to examine how executive leaders in presidential and parliamentary systems use narrative structures to navigate the two-step communication process, consistent with the principles of the Two-Step Flow Theory.

RQ3: How does repetition vary between presidential and parliamentary executive speeches?

Repetition is a key rhetorical strategy executive leaders use to emphasize priorities, create urgency, and enhance message retention. Pieniążek-Niemczuk (2021) demonstrates that such repetition enhances persuasiveness in political discourse. Sallomi and Chiad (2020) extend this by showing how executive leaders in different political contexts use repetition for emotional emphasis and message reinforcement, while Grube (2012) highlights the institutional constraint on prime ministers who must balance rhetorical impact with institutional decorum, leading to different patterns of repetition in their parliamentary address. However, comparative research examining how institutional structures shape repetition in executive speeches requires further study.

To address this gap, we analyze repetition through four quantitative measures repeated n-gram type-token ratios (proportion of bigrams and trigrams appearing three or more times), entropy of repeated n-gram frequency distributions measuring predictability of repetition patterns, Zipf's law slopes analyzing frequency distributions of repeated phrases, and repetition

density frequency of repeated n-grams within 100-word sliding windows. These variables collectively allow us to examine how executive leaders in presidential and parliamentary systems use repetition strategies to shape perceptions of majority opinion and suppress dissenting views, consistent with the principles of Spiral of Silence Theory.

RQ4. How does linguistic complexity vary between presidential and parliamentary executive speeches?

Institutional differences between presidential and parliamentary systems suggest potential variations in the linguistic complexity of executive speeches. While presidents engage directly with the public, parliamentary leaders must navigate communication with informed legislators, party members, and the electorate, potentially shaping their linguistic choices. Bischof and Senninger (2021) establish that simple political messages improve comprehension and retention of political messages. Schoonvelde et al. (2018) further corroborate these findings by demonstrating that conservative politicians use simpler language than liberals across multiple countries and political contexts. Spirling (2016) extends this by adding historical context showing how Britain's Second Reform Act led parliamentary ministers to reduce linguistic complexity when addressing a broader electorate. However, comparative research analyzing linguistic complexity in executive speeches between the two systems is an unexplored area that requires more systematic studies.

To address this gap, we analyze linguistic complexity through structural complexity using Type-Token Ratio (TTR), moving average TTR (MATTR), average sentence length, clauses per sentence, parse tree depth, mean dependency distance, and Flesch-Kincaid Grade Level scores, semantic complexity through lexical sophistication indices, average cosine distances between word vectors, vector space density, and semantic clustering coefficients. Additionally, we assess

contextual complexity using BERT's contextual embedding variations, attention pattern complexity, and cross-sentence semantic coherence measures. These variables collectively allow us to examine how executive leaders in presidential and parliamentary systems adjust linguistic complexity to satisfy audience needs for comprehension, information processing, and decision-making support, consistent with the principles of Uses and Gratifications Theory.

RQ5: How does agenda-setting vary between presidential and parliamentary executive speeches?

Institutional differences between presidential and parliamentary systems suggest different approaches to agenda-setting in executive speeches. While presidents can directly frame policy priorities through rhetoric, prime ministers must balance parliamentary dynamics and coalition interests. Vercesi (2020) demonstrated that parliamentary executives operate within a complex setup requiring strategic issue selection with changing political contexts and coalition relationships further shaping these choices (Turnbull, 2017). In contrast, Edwards III and Wood (1999) find that executives in presidential systems can more independently focus on specific issues through speeches. However, comparative studies analyzing how agenda-setting varies between presidential and parliamentary systems have not been conducted before.

To address this gap, we use BERTopic with topic per class to examine topics and their weight within each system, speech-level topic assignment, and topic distributions over time. These variables collectively allow us to examine how executive leaders in presidential and parliamentary systems shape public perceptions of policy importance and priority, consistent with the principles of Agenda-Setting Theory.

RQ6: To what extent can linguistic patterns in executive speeches predict electoral outcomes?

Executive rhetoric may contain linguistic hints to signal electoral prospects. Conway et al. (2012) find that shifts in rhetorical complexity can indicate electoral vulnerability, while Grube (2011) demonstrates how leaders adapt their rhetoric based on institutional position and electoral timing. Studies have further established that hostile and emotional campaign rhetoric correlates with increased media attention and electoral success (Gerstlé & Nai, 2019). However, research examining the predictive relationship between executive speeches and electoral outcomes using combined data from different institutional contexts has not been attempted before.

1.7 Summary

This study makes several key contributions to data science: (1) the creation of a first-of-its-kind multilingual corpus of executive speeches for computational research; (2) developing novel methods for cross-lingual analysis of political texts; (3) demonstrating applications of advanced NLP techniques to comparative political communication; and (4) establishing a computational framework for analyzing institutional influences on communication patterns. Through this interdisciplinary approach, the dissertation aims to advance our understanding of how institutional structures influence political communication while developing innovative computational methods for cross-linguistic political text analysis. The next chapter will review relevant literature, examine theories and empirical findings while identifying the research gaps this study addresses.

2

Literature Review

Executive leaders use speeches as an essential tool for political rhetoric. The language used by executive leaders in their speeches helps shape public opinion, set policy agendas, and mobilize support (Cohen, 1995; Eshbaugh-Soha, 2010), while the institutional setup and cultural contexts influence leaders' communication styles and strategies (Lim, 2008). These institutional and cultural elements combine to form fundamental aspects of political leadership and persuasion (Charteris-Black, 2005). Based on this understanding, executive leaders use personal pronouns, emotional language, storytelling, repetition, and policy framing to effectively convey their messages and connect with their audiences (Brader, 2005; Green & Brock, 2000; Tannen, 1987). A comparative analysis of these rhetorical strategies in executive speeches between presidential and parliamentary systems using data driven analytical approaches can shed light on how institutional factors influence executive communication styles and their linguistic choices.

The emergence of artificial intelligence, particularly natural language processing (NLP) and deep learning techniques, has led to novel ways of analyzing a large corpus of text data, including political texts (Grimmer & Stewart, 2013). The data science pipeline encompassing

data collection, preprocessing, feature extraction, modeling, and visualization can transform unstructured political speeches into quantifiable insights using NLP-based text analytics and visualization (Katre, 2019). Utilizing these methodological advances in data science, NLP techniques such as part-of-speech (POS) tagging, sentiment analysis, topic modeling, and predictive modeling have been applied to understand the language and rhetoric of political communication (Sim et al., 2013), issue framing (Tsur et al., 2015), and political polarization (Gentzkow et al., 2019). These developments in data science and computational text analysis offer enormous possibilities for analyzing a large corpus of political texts to gain insights into executive minds and governance styles across political systems (Glavaš et al., 2019) helping create generalized, reproduceable and scalable research framework for more advanced comparative studies in future.

The first section of the literature review focuses on exploring the intersection of communication theory and political communications. The second section highlights rhetorical and linguistic strategies political actors use to advance their agendas. The third section of the literature review explores the application of machine learning and natural language processing (NLP) techniques in predictive analysis using political text.

2.1 Communication Theories

Political communication and institutional discourse theories set the foundation for a comparative analysis of executive speeches in presidential and parliamentary systems. These theories collectively explain how differences in accountability structures, decision-making processes, executive relationships with legislatures, and message dissemination produce communication patterns distinct from presidential and parliamentary systems. The application of data science driven approach to a large speech corpus would enable us to quantitatively measure

and validate these theoretical frameworks in different political contexts, offering empirical evidence that were previously unavailable.

Executive leaders influence public discussion through discourse and emphasis choices, with their institutional setups determining their agenda-setting power. Agenda-Setting Theory (McCombs & Shaw, 1972), with its roots in mass communication, has been extended to political communication to explain how executive leaders influence public priorities by repeatedly emphasizing specific issues in their rhetoric. The more attention an issue receives in executive speeches, the more likely it is to be perceived by the public as important. Building on this understanding, we extend the theory to examine how leaders make specific issues more prominent in public debate leveraging their institutional freedom. This theory is relevant to our research as presidents enjoy more agenda-setting freedom due to their direct mandate and separation of powers, unlike prime ministers, who require legislative and party support. Despite the theoretical significance, few studies have systematically compared agenda-setting patterns across presidential and parliamentary systems using quantitative methods. Our study addresses this research gap by systematically comparing how these institutional constraints shape agenda setting in executive communication across diverse political systems contributing to our understanding of how institutional structures directly shape communication strategies.

Executive communication reaches stakeholders through various intermediaries and channels, with institutional structures influencing message transmission patterns. Two-Step Flow Theory (Katz & Lazarsfeld, 1955) proposes that political messages reach the public indirectly, first influencing key opinion leaders, who interpret and relay those messages to the broader audience. This theory connects to our study as it helps us understand how presidents, with their direct appeal, and prime ministers, with collective decision-making, foster relationships with

their intermediaries. However, more studies are needed to understand how these transmission patterns vary between presidential and parliamentary systems. Analyzing how institutional setups affect message transmission using quantifiable data driven analytical approach will reveal how these governance systems shape linguistic choices and narrative approaches in executive speeches contributing to our understanding on how institutional structures influence flow of communication across different democratic systems.

Executive leaders tailor their messages to specific audiences based on their institutional settings. Uses and Gratifications Theory (Katz et al., 1973) shifts the focus from the message to the audience, proposing that individuals actively seek out content that meets their needs for information, identity, social connection, or decision-making. This theory directly relates to our comparative analysis of presidential and parliamentary systems and how executive leaders adapt their rhetoric to meet broad audience aspirations. A direct electoral mandate may lead presidents to address broader public audiences while prime ministers' resort to more targeted communication with the legislature, party supporters, and coalition partners. However, studies comparing how this theory applies across different institutional setups remain limited. Comparing linguistic patterns in executive speeches using a large corpus of executive speeches aided by analytical approach will reveal how institutional structures shape language focused on the audience and how it varies between systems, thereby addressing this research gap.

Executive leaders' communication styles reflect their institutional roles and constraints. Rhetorical Presidency Theory (Tulis, 1987) argues that modern presidents increasingly rely on direct public communication, particularly through speeches, to assert leadership and influence public opinion. Despite its theoretical importance, the theory remains confined to the American presidency. Our study extends this theory to parliamentary systems backed by data driven

analytical approach to determine whether collective cabinet responsibility produces the same rhetorical patterns as presidential systems, thereby contributing to our understanding of how this theoretical understanding applies to parliamentary systems and executive rhetoric beyond the American presidential context.

Perceptions of public opinion influence executive communication strategies differently across institutional contexts. Spiral of Silence Theory (Noelle-Neumann, 1993) proposes that individuals are less likely to express their opinions when they perceive that their views are in the minority, for fear of social isolation or rejection. This theory connects to our study explaining how executives in both systems present their agendas as having majority support, potentially silencing opposing views. However, studies on the applicability of this theory in presidential and parliamentary systems are limited. Our study employs computational text analysis and natural language processing techniques to examine how presidential and parliamentary leaders employ different rhetorical strategies to create perceptions of consensus and majority support, based on their distinct institutional structures, revealing how consensus-building tactics vary across democratic systems.

These communication theories collectively lay a foundation for institutional, and audience influence on executive communication. Agenda-Setting and Rhetorical Presidency Theories explain how executive leaders project authority and shape public priorities. Two-Step Flow and Uses and Gratifications Theories describe how messages are filtered through intermediaries and tailored to audience needs. Spiral of Silence Theory shows how rhetorical strategies can create a sense of consensus and discourage dissent. Our comparative study of executive speeches between presidential and parliamentary systems applies data driven analytical approach using computational linguistics and natural language processing techniques to address

this gap by creating quantifiable metrics that can help us extend our understanding of these theories to diverse presidential and parliamentary systems worldwide, revealing how institutional and cultural differences shape executive communication patterns and linguistic choices.

2.2 Pronoun Usage

Pronoun usage in executive speeches varies across political systems, giving insights into leadership and institutional structures and emphasizing our belief that executive speeches in presidential systems will contain more first-person singular pronouns than those in parliamentary systems. Pennebaker's (2011) foundational work shows that personal pronouns reveal speakers' psychological states and relationships. Steffens et al. (2014) extends Pennebaker's findings to political communication, contending that using first-person plural pronouns fosters collective identity, audience connection, and unity. Even though these studies highlight the significance of pronoun usage it left a gap in the comparative understanding of how institutional structures might influence pronoun usage.

A comparative study of pronoun usage in executive speeches between presidential and parliamentary systems is an unexplored area. Existing research has focused on a single institutional context, and their findings are often contradictory. Slatcher et al. (2007) analyzed U.S. presidential candidates, revealing that candidates who used first-person singular pronouns are considered more candid and personal. This finding contrasts with Steffens and Haslam's (2013) findings that Australian prime ministers employing collective language are considered more charismatic and influential. While these studies hint at systematic differences, they are limited to U.S. and Australian contexts, lacking cross-system comparison and limiting their generalizability.

The institutional context in which executive communication occurs plays a vital role in pronoun usage in executive speech. Gunsch et al. (2000) highlighted the high usage of first-person singular pronouns in executive speeches in the U.S. presidential system. Hatab (2013) and Vuković (2012) build upon this research to different cultural and political contexts by analyzing speeches from Middle Eastern country leaders and in the Parliament of Montenegro. These studies collectively indicate that institutional context influences pronoun usage. However, these studies are limited to single countries and institutions, highlighting the need for a broader analysis of pronoun usage between presidential and parliamentary systems.

The use of personal pronouns in political speeches reflects power dynamics and status relationships. Chung and Pennebaker (2011) established that pronoun usage reflects social dynamics, with first-person singular pronouns often indicating lower status in interaction. Extending this to the political space, Alavidze (2016) demonstrated that in political discourse specifically, pronouns reveal speakers' attitudes and intentions—with "I" showing authority, "we" creating collective identity, and "you" establishing audience solidarity. Even though these studies highlighted the importance of pronoun usage in social and political contexts, the findings can further be extended to understand how different institutional setups shape executive communication patterns.

A comparative analysis of pronoun usage in executive speeches between presidential and parliamentary systems can offer unique insights unavailable through existing single-system studies. Understanding these differences is significant because pronoun choices reflect power, leadership style, accountability, and influence citizen trust. Leveraging advances in computational text analysis, we extend qualitative findings from earlier single-system studies through data science approaches to quantify pronoun usage patterns in executive speeches

between the presidential and parliamentary systems. Our methodology employs advanced natural language processing algorithms to analyze how institutional structures influence leaders' pronoun choices across a large-scale corpus of executive speeches. Building on Rhetorical Presidency Theory emphasize on personalized executive communication, we focus on variables such as the frequency of first-person singular, first-person plural, second-person, and third-person pronouns; shifts between individual and collective pronouns; and pronoun–verb combinations across past, present, and future tenses to capture differences in how leaders project authority in presidential and parliamentary systems. These methodological innovations will reveal whether the concentrated power in presidential systems leads to more individualistic language (I, me), while the collective decision-making in parliamentary systems results in more communal rhetoric (we, us). Further, these linguistic patterns will provide empirical evidence of how institutional constraints shape executive communication styles, helping speechwriters, policy communicators, and political consultants adapt their approach to different political systems.

2.3 Repetition

A comparative study of executive speeches between presidential and parliamentary systems can reveal how institutional structures influence repetition patterns. Various forms of repetition in political speeches, from anaphora, epistrophe, mesodiplosis, and epizeuxis, serve different rhetorical purposes (Fahnestock, 2011). While Fahnestock's work sets the foundation for repetition usage in political speeches, the effectiveness of repetition across different institutional setups requires further study.

The psychological impact of repetition, even though well recorded, could vary across institutional contexts. Alter and Oppenheimer (2009) demonstrate how repetition enhances

information processing and shapes audience perceptions. Fazio et al. (2015) extend this finding by establishing repetition's role in the "illusion of truth" effect. These findings align with Crowder's (2014) and Cacioppo et al.'s (1989) study that reinforces the role of repetition in memory and emotional impact. However, the experimental nature of these studies leaves unexplored the practical application of repetition in political speeches and how it varies between different presidential and parliamentary systems.

Studies examining repetition patterns in executive speeches suggest systematic differences between political systems. Tannen's (2007) work on the presidential system demonstrates how presidential systems rely on repetition for direct public persuasion and emotional resonance. This finding contrasts with Heritage & Greatbatch's (1986) findings that high repetition has a distinct function in parliamentary debates, primarily to build collective agreement and consensus. Benoit's (2001) analysis of U.S. presidential campaigns supports this institutional divide and provides evidence of how repetition serves personal leadership goals in presidential systems. However, all these studies are limited to single systems, limiting broader applicability and our understanding of how institutions shape repetition strategies.

Social media's emergence has introduced new dimensions to political repetition by highlighting continued institutional differences. Ott's (2017) examination of U.S. presidential tweets reveals leaders' systematic deployment of repetitive slogans for direct public appeal, contrasting with Ross & Rivers' (2018) findings of parliament members' more nuanced messaging approaches. However, these studies are limited to digital platforms and are confined to single institutional contexts and fail to explain how different institutional structures might drive these variations in repetition strategies.

A comparative analysis of repetition patterns in executive speeches between presidential and parliamentary systems will address these crucial research gaps identified in the reviewed literature. Understanding these differences is important because repetition is a key rhetorical strategy in shaping public opinion and reinforcing policy priorities. Our research makes explicit data science contributions by creating a computational framework for quantifying repetition and identifying system-specific patterns across political systems using natural language processing algorithms. Drawing on Spiral of Silence Theory's explanation that people are less likely to speak out when they believe the majority holds a different view, we examine how repetition in executive speeches can create the impression of widespread public support. We analyze variables such as bigram frequency, lexical bundle concentration, and entropy-based predictability, which capture how leaders reinforce core messages to signal dominant opinion. This data-driven approach will reveal institutional differences in repetition strategies and measure which system uses repetitive patterns more frequently, helping us advance our understanding of repetition in executive speeches beyond single-country, single-institutional contexts.

2.4 Storytelling

Executive leaders leverage storytelling in their speeches as a rhetorical technique to develop emotional bonds with citizens. The narrative paradigm theory (NPT) by Fisher (1984) sets the foundation for this belief by asserting that humans are natural storytellers and that narratives that share our ethos, experiences, and beliefs are more likely to be accepted. Jones and McBeth's (2010) extend this to political landscape by demonstrating how narratives systematically shape public opinion, influence decision-making, and rally support for political agendas. These studies, though, do not address the influence of institutional structures on

storytelling in executive speeches, leaving a gap in how narrative strategies might differ between presidential and parliamentary systems.

Studies on storytelling in executive speeches have demonstrated the power of storytelling in political communication. From a presidential context, Hammer (2010) analyzed Barack Obama's 2004 Democratic National Convention keynote address, revealing that his personal stories helped establish credibility, connect with the audience, and articulate a vision of shared values and common purpose. However, the single-system study limits storytelling's broader applicability and generalizability. Further investigating this, Franch (2018) studied the use of storytelling in U.S. presidents' speeches during wartime, observing that presidents used narrative techniques to reinforce the idea of America's duty to preserve freedom, even across ideologically different administrations. Even though it is limited to the U.S. context, these studies provide valuable information on storytelling in presidential rhetoric; however, they do not answer how these patterns could vary between executive speeches in presidential and parliamentary systems.

Social and political contexts influence storytelling in executive speeches. Salmon (2017) examined these contextual influences, demonstrating that politicians have used personal narratives and anecdotes to connect with voters and establish credibility across various political contexts. Advancing this narrative understanding, Chong and Druckman (2007) observed that storytelling had been widely used in political campaigns to craft compelling narratives that resonate with voters' minds. However, their research does not focus on how political systems shape narrative choices. Building on this understanding of political rhetoric, Jones and McBeth (2011) showed how advocacy groups strategically frame issues and mobilize support through storytelling. These align with Polletta's (1998) findings, that social movements in political history had used narratives in their communication to foster solidarity and urge collective action.

While these studies collectively highlight the effectiveness of storytelling in different contexts, they failed to explain how institutional differences between presidential and parliamentary systems influence storytelling.

Storytelling techniques in executive speeches can heighten audience engagement and persuasion. Focusing on audience engagement, Appel and Richter (2007) demonstrated that narratives can transport individuals into a world of stories, fostering greater emotional engagement and empathy. Building on this understanding of narrative impact and extending to political domain, Carpini and Keeter (1996) showed that storytelling can make complex political issues more accessible and relatable to the general public, facilitating comprehension and eliciting emotional responses. However, these studies do not answer how storytelling varies across institutional setups. Advancing this understanding Entman (1993) and Chong and Druckman (2007) noted that narratives can shape public perception by selectively highlighting certain aspects of an issue and downplaying others. Extending this further, Lerner et al. (2015) found that emotionally charged narratives can trigger specific emotional responses, influencing decision-making and policy preferences. However, these findings on storytelling remain generalized and do not explore how institutional variations between presidential and parliamentary systems influence storytelling. While these studies collectively establish a cause for storytelling's persuasive power, they remain short of how institutional differences between presidential and parliamentary systems shape narrative storytelling in executive speeches.

Current research on executive communication overlooks how institutional structures shape storytelling approaches and their variation between systems. Understanding these differences is crucial because of storytelling's power to create emotional connections and shared national narratives. Our study addresses this research gap by investigating how institutional

frameworks influence storytelling in executive speeches. Drawing on Two-Step Flow Theory's understanding of intermediaries in message transmission, we examine how executive leaders in presidential and parliamentary systems craft narratives through staging, plot progression, and cognitive tension for effective message transmission. This data-driven comparison of presidential and parliamentary systems allows us to understand how institutional structures influence storytelling effectiveness, emotional connection, and public engagement across different political contexts.

2.5 Agenda Setting-

Executive leaders use rhetoric to shape national priorities and policy debates, with agenda-setting taking different forms in presidential and parliamentary governments. While Cohen (1995) demonstrates agenda-setting's impact on public opinion formation, his work does not consider the institutional perspective. Eshbaugh-Soha et al. (2005) expand this analysis by showing how presidential systems enable more substantial agenda-setting capabilities through executive speeches. However, these studies do not address how institutional differences between presidential and parliamentary systems shape agenda-setting in executive speeches.

Empirical evidence demonstrates the impact of agenda-setting in executive speeches. Analyzing agenda setting in the presidential system, Peake and Eshbaugh-Soha (2010) discovered that U.S. presidents' focused attention on specific issues in speeches significantly increased public perception of those issues' importance. Rutledge and Larsen Price (2014) extend these findings to legislative agenda setting, showing that policy priorities in presidential speeches directly influenced Congress's legislative agenda, with highlighted issues receiving increased attention in congressional hearings and bill introductions. While these studies establish apparent

agenda-setting effects in presidential systems, they lack comparative insight into how such effects might differ in parliamentary systems.

Comparative studies of executive speeches between presidential and parliamentary systems reveal distinct agenda-setting institutional capabilities. A comparative study of the U.S. and U.K. systems by Binzer Hobolt and Klemmensen (2008) revealed that U.S. presidents enjoy greater freedom in setting policy agendas while U.K. prime ministers remain constrained by party and coalition priorities. Even though novel, this study was limited to two countries. Habel (2012) extended this finding by including the Canadian system, showing that U.S. presidential speeches address a broader range of policy issues to appeal to diverse constituencies. In contrast, parliamentary executives in Canada and the U.K. remain bound to their party's legislative agenda. Even though these studies attempted cross-system comparisons, these studies were limited to a few Western democracies, lacking a broader generalizable cross-national, cross-cultural analysis of how institutional differences shape agenda-setting strategies in executive speech between presidential and parliamentary systems.

Legislative independence of presidents offers them enhanced agenda-setting capabilities compared to prime ministers. Wood's (2007) work demonstrates this by showing how direct election and separation of powers enable presidents to set independent national agendas. However, this study does not consider how party agendas influence prime ministers' agenda-setting in parliamentary systems. Building on this institutional perspective, Eshbaugh-Soha and Peake (2005) and Eshbaugh-Soha (2010) reveal how media coverage amplifies presidential influence and how this effect peaks during early presidential terms. However, these studies are limited to the U.S. context and do not account for other presidential or parliamentary systems,

lacking a systematic comparison of how institutional differences affect agenda-setting capabilities in executive speeches across diverse presidential and parliamentary systems.

While existing research suggests agenda-setting capabilities differ between presidential and parliamentary speeches, current studies are limited to a few Western democracies focusing primarily on policy content rather than how agenda-setting varies in executive speeches between the two systems, leaving a critical gap. Understanding these differences is crucial because agenda-setting directly shapes policy priorities and public discourse, determining which problems receive attention. Our research addresses this research gap and contributes to data science by utilizing advances in natural language processing to identify and quantify agenda-setting patterns and policy priorities in executive speeches between presidential and parliamentary systems. Extending Agenda-Setting theory, which states that political actors influence public priorities by emphasizing specific issues, we measure agenda-setting through variables such as topic weight, topic concentration, and temporal trends derived from topic modeling techniques. This data science approach provides quantifiable evidence about how institutional structures help executives set national priorities, providing empirical insights previously unavailable through traditional qualitative analysis. This data-driven comparative analysis enhances our understanding of how institutional setups influence executive communication strategies and their ability to set national agendas.

2.6 Linguistic Complexity

Institutional differences between presidential and parliamentary systems influence the linguistic complexity of executive speeches. While studies document a general decline in political language complexity over time (Diermeier et al., 2011), these findings are often based

on single political systems. In the presidential context, an analysis of U.S. presidential rhetoric reveals consistent patterns of declining complexity (Conway & Zubrod, 2022) while studies on the parliamentary system by Treimane (2011) and Tsakona (2012) demonstrate how institutional design, political culture, and the need for deliberation and consensus affect linguistic complexity. However, these studies are limited to specific systems and fail to provide comparative insights into linguistic complexity between presidential and parliamentary systems.

Linguistic complexity is a rhetorical strategy that executives use to communicate and establish authority. An analysis of U.S. presidents from 1981 to 2021 (Urbaniak, 2022) shows how presidents use linguistic complexity in their inaugural addresses to signal competence. According to Lim (2008), this decline in linguistic complexity is a deliberate choice to simplify discourse and manage public opinion. While these studies highlight the significance of linguistic complexity, their limited scope in the U.S. presidential system limits their generalizability and applicability to other presidential and political systems. Electoral studies further complicate this understanding; a study by Thoemmes and Conway III (2007) demonstrated that winning presidents strategically decreased complexity closer to reelection, while losing candidates reduced complexity too early in their terms, contrasting with Kayam (2018) that presidential candidates gain more support using more straightforward language. While these studies highlight the nuances of linguistic complexity, their limited focus on the U.S. system leaves unexplored how they unfold in other presidential and political systems.

Executive heads in parliamentary systems balance their linguistic complexity to adjust to institutional needs, party agendas, and legislative audiences. Focusing on audiences, Spirling (2016) demonstrated that British parliamentary leaders varied complexity between elite and public audiences, while Lin and Osnabrügge (2018) demonstrated how parliamentary leaders

adjusted their speech complexity to match their constituents' linguistic skills. However, both studies analyzed linguistic complexity from an audience comprehension perspective in a parliamentary system, overlooking the influences of institutional setups. From an institutional perspective, a study by Bischof and Senninger (2018) showed how Austrian parliamentarians simplified language during consensus-building. Extending this to an ideological angle, Schoonvelde et al. (2019) revealed systematic variations in linguistic complexity between right and left-wing parties across Western Europe. Even though these studies provided interesting insights, these studies were again limited to specific geographies and parliamentary setups, overlooking how institutional variations influence linguistic complexity in executive speeches between presidential and parliamentary systems.

While existing research provides information on linguistic complexity in specific political systems, a comparative analysis between presidential and parliamentary systems remains limited. Understanding differences in linguistic complexity is important as it impacts public comprehension of policies and citizen engagement in the democratic process. Our research addresses this gap by adopting a data science approach to quantify and compare linguistic complexity across institutional contexts using a large corpus of executive speeches from diverse political systems. Leveraging Uses and Gratifications Theory, which holds that audiences actively seek and engage with messages that meet their informational and cognitive needs, we measure linguistic complexity through lexical, semantic, and syntactic features. This data-driven approach will help us quantify and reveal patterns previously undetectable through qualitative analysis and thus enhance our understanding of how governance structures influence political discourse and communication strategies, with implications for democratic transparency and effective civic participation.

2.7 Prediction using NLP

Linguistic choices in executive speeches of presidential and parliamentary systems are linked to electoral outcomes. A foundational work by Chatsiou and Mikhaylov_(2020) demonstrated how Machine learning and natural language processing techniques can extract predictive insights from diverse data sources, including political speeches. This general study though did not factor in institutional differences. Liu et al. (2021) and Khan et al. (2021) addressed some of these limitations by developing integrated time series analysis with machine learning and ensemble methods to predict specific institutional contexts. However, these studies failed to consider how linguistic variations between presidential and parliamentary systems can be harnessed to create a predictive model that can work across the system.

AI and NLP models have evolved to adapt to diverse political systems and institutional setup complexities. Addressing this complexity, Feng et al. (2023) developed a two-tier architecture combining sentiment analysis with institutional metrics, achieving high predictive accuracy. However, their approach was limited to specific institutional contexts. Ali et al. (2022) expanded these capabilities by implementing deep learning models that capture complex patterns in political text for prediction but fell short of exploring how variations in political systems might require distinct modeling approaches. Even though these studies advanced the field of AI/NLP by using political text for predictive purposes, further research is needed to create a unified predictive model that can factor in institutional differences and be deployed for electoral prediction, irrespective of the system.

The linguistic relationship between executive speeches across varying institutional setups and electoral outcomes can open new possibilities in predictive electoral analysis.

Acknowledging this, Perera and Karunanayaka (2022) demonstrate new possibilities in predictive electoral analysis through the effectiveness of fuzzy-rough set classifiers in political discourse analysis, though their study was limited to a single system. Brito and Adeodato (2023) and Elkink and Farrell (2021) built upon this foundation by integrating traditional polling data with computational analysis, showing improved prediction accuracy. While these studies introduced innovative approaches to the field, they failed to propose a unified predictive model that can effectively account for institutional variations between presidential and parliamentary systems and how executive speeches can influence voter response.

The choice of predictive models for electoral prediction requires adaptation to institutional contexts. Brito et al. (2021) established fundamental computational frameworks for analyzing political discourse, but their work did not fully explore how these system-level differences might necessitate distinct architectural choices. Santos et al. (2021) advanced this understanding by highlighting how data quality and model bias affect predictive accuracy. However, these studies did not address how differences in institutional setups affect prediction and how a unified model can address those challenges.

A unified predictive model that can analyze executive speeches from presidential and parliamentary systems for electoral forecasting would address a critical research gap as current prediction models are system-specific and limit cross-system applicability. Our research contributes to data science by developing a novel deep-learning model that maintains predictive accuracy across presidential and parliamentary systems without requiring separate modeling approaches. Such a unified generalizable model would significantly advance electoral forecasting methodology by overcoming system-specific limitations, providing a generalizable

tool for psephologists, media analysts, policymakers, and campaign strategists to predict electoral outcomes consistently across diverse political environments.

2.8 Machine Translation

A comprehensive analysis of executive speeches across multiple geographies and languages necessitates a practical approach to large-scale translation that preserves critical linguistic cues essential for reliable cross-linguistic comparison. Windsor et al. (2019) provide foundational evidence that machine translation using Google Translate can produce linguistic outputs that closely mirror human translations across multiple languages and that it can preserve psychological and summary language variables across most LIWC categories. Building on this foundation, Boot (2021) demonstrates that machine translation using tools like LIWC can effectively preserve word categories and psychological indicators sufficiently for reliable cross-linguistic comparison. Extending these findings specifically to political discourse, Licht and Lind (2023) confirm that machine translation is a viable tool for analyzing multilingual political texts, demonstrating that core meanings, issue positions, and topical structures are preserved across languages. Complementing this, Meier et al. (2021) demonstrate that psychological meaning is well retained in translated speeches, even when the linguistic form changes. These findings support machine translation as a methodologically sound approach for rhetorical analyses of executive speeches, facilitating cross-linguistic analysis while preserving the linguistic cues necessary for meaningful comparative insights.

2.9 Analytical Framework

An NLP-based approach using spaCy for pronoun analysis in political discourse can reveal power dynamics and leadership styles. Qi et al. (2020), through their comparative studies of natural language processing toolkits, highlighted spaCy's strengths in POS tagging and dependency parsing critical for pronoun identification. Raut and Mahender (2022) built upon this by showcasing spaCy's superior performance in part-of-speech tagging and more reliable pronoun categorization for complex political texts. Anwar et al. (2023) further confirmed spaCy's exceptional accuracy in tagging parts of speech across diverse text types, providing the reliability necessary for comparative analysis of pronoun usage patterns. Richter et al. (2023) applied spaCy's capabilities to large-scale political discourse analysis using 920,000 German parliamentary speeches spanning seven decades, establishing the robustness of spaCy- for large-scale analysis. Despite these studies highlighting the robustness of spaCy, a critical gap remains in the limited use of spaCy for comparative analysis of pronoun usage patterns in executive systems between presidential and parliamentary systems.

Quantifying narrative arcs in executive speeches reveal how political leaders structure policy stories across governance systems. Krebs (2015) established that political narratives achieve dominance in discourse by mixing current challenges, past failures and triumphs, and potential futures into a coherent tale that defines legitimate politics and shapes policy priorities. Building on this narrative understanding Boyd et al. (2020) used Pennebaker et al.'s (2015) LIWC approach to develop a computational method that identifies three core narrative elements: (1) staging (who, when, where), (2) plot progression (how the story moves forward), and (3) cognitive tension (conflict, dilemma, or decision). However, applying the narrative arc in the political domain for a comparative analysis of executive speeches between presidential and parliamentary systems requires further study.

Repetition patterns in executive speeches can reveal quantifiable linguistic patterns across different governance structures and institutional systems. Manning and Schütze (1999) set the theoretical foundation by establishing that the "frequency of a word type is inversely proportional to its rank" through Zipf's law. Building on this foundation, Bird et al. (2009) developed the Natural Language Toolkit (nltk), creating a computational framework for extracting n-grams that capture insights into word relationships for measuring repetitive patterns. In advancing this methodological approach, Bentz et al. (2017) applied information theory to linguistic analysis by showing how entropy calculations effectively measure the predictability and randomness of word sequences while addressing the challenges of text size variance. Despite these methodological advances, a significant research gap exists in applying these advancements to compare executive speeches between different institutional setups.

Institutional structures fundamentally shape how executive leaders frame policy priorities through speeches, requiring advanced computational methods to analyze complex communication patterns. Grootendorst (2022) introduced BERTopic to generate document embeddings and topic representations using transformer-based language models with class-based TF-IDF procedures for more precise capture of semantic relationships in political texts. Pinecone (2023) extended this by demonstrating BERTopic's superior computational advantages for processing complex data through UMAP and HDBSCAN integration. Practical applications of BERTopic by Achmann and Wolff (2023) revealed its effectiveness in distinguishing between policy-focused and documentation-focused political content during election campaigns and its superior performance over traditional methods like LDA for analyzing political communications (Mendonça & Figueira, 2024). Even though BERTopic has proven its robustness and utility in

the political domain, a research gap exists in applying BERTopic for comparative analysis of agenda-setting patterns across different political systems.

Measuring linguistic complexity in executive speeches in presidential and parliamentary systems can quantify linguistic complexity patterns. The measurement of lexical complexity has advanced with Covington and McFall's (2010) development of the Moving-Average Type-Token Ratio (MATTR), which calculates lexical diversity across overlapping word windows to provide a stable estimate unaffected by text length. Advancing to syntactic analysis, Honnibal and Johnson (2015) developed neural dependency parsing methods that achieved accuracy while maintaining computational efficiency suitable for processing large corpora. The emergence of transformer models added new analytical dimensions. Devlin et al. (2019) demonstrated that BERT's bidirectional training produces representations that fuse left and right context, allowing it to model deep interactions between words in both directions. Building on Devlin et al. (2019) findings, Clark et al. (2019) discovered that different attention heads specialized in distinct linguistic phenomena, with some heads focusing predominantly on syntactic relationships while others capture broader semantic patterns. Despite these methodological advances, the application of these findings for a comparative analysis of linguistic complexity across political systems remains further studies.

Executive speeches from presidential and parliamentary systems may contain linguistic clues to predict electoral outcomes. Chen and Guestrin's (2016) XGBoost implementation provided a practical gradient-boosting framework that offered improved handling of complex feature interactions and non-linear relationships in predictive modeling. Applying these ensemble learning advances to political forecasting, Clark et al. (2018) successfully implemented

random forest algorithms to predict electoral outcomes using e-petition data, demonstrating ensemble learning effectiveness in capturing voting patterns based on textual analysis. Sinha et al. (2024) expanded this methodological approach by comparing multiple machine learning techniques, finding that gradient boosting and random forest models achieved superior accuracy in predicting U.S. presidential election outcomes by effectively capturing the complex relationships between economic indicators and voter preferences. However, research remains limited on how linguistic variations in executive speeches between presidential and parliamentary systems can provide valuable insights into predicting electoral outcomes.

Despite significant advances in computational linguistics and natural language processing, a critical research gap exists in applying these techniques for comparative quantitative analysis of executive communication across different institutional systems. Our approach integrates spaCy for pronoun analysis, Boyd's narrative arc framework, entropy-based repetition measurement, BERTopic for agenda-setting analysis, and transformer-based linguistic complexity assessment to analyze and quantify political texts at scale. The contributions of such an approach include quantitative metrics for measuring institutional influences on communication, reproducible models for cross-system political text analysis, and a demonstration of how data science techniques can contribute to large-scale quantitative analysis in political communication research.

2.10 Conclusion

This comprehensive literature review demonstrates how rhetorical strategies in political communication vary based on institutional structures. Understanding these rhetorical differences is crucial because they fundamentally shape how executive leaders communicate policy, build

public consensus, establish authority, and navigate institutional constraints across different governance systems. While existing research provides valuable insights into pronoun usage, repetition, storytelling approaches, and linguistic complexity in political communications, studies have remained primarily confined to single geographies or institutional setups from Western democracies. This limitation presents an exciting opportunity for a broader comparative analysis of executive speeches between presidential and parliamentary systems using advanced data science techniques.

From a data science perspective, this research on executive speeches between presidential and parliamentary systems would advance natural language processing capabilities through algorithms that detect and quantify system-specific rhetorical patterns. This approach extends current computational linguistics approaches and contributes to comparative political text analysis by applying advanced NLP techniques to analyze how institutional setups shape executive communication across different governance systems worldwide.

3

Methodology

Using computational text analysis and quantitative methods, we examine how institutional structures shape executive communication strategies and rhetoric across different institutional setups. This study investigates six research questions about how pronoun usage, storytelling patterns, rhetorical repetition, linguistic complexity, and agenda-setting varies between presidential and parliamentary systems and how executive speeches can predict electoral outcomes. The upcoming section of the chapter describes the data collection and text preprocessing steps, followed by the computational and quantitative methods used to examine these questions. Finally, we address methodological limitations and validity concerns in comparing executive communication across different institutional contexts.

Our methodology employs specific models to answer each research question. We use spaCy's dependency parser to analyze pronoun usage patterns (RQ1), Boyd et al.'s narrative arc framework to quantify storytelling structures (RQ2), and BERT transformer architecture with spaCy to measure linguistic complexity (RQ3). NLTK enables detection of repetition patterns through n-gram analysis (RQ4), while BERTopic combined with UMAP (Uniform Manifold Approximation and Projection), a non-linear dimensionality reduction technique that preserves

local and global text relationships, and HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise). This clustering algorithm identifies dense topic groupings without requiring a predefined number of clusters (RQ5). For electoral outcome analysis (RQ6), gradient boosting and random forest algorithms determine whether linguistic patterns reliably predict legislative majority attainment across different institutional contexts.

3.1 Data Collection

Our study analyzed executive speeches from 16 countries evenly divided between presidential and parliamentary systems. Countries were selected to ensure representation across diverse geographic regions, multiple language families, and cultural contexts, following comparative frameworks in executive speech analysis (Baturu et al., 2018). We extracted speeches from various official sources, including presidential/prime ministerial websites, national archives and digital libraries, government information services, and parliamentary records. Our comprehensive speech collection covered speech categories ranging from formal addresses to the nation, legislative speeches, diplomatic speeches, crisis communications, and ceremonial speeches and remarks, spanning multiple administrations and electoral cycles across most sampled countries.

For web-accessed repositories, we developed custom web scraping scripts using Python libraries BeautifulSoup4 v4.12.3 (Richardson, 2007), Selenium v4.17.2 (Selenium Project, 2024), and Requests v2.32.3 (Reitz, 2024) adapted to each country's website architecture. Our approach varied based on the technical requirements of repositories. For standard websites, we utilized HTTP requests with BeautifulSoup4 parsing; JavaScript-heavy websites required headless browser automation with custom wait functions; dynamically loaded content

necessitated specialized Selenium-based extraction with scroll simulation; PDF repositories needed text extraction using libraries like pdf plumber; and multilingual content required language detection and specialized handling for non-Latin scripts. Throughout the collection process, we implemented ethical web scraping practices: consistent delays between requests (3-5 seconds), standardized user-agent strings for identification, robust retry mechanisms with increasing backoff periods, sequential URL processing to minimize server load, and progress tracking with incremental saving mechanisms to prevent data loss.

The following sections detail the specific collection process for each country, highlighting unique challenges and solutions encountered during data extraction.

Presidential Systems

United States

A Python-based web scraping script was used to extract executive speeches from the American Presidency Project at UC Santa Barbara's repository. The scripts collected weekly addresses (2010-2018), farewell addresses (2009-2024), state dinner remarks (2010-2024), and State of the Union addresses (2010-2024), navigating the website filtering and pagination systems. The script uses Python libraries, including BeautifulSoup4 for HTML parsing, Requests for HTTP communications, and DateTime for handling temporal data. Technical features included CSS selector-based targeting (e.g., "div.field-docs-content"), session management, regular expression pattern matching, and recursive pagination handling through "pager-next" element detection. The script handled challenges like HTML structure complexity, network reliability issues, and formatting inconsistencies across administrations (resolved through UTF-8 encoding and data normalization). Ethical scraping practices were adhered to by implementing strategic delays between requests (1-5 seconds) and comprehensive logging for monitoring and

process recovery. A total of 1,607 speeches were extracted with a total of 4,043,286 words. The shortest speech was 53 words and the longest had 23,297 words with an average speech length of 2,516 words. The resulting dataset contains speeches of US presidents from various occasions, which helps our comparative analysis of executive speeches in different institutional contexts.

Argentina

A Python-based web scraper scraped Argentina's president's presidential speeches from the country's government website (Casa Rosada) and the Internet Archive's Wayback Machine. The script identified speech links by detecting keywords like "Palabras," "discurso," "declaraciones," and "mensaje" in page content and then extracted both metadata and full speech text. The Python script used libraries including BeautifulSoup4 for HTML parsing, Requests with custom headers for HTTP communications, pandas for data structuring, and Regex for pattern matching. The web scraping script included content container detection using multiple CSS classes, president identification through year-based logic and text pattern analysis, and Spanish data extraction using regular expressions with month name mapping. Key challenges during scraping included handling inconsistent website structures across archived snapshots, extracting dates from Spanish-language text formats, and navigating Internet Archive's limitations. The script incorporated error handling and inbuilt delays to respect server resources. The dataset contained Argentine presidential speech data from 2007 to 2024 across multiple administrations with a total of 443 speeches extracted with a total of 871,897 words. The shortest speech was 124 words and the longest had 10,136 words with an average speech length of 1,968 words enabling a comprehensive comparative analysis.

Brazil

A Python-based web scraping script extracts executive speeches from the official Brazilian government website (<https://www.gov.br/mre/en/content-centers/speeches-articles-and-interviews/president-of-the-federative-republic-of-brazil/speeches>) . The script identifies speech links by searching for elements with the "summary url" class and then extracts metadata, including titles and dates. The implementation excludes press statements and focuses solely on formal speeches. The script utilizes Python libraries, including BeautifulSoup4 for HTML parsing, Requests with custom headers for HTTP communications, and regex for date extraction. The script handled pagination through incremental URL parameters (b_start), parent-child relationship navigation in the HTML structure, and date standardization through datetime parsing and formatting. Key challenges during scraping included handling incomplete or improperly formatted dates, filtering out non-speech content, and managing the website's pagination system. The script provides detailed logging for troubleshooting, implemented error handling, and delays between page requests to ensure ethical web scraping. We also complemented our web-scraped data set with data from the Harvard Dataverse repository "Brazilian Presidential Speeches from 1985 to July 2020." The resulting combined dataset contained a total of 3,708 speeches with a total of 7,365,922 words. The shortest speech was 83 words and the longest had 6,131 words with an average speech length of 1,987 words enabling a comprehensive comparative analysis with speeches from other countries.

Egypt

We implemented a more advanced web scraper for the Egyptian presidential speeches, leveraging Selenium WebDriver to handle dynamic content on the Arabic-language presidency website. This approach was necessary because the Egyptian presidency portal (<https://www.presidency.eg/AR/>) used JavaScript content loading, which requires browser

automation compared to simple HTTP requests. The script utilized specialized components like Chrome WebDriver with Arabic language support, ActionChains for simulating human-like interactions, and a scrolling mechanism to trigger lazy-loaded content. Key challenges included handling Arabic text encoding and extracting content from dynamically loaded elements. The script implemented error handling with retry mechanisms, with varied timing delays to simulate human-like browsing behavior and respect server resources. The resulting dataset contained a total of 84 speeches Egyptian presidential speeches with a total of 60,100 words. The shortest speech was 15 words and the longest was 1,677 words with an average speech length of 716 words for our comparative analysis with other systems.

South Korea

South Korean presidential speeches were collected using a two-phase approach implemented using Selenium WebDriver. In the first phase, we developed a script to extract speech URLs from the Presidential Archives website (<https://www.pa.go.kr>). This script navigated the archive's search page by clicking the speech tab ("연설문"), setting a comprehensive date range (2003-2025), configuring the display to 100 items per page, and executing the search. It then processed 34 results pages, extracting structured metadata (ID, president name, title, URL, and date) for approximately 3,400 speeches and saving this information to JSON files. In the second phase, a script processed these collected URLs to extract the full speech content. This implementation used headless Chrome to visit each URL and extract the text using targeted CSS selectors ("td.content"). The script used timing delays between requests to respect server resources, incremental saving to prevent data loss during long processes, and tracking of extracted speeches to enable resumption if interrupted. The implementation's error handling ensured the process continued despite individual extraction

failures, resulting in a comprehensive dataset of South Korean presidential speeches with 3,451 speeches with a total of 3,012,948 words spanning multiple administrations. The shortest speech was 733 words and the longest had 21,076 words with an average speech length of 2,143 words contributing to our comparative analysis of presidential communication across different institutional contexts.

Seychelles

For the Seychelles presidential speeches, we implemented a scraper that uses Selenium WebDriver to handle the website's JavaScript-dependent content and pagination system from the Seychelles State House website (<https://www.statehouse.gov.sc/speeches>). For each page in the webpage (pages 1-20), the script first collected all valid speech links by filtering out non-speech content, pagination elements, and category pages. It then processed each speech by visiting the individual URL and extracting the full content. The script handled the website's structure by using extended wait times of up to 2 minutes for main content sections to load, additional fixed delays of 10 seconds after initial loading, repeated scrolling operations specifically designed to trigger lazy-loading elements, and comprehensive CSS selector targeting (".u-section-1 .u-text, p") to capture all relevant text elements. Key challenges during scraping included handling dynamically loaded content, distinguishing speech links from navigation elements, and addressing inconsistent content structure across speeches. The script improved reliability by implementing strategic delays between operations, filtering out copyright notices, and ensuring data quality during extraction through length verification and flagging speeches with unusually short content (under 500 characters). This approach allowed us to collect Seychelles presidential speeches in English, French, and Creole, adding linguistic diversity to our comparative analysis of executive communication across different institutional contexts. The dataset contained a total

of 257 speeches with a total of 558,779 words. The shortest speech was 359 words and the longest was 10,750 words with an average speech length of 2,710 words.

Indonesia

A Python-based web scraper scraped Indonesian presidential speeches from the Cabinet Secretariat website (setkab.go.id). The scraper used the requests library and BeautifulSoup for HTTP communication and HTML parsing; the script systematically processed presidential speeches by navigating through monthly archives of a specific speech category (cat=87) from 2014 to 2024. The script used features to handle date extraction from URLs and titles using regex pattern matching with Indonesian month names, content filtering to remove unwanted elements while preserving speech text, duplicate URL detection, and error handling with retry mechanisms. Key technical challenges during scraping included extracting dates from Indonesian language text formats, navigating pagination structures, cleaning extracted content by removing unwanted elements like scripts and copyright notices, handling inconsistent HTML structures across years and incremental JSON saving to prevent data loss, and a test function to validate extraction before running the complete scrape. This approach enabled the collection of a total of 3,339 Indonesian presidential speeches spanning multiple administrations with a total of 3,410,253 words. The shortest speech was 52 words and the longest had 615,875 words with an average speech length of 1,021 words providing valuable data for comparative analysis with speeches from other countries.

Kenya

The Kenyan presidential speeches were collected using a Python-based web scraper from the official President of Kenya website (<https://www.president.go.ke/speeches>). Unlike many previous scrapers that worked with HTML content, this implementation required handling PDF

documents. The script used requests for HTTP communications, BeautifulSoup for HTML parsing, and pdfplumber for extracting text from PDF files. The script handled automatic pagination for the website's 25-page archive, PDF link detection, and direct processing of binary PDF content through in-memory file handling with BytesIO to avoid the need for temporary file storage while maintaining the ability to process multiple pages of PDFs. Key challenges during scraping included handling binary PDF content, maintaining SSL security with an option to bypass certificate verification when needed, and processing potentially complex PDF layouts that may consist of headers, footers, and multi-column text. The script included error handling to address failures and controlled delays between requests to respect server resources. The resulting dataset contained 237 Kenyan presidential speeches with a total of 435,556 words. The shortest speech was 156 words and the longest had 46,878 words with an average speech length of 3,496 words for our comparative analysis and added an African perspective to our cross-institutional study of executive communication patterns.

Parliamentary Systems

Australia

The Australian prime ministerial speeches were collected using a Python-based web scraper from the official Prime Minister Transcripts website (<https://pmtranscripts.pmc.gov.au/>). This implementation leveraged Selenium WebDriver with Chrome in headless mode, to process large volumes of content efficiently. The script was designed to handle speeches between January 2005 and January 2025, with configurable pagination settings supporting up to 200 items per page. Technical features included multiple fallback content selectors to ensure reliable data capture across inconsistent page layouts, incremental JSON-based progress saving to prevent data loss during extended collection sessions, and parallel metadata and full speech content

extraction. Key challenges addressed included handling inconsistent HTML structures across different administrations and maintaining session stability throughout the collection process. The resulting dataset contained a total of 300 Australian prime ministerial speeches with a total of 630,820 words. The shortest speech was 219 words and the longest had 10,775 words with an average speech length of 2,103 words contributing to our comparative analysis of executive speeches between presidential and parliamentary systems.

Canada

The Canadian prime ministerial speeches were collected using a Python-based web scraping script from the official Prime Minister of Canada website (<https://www.pm.gc.ca/en/news/speeches>). The script used Selenium WebDriver with Chrome in headless mode for efficient content extraction with content selectors to handle potential page structure variations and parsing logic to identify speech dates within various text formats. Technical features included automated pagination handling for up to 34 archives pages, intelligent speech link detection that filters for the '/en/news/speeches/' URL pattern, and parallel metadata extraction and full speech content extraction. The implementation addressed challenges such as date format variations by attempting to standardize dates to the 'YYYY-MM-DD' format and efficiently processes large text volumes by adjusting CSV field size limits to accommodate extensive speech content. The resulting dataset is saved in CSV and JSON formats with proper UTF-8 encoding and contains a total of 326 speeches with a total of 403,451 words. The shortest speech was 365 words and the longest had 3,264 words with an average speech length of 1,238 words enabling comprehensive comparative analysis of executive speeches.

Israel

The Israeli prime ministerial speeches were collected using an asynchronous Python-based web scraper from the official Israeli government search portal (<https://searchgov.gov.il/govil/generalsearch>). This script leveraged Playwright with stealth techniques to interact with dynamic JavaScript-rendered content while avoiding detection as an automated scraper. The script followed a sophisticated bilingual approach, independently collecting speeches in Hebrew and English. The script used asynchronous browser control for improved performance, multiple date extraction methods that handle formats in Hebrew and English, and content collection through browser DOM evaluation rather than simple HTML parsing. The implementation addresses significant challenges, including language-specific processing requirements and variable date formats in Hebrew and English contexts. The resulting dataset contained a total of 172 speeches with a total of 1,678,799 words. The shortest speech was 42 words and the longest was 6,240 words with an average speech length of 1,292 words, helping us in our comparative analysis of executive speeches between different institutional setups.

Japan

Prime ministerial speeches of the Japanese prime minister were collected from the official Kantei (Prime Minister's Office) website (<https://www.kantei.go.jp>) using a Python-based web scraper. The script used requests and BeautifulSoup to efficiently process HTML content across multiple administrations from 2010 to 2024, including the Kan, Noda, Abe, Suga, and Kishida governments. The scrapping script used language-specific speech detection through Japanese keywords (演説, 談話, 会見, etc.) and date extraction that handles both traditional Japanese era format (平成/令和年月日) and Western date formats. The script addressed challenges like Japanese language processing, era-based date conversion from Heisei/Reiwa

years to the Western calendar, controlled delays between requests to respect server resources, and navigating the distinct website structures for each prime minister's administration. The resulting dataset contains Japanese prime ministerial speeches with a total of 633 speeches with a total of 882,610 words. The shortest speech was 80 words and the longest had 8,948 words with an average speech length of 1,394 words contributing to our comparative analysis of executive speeches across different institutional setups.

India

The Indian prime ministerial speeches were collected using web scraping from multiple official government websites spanning two administrations. For Dr. Manmohan Singh's speeches, a Python-based scraper extracted the speeches from the PMO website (<https://archivepmo.nic.in/drmanmohansingh/>) for years spanning from 2004 to 2014, utilizing requests for HTTP communications and BeautifulSoup for HTML parsing. This script identified speech links containing "speech-details.php" within the URL structure, and date information from the specialized 'lft' class divisions and full speech content from the corresponding 'rt' class divisions on individual speech pages.

For Prime Minister Modi's speeches, a more sophisticated two-phase approach was necessary to handle the heavy JavaScript architecture of the current PMO website (<https://www.pmindia.gov.in/>). The first phase employed Selenium WebDriver with Chrome in headless mode to handle the site's infinite scrolling mechanism, systematically loading and extracting speech URL links through an optimized scrolling algorithm to extract all speech URLs from 2014 to 2024. The second phase processed each URL collected from phase 1, focusing on the 'left-content' div class, with specialized regex-based date extraction and content cleaning procedures. This implementation also incorporated language detection to separate Hindi and

English speeches. The combined dataset spans multiple administrations of Dr. Manmohan Singh and Prime Minister Modi from 2004 to 2024, providing a comprehensive corpus of 3,114 speeches with a total of 1,275,348 words. The shortest speech was 11 words and the longest was 24,185 words with an average speech length of 2,386 words contributing to our comparative analysis of executive speeches between presidential and parliamentary systems.

Norway

The Norwegian prime ministerial speeches were collected using a Python-based web scraper from the official Norwegian Government website (<https://www.regjeringen.no/>). This script used a dual-strategy approach to handle structural differences between the government's archive system (2010-2021) and its newer website format (2022-2024). Selenium WebDriver is utilized for browser automation to navigate paginated content and handle dynamic website elements across both periods. The script used parallel extraction of speeches in both Norwegian and English by systematically detecting and accessing language-specific URLs, sophisticated metadata extraction through regex pattern matching for dates and speakers in multiple formats, and multi-selector content identification that adapts to variations in page structure. The scraping script addressed challenges of bilingual content processing, archive system navigation across various years, and structural differences between the old and new website architectures. The resulting dataset contains a total of 914 speeches with a total of 1,102,965 words. The shortest speech was 41 words and the longest had 6,785 words with an average speech length of 1,207 words contributing to our comparative analysis of executive speeches between different institutional setups.

Mauritius

The Mauritian prime ministerial speeches were collected from the official Prime Minister's Office website (<https://pmo.govmu.org/>) using a Python-based web scraper that handles PDF documents. The script uses requests for HTTP communications, BeautifulSoup for HTML table parsing, and, crucially, pdfplumber for extracting text from PDF files. The script automated table navigation to identify speech links, direct processing of binary PDF content through in-memory file handling with BytesIO, and sophisticated language detection that categorizes speeches into English, French, and Creole based on linguistic pattern analysis. The script addressed Mauritius's three official languages, SSL certificate verification handling, multilingual content identification across three languages, and complex PDF text extraction that preserves formatting and special characters. Each speech is systematically processed to extract the complete text content before being categorized by language. The resulting dataset contains a total of 58 speeches with a total of 63,693 words. The shortest speech was 372 words and the longest had 2,248 words with an average speech length of 989 words aiding our comparative analysis of executive speeches.

United Kingdom

The UK prime ministerial speeches were collected using a Python-based web scraper from the official UK government website (gov.uk). This scraper used requests for HTTP communications, BeautifulSoup for HTML parsing, and pandas for data organization and export. The scraper handled automated pagination, extraction of speech metadata (including titles, dates, and URLs), and comprehensive content extraction from the standardized "go speak" content divisions that contain the official speech text. The scrapping script addressed specific challenges, including handling relative URLs, maintaining consistent browser-like request headers to avoid blocking, and implementing appropriate error handling at multiple levels. The scraper

systematically processed speeches from six different UK Prime Ministers (David Cameron, Theresa May, Boris Johnson, Elizabeth Truss, Rishi Sunak, and Keir Starmer), with specialized filtering for Elizabeth Truss to focus only on speeches delivered during her brief tenure as Prime Minister. The resulting dataset contains a total of 1,016 speeches with a total of 1,956,185 words. The shortest speech was 46 words and the longest had 10,632 words with an average speech length of 1,925 words adding to our comparative analysis of executive speeches between presidential and parliamentary systems.

Table 1

Corpus Summary

Country	Time Period	Number of Speeches	Total Words	Min Words	Max Words	Avg Words
Seychelles	2010-2024	257	558,779	359	10750	2710
USA	2010-2024	1607	4043286	53	23297	2516
Korea	2003-2024	3451	3,012,948	733	21076	2143
Kenya	2022-2024	237	435,556	156	46878	3496
Indonesia	2014-2025	3339	3410253	52	615875	1021
Egypt	2023-2024	84	60100	15	1677	716
Brazil	2005-2024	3708	7365922	83	6131	1987
Argentina	2007-2024	443	871897	124	10136	1968
Canada	2015-2024	326	403451	365	3264	1238
India	2004-2025	3114	1275348	11	24185	2386
Israel	2005-2024	172	1678799	42	6240	1292
Japan	2017-2024	633	882610	80	8948	1394
Mauritius	2020-2024	58	63693	372	2248	989
Norway	2005-2024	1226	1102965	41	6785	1207
Australia	2005-2024	300	630820	219	10775	2103
United Kingdom	2010-2024	1016	1956185	46	10632	1925

3.2 Data Preprocessing and Cleaning

The extraction process involved content parsing techniques customized for each national context using standardized markers to preserve paragraph structures. We implemented automated language identification for sources with multilingual content using the `langdetect` library's `detect()` function to identify speech language automatically, separating content into language-specific datasets for appropriate processing. Each extracted speech underwent systematic verification with our code explicitly checking for completeness of critical components (title, date, content) and only accepting items where all required elements were successfully retrieved, implementing multi-retry approaches for problematic URLs with increasing wait times between attempts. Content cleaning pipelines remove extraneous elements like social media buttons and navigation components to isolate the actual speech text, followed by systematic preprocessing following established computational linguistics protocols (Bird et al., 2009). The data cleaning included removing HTML tags, URLs, and special characters and then standardizing data through lowercase conversion and format normalization.

For non-English speeches, we used the Google Cloud Translation API (Google, 2023), tagging untranslatable words as "[NONENGLISH]." A growing body of research supports the use of machine translation for cross-linguistic political text analysis. Windsor et al. (2019) and Boot (2021) show that tools like Google Translate retain key linguistic and psychological features across languages, enabling reliable LIWC-based analysis. Chew et al. (2023) further validate that Google Translate preserves semantic content effectively for computational analysis. Extending these findings to political texts, Licht and Lind (2023) demonstrate that translated speeches maintain issue positions and rhetorical structure. Meier et al. (2021) also confirm that translated texts preserve core psychological meanings even when linguistic forms shift. These findings justify the use of translated executive speeches in our analysis and address concerns

about linguistic reliability beyond simple unigram analysis. Metadata extraction utilized regular expression patterns tailored to each country's date formatting conventions, with specialized systems for multilingual date recognition, normalizing diverse data representations into a consistent ISO format (YYYY-MM-DD) for cross-national analysis. For morphological analyses, we created stemmed and lemmatized versions of the corpus using WordNet (Fellbaum, 1998) and the Porter algorithm (Porter, 1980), preserving the original text structure for complexity measures that depend on grammatical relationships.

The resulting corpus comprises of speeches from 16 countries facilitating a comprehensive analysis of executive speeches between presidential and parliamentary systems. The speech categories include formal addresses, legislative speeches, diplomatic engagements, crisis communications, ceremonial events, and partisan addresses enabling a meaningful comparison despite institutional differences between presidential and parliamentary systems. The speech corpus consists of 13 languages with English translations for non-English speeches for comparative analysis while preserving originals for language-specific research. We acknowledge methodological limitations like digital availability bias favoring recent speeches and cross-system variations in speech classification conventions. The dataset's structure enables rigorous comparative analysis while maintaining sensitivity to linguistic and cultural contexts, establishing the foundation for novel insights into executive communication patterns across diverse political systems.

3.3 Pronoun Analysis

Pronoun usage patterns between presidential and parliamentary systems and the influence of institutional structures are analyzed using spaCy's natural language processing pipeline

(Honnibal et al., 2020). We process each speech text through spaCy after applying contraction normalization and handling non-standard contractions. For each speech, we then identify and categorize personal pronouns (PRON) and possessive determiners (PRP\$) using custom rules into first-person singular, first-person plural, second-person, third-person singular, and third-person plural. The raw counts are normalized as percentages of total speech length to ensure consistent, comparable analysis.

We examine pronoun-verb collocations by extracting all pronoun-verb pairs and categorizing them by temporal orientation. The analysis identifies past-tense verbs (VBD, VBN), present-tense verbs (VBG, VBZ, VBP), and future-oriented constructions (pronoun + "will" + verb). This temporal categorization is applied separately to individual and collective frames to yield six distinct temporal metrics. We normalize the relative frequencies of each pronoun category and temporal verb combinations as percentages of total speech length to ensure consistent, comparable analysis.

We track transitions between individual ("I") and collective ("we") framing by monitoring consecutive pronoun usage throughout each speech. We detect frame shifts when speakers transition from individual to collective framing or vice versa, with shift rates normalized to per 100 sentences to factor in varying speech lengths.

Higher usage of first-person singular pronouns may indicate a more personalized leadership style of presidential systems. In contrast, greater use of first-person plural pronouns could suggest the more collective leadership style expected in parliamentary systems (Baturu & Mikhaylov, 2018). Further, we also explore alternative interpretations of other pronoun categories, pronoun-verb collocations, and the directional shifts from individual to collective

framing and vice versa. We calculate group means, standard deviations, and 95% confidence intervals using the Percentile Bootstrap Method with 10,000 random resamples, and compute Cohen's d effect sizes by dividing group mean differences by pooled standard deviations to quantify differences between presidential and parliamentary systems. This multidimensional analysis of pronoun usage, frame shifting, and temporal contextualization provides insights into how different democratic institutional setups shape executive communication strategies and leadership performance.

3.4 Linguistic Complexity

Linguistic complexity is analyzed using traditional metrics and modern NLP approaches. To compare the structural complexity of executive speeches in presidential and parliamentary systems, we measure the Type-Token Ratio (TTR) and moving average TTR (MATTR) for vocabulary diversity to account for text length differences. We use spaCy's dependency parser (Honnibal et al., 2023) to (1) calculate average sentence length by tokenizing each text into sentences and dividing the total token count by the number of sentences. (2) we measure clauses per sentence by identifying tokens with dependency labels indicating clause heads (ROOT, ccomp, xcomp, advcl, acl, relcl) and averaging the count across all sentences. (3) we determine parse tree depth by measuring the maximum path length from root to leaf node in each sentence's dependency tree, then averaging these depths. (4) we calculate mean dependency distance by measuring the linear distance (absolute difference between token positions) between each dependent token and its head, averaging first within sentences and then across the entire text. Together with the Flesch-Kincaid Grade Level formula (Brewer, 2018), these metrics provide a comprehensive assessment of the structural complexity in executive speeches.

For semantic complexity analysis, we employ spaCy's pre-trained word vectors to evaluate dimensions like lexical sophistication through average cosine distances between word vectors, vector space density of unique terms, and semantic clustering coefficients. Additionally, we calculate a lexical sophistication index by combining rare word usage and vector norm scores to measure vocabulary sophistication.

Contextual complexity assessment uses BERT's transformer architecture (Devlin et al., 2019) across three dimensions: (1) Contextual embedding variations: This measures how a word's meaning changes based on context by calculating the difference between its representation in BERT's first and final layers, with larger differences indicating more context-dependent meanings. (2) Attention pattern complexity: This analyzes how widely BERT distributes focus across words in a sentence by measuring attention weights' entropy (or spread), with high entropy suggesting more complex word relationships. (3) Cross-sentence semantic coherence: This determines how smoothly ideas flow between sentences by processing speeches in sliding window chunks (512 tokens with 128-token stride) and measuring similarity between consecutive chunk representations using the special [CLS] token embeddings, which capture whole-sentence meaning in BERT models. Higher similarity indicates stronger thematic consistency between consecutive sentences.

All metrics undergo normalization to factor in differences between presidential speeches and parliamentary addresses. We calculate group means, standard deviations, and 95% confidence intervals using the Percentile Bootstrap Method with 10,000 random resamples, and compute Cohen's d effect sizes by dividing group mean differences by pooled standard deviations to quantify differences between presidential and parliamentary systems.

3.5 Repetition

Repetition patterns in executive speeches are analyzed systematically using bigrams (two-word sequences) and trigrams (three-word sequences). Using NLTK (Bird et al., 2009), we focus specifically on repeated n-grams (those appearing three or more times within each speech) to capture meaningful rhetorical patterns.

We analyze repetition through four quantitative measures applied to both bigrams and trigrams. First, we calculate repeated n-gram type-token ratios by measuring the proportion of n-grams that appear three or more times relative to total n-grams. Second, we compute the entropy of repeated n-gram frequency distributions to measure predictability, where higher entropy indicates more diverse repetition patterns. Third, we apply Zipf's law slopes to analyze frequency distributions and examine the distribution patterns of repeated phrases across the discourse. Fourth, we measure repetition density by calculating how frequently repeated n-grams occur within 100-word sliding windows, capturing local concentrations of rhetorical repetition.

Additionally, we identify frequently used bigrams and trigrams by extracting n-grams that appear three or more times within speeches, revealing characteristic phrasal patterns and institutionalized expressions used across different democratic systems.

We employed the bootstrap method previously described to identify significant differences in repetition measures between presidential and parliamentary systems, controlling speech length and other quantifiable variables. This analysis reveals whether institutional structures influence rhetorical strategies: higher repetition patterns may indicate message reinforcement techniques in presidential systems, while different patterns in parliamentary systems might reflect their distinct institutional communication needs.

3.6 Story Telling

Storytelling patterns in presidential and parliamentary speeches are analyzed using a computational linguistic approach adapted from Boyd et al.'s (2020) narrative arc framework. This methodology quantifies narrative structures using function word usage and cognitive markers. We will analyze three fundamental narrative processes: (1) staging, measured through the relative frequency of articles and prepositions which typically appear at higher rates early in narratives; (2) plot progression, tracked via pronouns, auxiliary verbs, conjunctions, negations, and non-referential adverbs that increase as speeches develop; and (3) cognitive tension, analyzed through cognitive process words (e.g., "think," "understand," "believe") that typically form an inverted U-shape pattern in traditional narratives. We standardize all language dimensions by converting raw word frequencies into comparable metrics within each speech. This standardization enables us to analyze how narrative structures unfold while controlling for each system's baseline linguistic tendencies.

To identify significant differences between presidential and parliamentary storytelling patterns, we will employ bootstrapped mean differences (Efron & Tibshirani, 1994; Cumming, 2014), providing more interpretable measures of effect size while requiring fewer assumptions about data distributions. Following Boyd et al.'s findings that narrative structures differ between traditional stories and fact-driven texts, we will examine whether presidential speeches resemble traditional narrative structures compared to parliamentary addresses, potentially reflecting differences in personalized versus collective leadership approaches.

3.7 Agenda-Setting Analysis

Agenda-setting patterns in executive speeches of presidential and parliamentary systems are analyzed using BERT-based dynamic topic modeling through BERTopic (Grootendorst, 2022) with topics per class functionality to examine system-specific topic distributions. Using preprocessed, tokenized executive speeches, BERTopic computes contextual embeddings using BERT's pre-trained neural network, capturing word meanings based on their surrounding contexts from both preceding and following words (bidirectional understanding). These dense document embeddings are then dimensionally reduced through Uniform Manifold Approximation and Projection (UMAP), which preserves local neighborhood structures while reducing computational complexity, and clustered using Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) to identify distinct topic groups based on density variations in the embedding space.

We apply extensive text cleaning and noise filtering to remove ceremonial language, named entities (NORP, FAC, GPE, LOC), and system-specific procedural terms that could bias topic discovery. Custom stop words and standard English stop words filter out common parliamentary and presidential procedural language. BERTopic is trained on the combined corpus of both systems to ensure comparable topic identification across institutional contexts, generating topics per class to examine system-specific distributions.

To compare agenda-setting patterns between systems, we focus on three key metrics: (1) Topic Weight: The proportion of speeches containing a specific topic within each system, calculated as the frequency of each topic divided by total frequency within that system, indicating policy area prevalence. (2) Speech-level Topic Assignment: Each speech receives a dominant topic assignment with associated probability scores, enabling analysis of how confidently speeches align with specific policy domains. (3) Topic Distributions Over Time: We

examine how topic prevalence evolves across years within each system, revealing temporal patterns of policy emphasis and institutional responsiveness to external events.

Our comparative analysis examines policy priorities that dominate each system and topics unique to each institutional context. We calculate topic frequencies and their distributions to understand how institutional structures influence agenda-setting patterns. We focus on selected high-frequency topics that demonstrate apparent institutional differences, visualizing topic prevalence and temporal trends separately for presidential and parliamentary systems to reveal agenda-setting priorities and policy focus stability over time.

3.8 Electoral Prediction

We develop a predictive model examining how linguistic features in executive speeches in presidential and parliamentary systems correlate with legislative outcomes in subsequent elections. We leverage results from our earlier analysis of pronoun usage patterns, linguistic complexity measures, repetition structures, and narrative arcs and pair these with a binary outcome variable indicating whether the executive's party or coalition secured a legislative majority.

The models will be trained on a combined dataset of presidential and parliamentary systems to determine whether the language used in speeches could serve as a predictive signal for electoral success. The Results variable will serve as the binary target (1 = Win, 0 = Loss). Electoral outcomes will be classified as a Win or a Loss by mapping publicly available election results to the corresponding executive's tenure. For presidential systems, win means that the president's party retained or gained a majority in the Senate (USA, Brazil, Argentina), the DPR (Indonesia), or the National Assembly (South Korea, Seychelles). For parliamentary systems, a

win will mean that the ruling party/coalition retained a majority in the lower house (e.g., India, Japan, UK, Australia, Canada) or the national legislature (e.g., Israel, Mauritius, Norway).

A machine learning approach using gradient boosting, support vector machines and random forest algorithms is used as they are well-suited for handling complex, non-linear relationships likely present between speech patterns and legislative majority outcomes. The dataset will be split into training (80%), and test (20%) sets, with cross-validation employed to minimize overfitting. By analyzing the relative importance of features within the model, we will identify which linguistic dimensions most reliably predict majority attainment while accounting for the institutional differences between presidential and parliamentary systems. This approach allows us to determine the predictive power of speech characteristics alone, providing insights into how communication strategies may influence voter perceptions, and coalition-building success.

3.9 Conclusion

This chapter outlined the methodological approach this study uses to compare executive speeches between presidential and parliamentary systems across 16 countries and 13 languages. In the chapter, we detailed the data science and computational linguistics approaches for examining pronoun usage, linguistic complexity, repetition patterns, storytelling techniques, agenda-setting strategies, and predicting electoral outcomes.

Creating a novel multilingual corpus of executive speeches from presidential and parliamentary systems opens new possibilities for more exhaustive cross-system political communication analysis. The data science contributions of this study include developing custom NLP pipelines for data extraction, cross-institutional text analysis, and implementing advanced

NLP and AI models for text analysis complemented with statistical techniques to identify system-specific patterns. This data-driven analytical approach helps quantify patterns across diverse contexts while acknowledging limitations in digital availability, translation effects, and contextual variations. We mitigate these concerns through bootstrapped statistics and normalization procedures to ensure robust comparative findings.

In the upcoming chapter, we present the results of these analyses, examining how institutional differences between presidential and parliamentary systems shape executive rhetoric, and addressing our research questions.

4

Results

This chapter presents the results of computational text analysis examining how institutional structures shape executive rhetorical strategies across presidential and parliamentary systems. Based on methods described in earlier chapter, the analysis covers quantitative evidence from executive speeches across six key dimensions: pronoun usage patterns, narrative complexity, agenda-setting patterns, linguistic sophistication, and predictability of electoral outcomes. These quantitative findings will help us understand systematic differences in how political leaders adapt their communication strategies to their respective institutional frameworks, contributing to our knowledge of the relationship between institutional structures and political discourse.

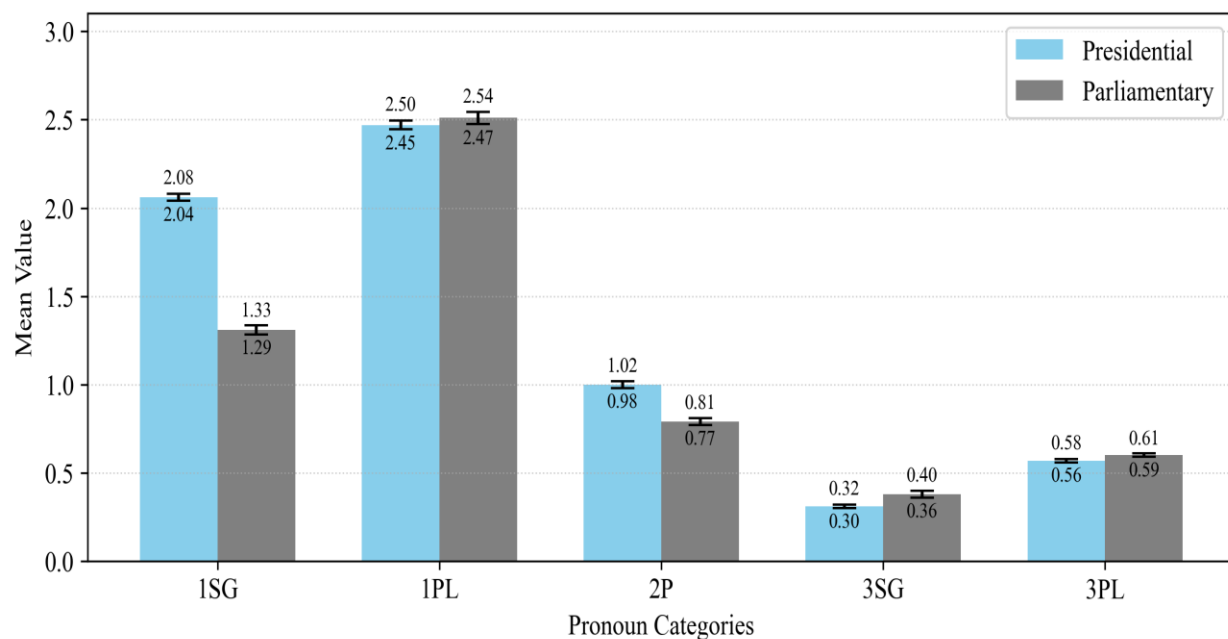
4.1 Pronoun Analysis

The first set of analysis quantified pronoun usage in executive speeches using spaCy-based part-of-speech tagging to identify pronouns in presidential and parliamentary systems with contractions expanded to ensure accurate identification and custom rules to categorize pronouns into five categories: First Person Singular, First Person Plural, Second Person, Third Person Singular, and Third Person Plural. I calculated group means, standard deviations, and 95% confidence intervals using the Percentile Bootstrap Method, with 10,000 random resamples, and

derived confidence intervals from the 2.5th and 97.5th percentiles of each bootstrap distribution. The study calculated Cohen's d by dividing the difference in group means by the pooled standard deviation to provide a standardized measure of effect size between presidential and parliamentary systems.

Figure 1

Pronoun usage by Political System



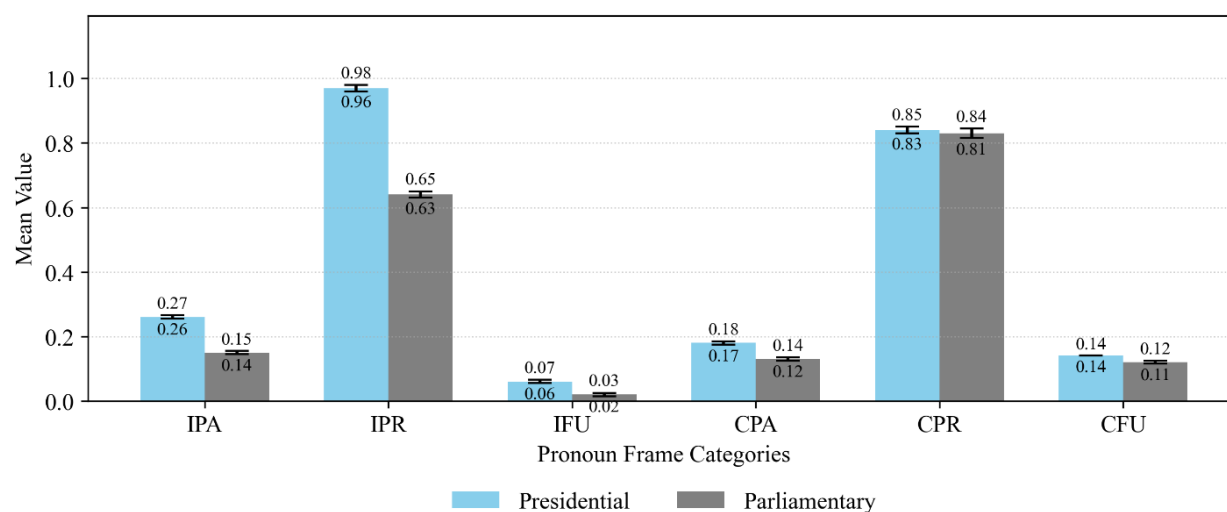
Note. 1SG=First Person Singular, 1PL=First Person Plural, 2P=Second Person, 3SG=Third Person Singular, 3PL=Third Person Plural, Error Bar=95% CI

I observed that presidents used more first-person singular pronouns ($M = 2.06\%$, $SD = 1.17$) than prime ministers ($M = 1.31\%$, $SD = 0.93$, $d = 0.69$) and roughly the same rate of first-person plural pronouns ($M = 2.47\%$, $SD = 1.33$) compared to prime ministers ($M = 2.51\%$, $SD = 1.39$, $d = -0.04$). For second-person pronouns, I found that presidents used more ($M = 1.00\%$, $SD = 1.07$) than prime ministers ($M = 0.79\%$, $SD = 0.90$, $d = 0.21$). For third-person pronouns, presidents used slightly fewer singular pronouns ($M = 0.31\%$, $SD = 0.56$) than prime ministers

($M = 0.38\%$, $SD = 0.73$, $d = -0.12$) and roughly the same number plural pronouns ($M = 0.57\%$, $SD = 0.58$) compared to prime ministers ($M = 0.60\%$, $SD = 0.53$, $d = -0.06$).

Figure 2

Pronoun-Verb usage by Political System



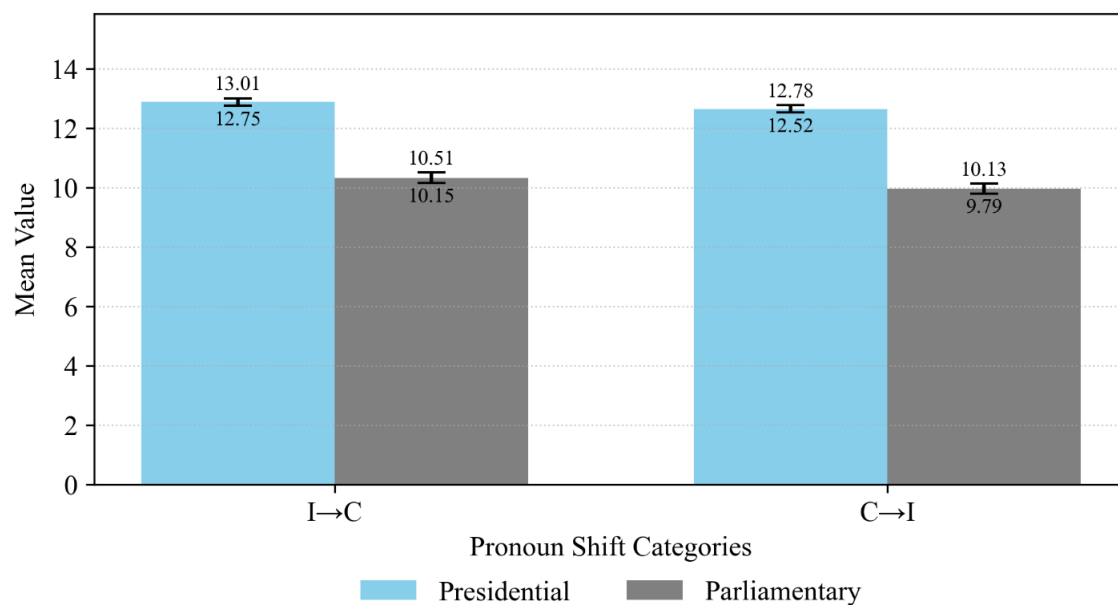
Note. IPA=Individual Past, IPR=Individual Present, IFU=Individual Future, CPA= Collective Past, CPR=Collective Present, CFU=Collective Future, Error Bar=95% CI.

Examining more complex constructions, I analyzed pronoun–verb collocations by identifying pronouns followed by verbs and categorizing them into six temporal frame categories, observing significant differences between systems in four of the six categories. Presidents used more Individual Pronoun + Past tense constructions ($M = 0.26\%$, $SD = 0.35$) than prime ministers ($M = 0.15\%$, $SD = 0.24$, $d = 0.34$). Presidents also used more Individual Pronoun + Present tense constructions ($M = 0.97\%$, $SD = 0.63$) than prime ministers ($M = 0.64\%$, $SD = 0.48$, $d = 0.57$). Presidents used more Individual Pronoun + Future tense constructions ($M = 0.06\%$, $SD = 0.14$) than prime ministers ($M = 0.02\%$, $SD = 0.08$, $d = 0.25$). Presidents used more Collective Pronoun + Past tense constructions ($M = 0.18\%$, $SD = 0.25$)

than prime ministers ($M = 0.13\%$, $SD = 0.19$, $d = 0.21$). The analysis revealed no meaningful differences between Collective Pronoun + Present tense (Cohen's $d = 0.02$) or Collective Pronoun + Future tense constructions ($d = 0.08$).

Figure 3

Pronoun switches by Political System



Note. I to C=Individual to Collective, C to I=Collective to Individual, Error Bar = 95% CI.

Next, I examined pronoun shifts by tracking transitions between individual (first-person singular) and collective (first-person plural) pronouns, with rates normalized per 100 sentences and means reflecting transitions per 100 sentences. Presidential speeches showed higher individual-to-collective transitions ($M = 12.88$, $SD = 7.78$) than parliamentary speeches ($M = 10.33$, $SD = 7.82$; $d = 0.33$) and higher collective-to-individual transitions ($M = 12.65$, $SD = 7.44$) compared to parliamentary speeches ($M = 9.96$, $SD = 7.44$; $d = 0.36$). This indicates that

presidential speeches contain approximately 13 individual-to-collective shifts and 13 collective-to-individual shifts per 100 sentences, compared to about 10 shifts of each type in parliamentary speeches.

Table 2

Descriptive Statistics and Effect Sizes for Pronoun Analysis by Political Systems.

	Presidential Language			Parliamentary Language				
	Mean	SD	95% CI	Mean	SD	95% CI	Sig	<i>d</i>
1SG	2.06	1.17	2.04-2.08	1.31	0.93	1.29-1.34	Yes	0.69
1PL	2.47	1.33	2.44-2.49	2.51	1.39	2.48-2.55	No	-0.04
2P	1.00	1.07	0.98-1.02	0.79	0.9	0.77-0.81	Yes	0.21
3SG	0.31	0.56	0.30-0.32	0.38	0.73	0.36-0.40	Yes	-0.12
3PL	0.57	0.58	0.56-0.58	0.60	0.53	0.59-0.61	Yes	-0.06
I to C	12.88	7.78	12.75–13.01	10.33	7.82	10.15-10.51	Yes	0.33
C to I	12.65	7.44	12.52–12.78	9.96	7.44	9.79–10.13	Yes	0.36
IPA	0.26	0.35	0.26–0.27	0.15	0.24	0.15–0.16	Yes	0.34
IPR	0.97	0.63	0.96–0.98	0.64	0.48	0.63–0.65	Yes	0.57
IFU	0.06	0.14	0.05–0.06	0.02	0.08	0.02–0.03	Yes	0.25
CPA	0.18	0.25	0.18–0.19	0.13	0.19	0.13–0.14	Yes	0.21
CPR	0.84	0.64	0.83–0.85	0.83	0.60	0.82–0.85	No	0.02
CFU	0.14	0.21	0.14–0.14	0.12	0.21	0.12–0.13	Yes	0.08

Note. 1SG=First Person Singular, 1PL=First Person Plural, 2P=Second Person, 3SG=Third Person Singular, 3PL=Third Person Plural, I to C=Individual to Collective, C to I=Collective to Individual, IPA=Individual Past, IPR=Individual Present, IFU=Individual Future, CPA=Collective Past, CPR=Collective Present, CFU=Collective Future, Sig=Significant based on non-overlapping mean confidence intervals and *d*=Cohen's *d*.

4.2 Story Telling Analysis

Storytelling patterns in executive speeches were analyzed using the Narrative Arc function in LIWC-22 (Boyd et al., 2020). Each speech was segmented into five chunks to quantify staging (who, when, where), plot progression (how the story unfolds), cognitive tension (conflict, dilemma, or decision points), and an overall narrativity score. The study calculated

group means, standard deviations, and 95% confidence intervals, and computed d as mentioned in the previous section, to provide a standardized measure of effect size between presidential and parliamentary systems.

Figure 4

Narrative Staging Progression

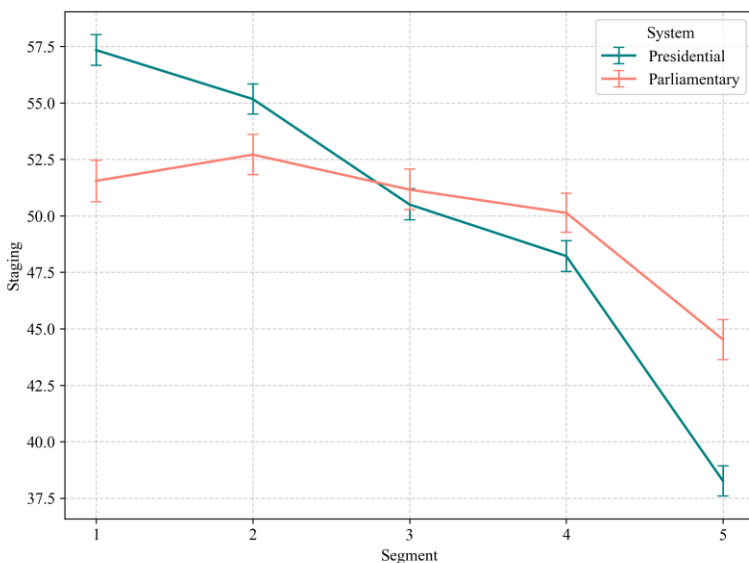


Figure 5

Narrative Plot Progression

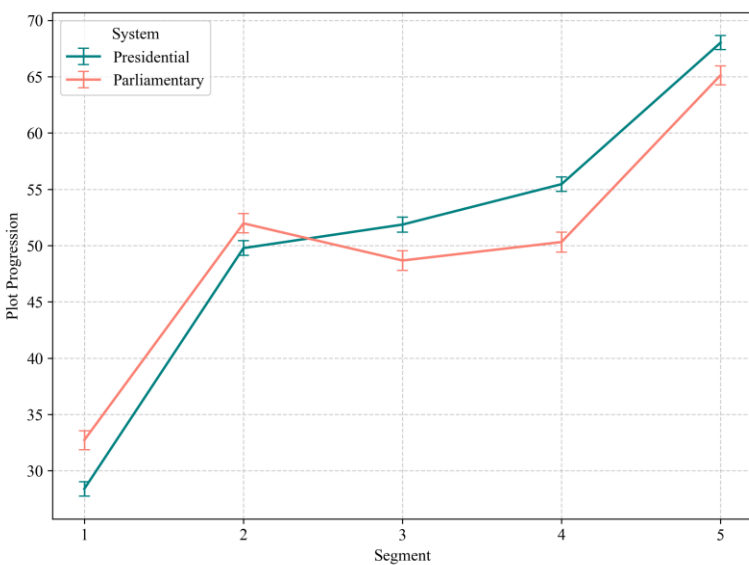
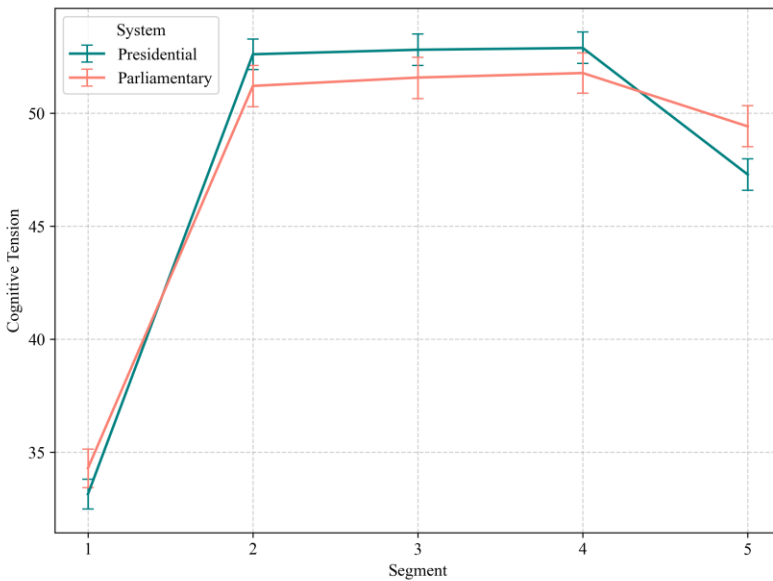


Figure 6*Narrative Cognitive Tension Progression*

Results revealed consistent and statistically significant differences in overall narrativity between presidential and parliamentary speeches. Presidential speeches exhibited significantly higher overall narrativity ($M = 25.23$, $SD = 28.13$) compared to parliamentary speeches ($M = 20.21$, $SD = 26.93$, $d = 0.18$). With large datasets statistical significance is not a practical metric, so here I focus on the effect size for interpretation. Rather than use generic cutoffs which fail to account for the variance and noise in text data, I compared to similar types of studies which routinely use a value of d around 0.10 as the smallest effect of interest (Gupta & Jordan, 2022; Boyd et al., 2020; Seraj et al., 2021). Breaking down the components of narrativity, presidential speeches also scored higher on staging ($d = 0.15$), plot progression ($d = 0.15$), and cognitive tension ($d = 0.08$) compared to parliamentary speeches. Means and CIs are shown in Table 2. For staging, the higher mean for presidential speeches suggests that they set up a more elaborate background. Despite high variability, the consistent difference points to a systematic difference in staging strategies. For plot progression, presidential speeches again showed higher mean

values, pointing to a more structured and coherent narrative build. The confidence intervals are again relatively narrow with minimal overlap, reflecting consistent differences across speeches. Although standard deviations vary considerably within each group, the consistent gap in average values suggests a system-specific rhetorical strategy in plot progression. For cognitive tension, a higher mean for presidential speeches indicated more references to conflict, uncertainty, or resolution. Non-overlapping confidence intervals suggest that this difference is still unlikely due to random variation. High variability and a steady mean difference reflect more tension-building elements in presidential rhetoric.

Table 3

Descriptive Statistics and Effect Sizes for Storytelling Analysis by Political Systems.

	Presidential Language			Parliamentary Language			Sig	<i>d</i>
	Mean	95% CI	SD	Mean	95% CI	SD		
Overall	25.23	24.73–25.73	28.13	20.21	19.57–20.86	26.93	Yes	0.18
Staging	35.6	34.91–36.32	39.22	29.68	28.73–30.64	40.46	Yes	0.15
Plotting	22.25	21.55–22.96	38.73	16.45	15.57–17.32	37.46	Yes	0.15
Cogten	17.81	17.06–18.56	41.5	14.5	13.52–15.46	41.47	Yes	0.08

Note. Overall=Narrativity Overall, Staging=Narrativity Staging, Plotting=Narrativity
Cogten=Narrative Cognitive Tension, Sig = Significant and *d*=Choen's *d*.

4.3 Repetition

Several metrics were used to quantify repetition. N-gram diversity in executive speeches was measured through type-token ratios (TTR) for bigrams and trigrams that appeared three or more times within each speech, normalized as percentages of total n-grams. TTR metrics were normalized as percentages of total n-grams. The bootstrap method previously described was used with Cohen's *d* calculated to provide standardized effect size measures.

Zipf slopes were determined by performing log-linear regression on rank-ordered n-gram frequencies to measure concentration patterns, reported as raw slope coefficients, indicating more concentrated frequency distributions in presidential language. Entropy measures were computed for repeated n-grams using Shannon entropy based on frequency distributions of n-grams appearing ≥ 3 times, with no additional normalization applied. Repetition density was calculated by identifying repeated n-grams (≥ 3 occurrences) within sliding 100-word windows and averaging across all windows in each speech, expressed as average repetitions per 100-word window. Across all measures of repetition, presidential speeches showed greater repetition than parliamentary speeches with ds ranging from 0.05 to 0.34. Additional analyses conducted with raw non-translated text (Appendix M) reveal comparable effect sizes, confirming that presidential speeches use more repetition than parliamentary speeches across languages. Further, these matching results validate the robustness of our findings and that machine translation preserves the linguistic cues necessary for repetition analysis in political speeches. Statistics for each metric are shown in Table 3.

Table 4

Descriptive Statistics and Effect Sizes for Repetition Analysis by Political Systems.

	Presidential Language			Parliamentary Language				
	Mean	95% CI	SD	Mean	95% CI	SD	Sig	<i>d</i>
Bigram TTR	3.43	3.40 - 3.45	1.43	3.18	3.15 - 3.21	1.24	Yes	0.17
Trigram TTR	1.01	1.00 - 1.03	0.80	0.78	0.77 - 0.80	0.65	Yes	0.30
Zipf Slope (Bigrams)	-0.29	-0.29 - -0.29	0.08	-0.27	-0.28 - -0.27	0.07	Yes	-0.16
Zipf Slope (Trigrams)	-0.13	-0.13 --0.13	0.06	-0.11	-0.11 - -0.11	0.05	Yes	-0.30

Bigram Entropy	4.60	4.57- 4.64	1.91	4.86	4.82 - 4.90	1.64	Yes	-0.14
Trigram Entropy	2.82	2.79 - 2.85	1.93	2.73	2.69 - 2.77	1.79	Yes	0.05
Bigram Repetition	15.16	15.02- 15.30	8.21	13.81	13.65-13.97	6.86	Yes	0.17
Trigram Repetition	4.04	3.97- 4.12	4.28	2.76	2.70-2.82	2.58	Yes	0.34

Note. TTR = Type Token Ratio, Sig = Significant, d=Cohen's *d* and higher negative Zipf slopes indicate higher complexity.

Next, I analyzed the top 15 most frequently used bigrams and observed differences in overall frequency between presidential and parliamentary systems. In presidential speeches, the most common bigrams included "of the" (151,065), "in the" (94,276), and "to the" (44,205). Parliamentary speeches also frequently used "of the" (81,698), "in the" (71,473), and "to the" (32,043), but with lower overall counts. Other frequently occurring bigrams in presidential speeches were "it is" (40,980), "we have" (39,887), and "and the" (38,869), while parliamentary speeches showed "it is" (25,959), "we have" (25,074), and "and the" (23,297) with consistently lower frequencies. A complete list of the top 15 bundles, along with their total counts in each system, is presented in Table 4.

Table 5

Top 15 Frequently Used Bigrams in Executive Speeches by System

Presidential Speeches	Frequency	Parliamentary Speeches	Frequency
of the	0.83	of the	0.72
in the	0.52	in the	0.63
to the	0.24	to the	0.28
it is	0.22	it is	0.23
we have	0.22	we have	0.22
and the	0.21	and the	0.21
that we	0.19	for the	0.2
we are	0.18	the country	0.17

for the	0.16	the world	0.16
to be	0.15	we are	0.15
and i	0.14	on the	0.15
want to	0.14	prime minister	0.13
we will	0.13	will be	0.13
going to	0.13	is a	0.13
that the	0.13	we will	0.12

Note: Frequencies calculated as percentage of total corpus words, Corpus size Presidential = 18524211, Parliamentary = 11341176.

Finally, I analyzed the most frequently used trigram in presidential and parliamentary speeches, revealing notable differences in both content and frequency. Presidential speeches showed higher overall frequencies, with dominant trigrams being "president of the" (17,598), "of the republic" (15,634), and "the republic of" (13,628). Parliamentary speeches featured "of the country" (5,784), "would like to" (4,065), and "I would like" (3,310) as the most frequent trigrams. Presidential speeches also commonly used phrases such as "I would like" (13,509), "would like to" (13,489), and "Republic of Indonesia" (10,021). Parliamentary speeches also featured phrases such as "we need to" (2,486), "in the world" (2,434), and "the prime minister" (2,041). A complete list of the top 15 trigrams, along with their total counts in each system, is provided in Table 5.

Table 6

Top 15 Frequently Used Trigrams in Executive Speeches by System

Presidential Speeches	Frequency	Parliamentary Speeches	Frequency
president of the	0.095	of the country	0.051
of the republic	0.084	would like to	0.036
the republic of	0.073	i would like	0.029
i would like	0.073	brothers and sisters	0.028
would like to	0.073	the people of	0.023

republic of indonesia	0.054	we need to	0.022
i want to	0.047	in the world	0.021
indonesia joko widodo	0.043	we have to	0.021
of indonesia joko	0.043	in the country	0.020
like to greet	0.034	the prime minister	0.018
the united states	0.033	i want to	0.017
a lot of	0.030	there is a	0.017
we have to	0.027	of the world	0.016
in the world	0.024	that is why	0.015
one of the	0.021	all of you	0.013

Note: Frequencies calculated as percentage of total corpus words, Corpus size- Presidential = 18524211, Parliamentary = 11341176.

4.4 Agenda Setting

Agenda-setting and policy priority patterns in executive speeches of presidential and parliamentary systems were analyzed using BERTopic's topics per class functionality with a single model trained on the combined corpus of both systems to ensure comparable topic identification across institutional contexts. Text preprocessing removed noise patterns through regex-based filtering, including audience reactions (applause, laughter, cheers), ceremonial language (welcome, pleased to, honor to), filler words (you know, I think, uh, um), parliamentary procedures (hon, speaker, assembly, motion, bill), and named entities (NORP, FAC, GPE, LOC locations and nationalities) to prevent bias in topic discovery. CountVectorizer was configured with min_df=10 to exclude unigrams occurring in fewer than 10 documents, max_df=0.95 to exclude terms appearing in more than 95% of documents to remove overly common words and stop words removal for both standard English stop words and custom parliamentary/presidential procedural terms. I used UMAP with n_neighbors=15 (number of neighboring points used in local approximations), n_components=5 (dimensionality of the reduced embedding space), min_dist=0.0 (minimum distance between points in the low-

dimensional representation), and cosine metric for measuring distances between high-dimensional data points. The model used a min_topic_size of 10 with automatic topic selection and probability calculation enabled. Post-processing focused on universal policies that represent policy domains relevant to both presidential and parliamentary systems, and excluding topics containing non-English content, personal names, country-specific references, ceremonial language, procedural terms, event-specific content, or very low-frequency occurrences that would not enable meaningful cross-system policy comparison. Temporal analysis focused on speeches from 2003 to 2025, identifying the top 5 topics and how they varied across the years.

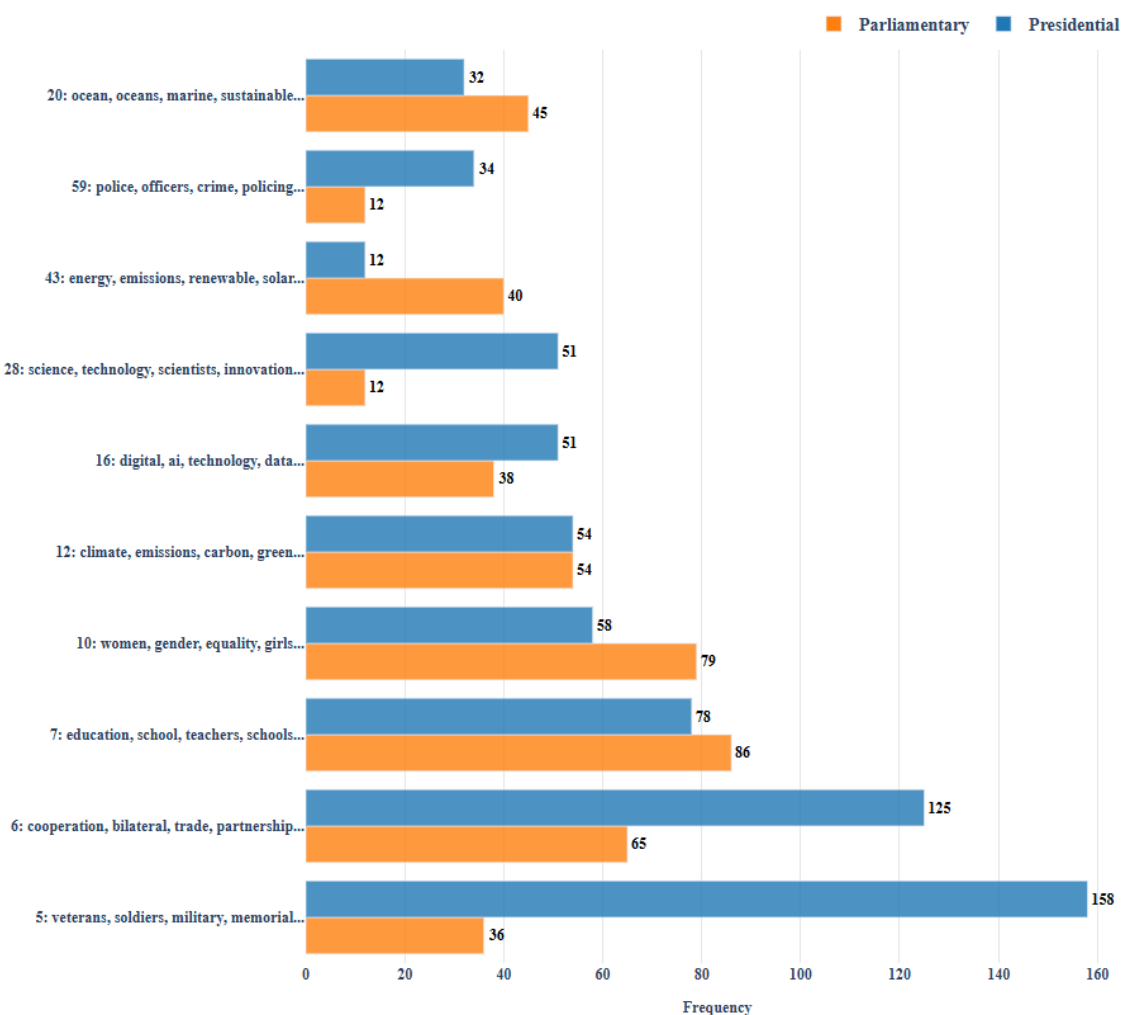
Analyzing executive speeches for the top 10 universal policy topics across both systems, I observed that presidential speeches showed stronger emphasis on military and veterans affairs (Topic 5: 158 vs 36), international cooperation and trade (Topic 6: 125 vs 65), science and technology (Topic 28: 51 vs 12), digital technology and AI (Topic 16: 51 vs 38), and law enforcement (Topic 59: 34 vs 12). While parliamentary speeches demonstrated greater focus on education policy (Topic 7: 86 vs 78, though relatively close), gender equality and women's issues (Topic 10: 79 vs 58), renewable energy policy (Topic 43: 40 vs 12), and ocean/marine policy (Topic 20: 45 vs 32). Climate change policy (Topic 12) showed equal emphasis across both systems (54 each). The focus and emphasis on topics varied, with no topic showing identical emphasis across systems.

As a follow-up analysis, I labeled each speech according to the most dominant topic, and here I compare the number of parliamentary speeches versus presidential speeches labeled as each topic, I observed that presidential speeches showed stronger emphasis on military and veterans affairs (Topic 5), with 158 presidential speeches compared to 36 parliamentary speeches, as well as international cooperation and trade (Topic 6: 125 vs 65), science and

technology (Topic 28: 51 vs 12), digital technology and AI (Topic 16: 51 vs 38), and law enforcement (Topic 59: 34 vs 12). While parliamentary speeches demonstrated greater focus on education policy (Topic 7: 86 vs 78), gender equality and women's issues (Topic 10: 79 vs 58), renewable energy policy (Topic 43: 40 vs 12), and ocean/marine policy (Topic 20: 45 vs 32). Climate change policy (Topic 12) showed equal emphasis across both systems (54 each). The focus and emphasis on topics varied, with only climate change policy showing equal emphasis across both systems.

Figure 7

Top 10 Common Topics Per Class



Next, I identified system-exclusive topics by ranking all topics by corpus weight, selecting the top 20 overall, and then determining which topics appeared exclusively in presidential or parliamentary speeches. Eight topics appeared exclusively in the parliamentary system, featuring institutional governance and public welfare including logistics and infrastructure (Topic 3: "logistics, bridge, roads, transport, construction"), sports and cultural events (Topic 11: "medal, medals, gold, olympic, sport"), public health infrastructure (Topic 15: "vaccine, vaccines, doses, vaccination, pandemic"), anti-corruption mechanisms (Topic 26: "corruption, anticorruption, transparency, vigilance, agencies"), children and family policy (Topic 27: "children, childrearing, child, childcare, millennium"), nuclear science and technology (Topic 29: "science, scientific, scientists, technology, nuclear"), healthcare services (Topic 34: "cancer, patients, organ, medical, hospital"), and judicial system governance (Topic 35: "court, judicial, courts, justice, judiciary"). In contrast, four topics were unique to the presidential system, emphasizing economic sector management including agriculture and rural development (Topic 60: "agricultural, agriculture, rural, farmers, areas"), fertilizer and farming policy (Topic 65: "fertilizer, tons, farmers, hectares, hectare"), fishing and fisheries management (Topic 75: "fish, fishermen, fishing, fisheries, fisherman"), and cultural arts policy (Topic 56: "cultural, film, culture, content, arts").

Table 7

System specific topics from Top 20

System	Topic	Words	Weight %
Parliamentary	8	nations, united, international, general, security	1.06
Parliamentary	27	children, childrearing, child, childcare, millennium	0.77
Parliamentary	15	vaccine, vaccines, doses, vaccination, pandemic	0.76

Parliamentary	11	medal, medals, gold, olympic, sport	0.59
Parliamentary	35	court, judicial, courts, justice, judiciary	0.47
Parliamentary	34	cancer, patients, organ, medical, hospital	0.40
Parliamentary	26	corruption, anticorruption, transparency, vigilance, agencies	0.33
Parliamentary	29	science, scientific, scientists, technology, nuclear	0.17
Parliamentary	3	logistics, bridge, roads, transport, construction	0.10
Presidential	56	cultural, film, culture, content, arts	0.31
Presidential	65	fertilizer, tons, farmers, hectares, hectare	0.31
Presidential	60	agricultural, agriculture, rural, farmers, areas	0.27
Presidential	75	fish, fishermen, fishing, fisheries, fisherman	0.27

Finally, I analyzed the temporal trends of the top 10 common topics, identified in Figure 4, across presidential and parliamentary speeches from 2003 to 2025. In the parliamentary system, the distribution was very volatile. Topic 12 (climate, emissions, carbon, green) demonstrated the most dramatic early spike, reaching approximately 4.2% in 2007 before declining and showing moderate fluctuations throughout the period. Topic 6 (cooperation, bilateral, trade, partnership) peaked around 2006 (3.6%) and showed cyclical patterns with notable increases in 2012 and 2024. Topic 7 (education, school, teachers, schools) exhibited significant growth, reaching its highest point around 2014 (3.2%) with sustained emphasis in recent years. Topic 10 (women, gender, equality, girls) showed variable presence with peaks in 2011 (2.5%) and 2018 (2.8%). Most notably, Topic 20 (ocean, oceans, marine, sustainable) surged dramatically around 2018-2019, peaking at nearly 4% before declining to moderate levels by 2025. Topic 5 (veterans, soldiers, military, memorial) maintained a consistently low presence across the timeline, rarely exceeding 1%. Topic 16 (digital, AI, technology, data) exhibited

modest increases after 2010, with a mild peak in 2018 (1.9%), and occasional mentions thereafter. Topic 28 (science, technology, scientists, innovation) showed occasional spikes, particularly around 2011 and 2018, but remained below 2% overall. Topic 43 (energy, emissions, renewable, solar) followed a relatively stable trend, peaking slightly in 2012 (2%) and again in 2018. Topic 59 (police, officers, crime, policing) had the weakest presence among the top 10, with minor mentions around 2011 and 2014.

Similar to parliamentary systems, the presidential system was also volatile. Topic 5 (military, veterans, soldiers, heroes) dominated early years, starting strong around 2003 (2.9%) and showing another significant peak in 2008 (2.7%) before gradually declining to lower levels in recent years. Topic 6 (cooperation, trade, bilateral, investment) showed moderate fluctuations with peaks around 2005 (1.9%) and 2014-2015 (1.7%). Topic 7 (education, teachers, school, schools) maintained a relatively stable presence with a notable peak in 2009 (1.4%). Topic 10 (women, gender, female, equality) exhibited the most dramatic surge around 2013-2014 (1.8%) before declining significantly. Topic 12 (climate, carbon, emissions, green) remained relatively low throughout most periods until a significant increase to 1.8% in 2021. Topic 16 (digital, AI, technology, data) exhibited intermittent attention, with a noticeable spike in 2022 (1.7%), suggesting an increasing relevance in more recent presidential discourse. Topic 20 (ocean, oceans, marine, sustainable) appeared sporadically, with minor peaks in 2019 and 2023, but never surpassing 1%. Topic 28 (science, technology, scientists, innovation) displayed brief peaks in 2013 and 2014, but its overall presence remained below 1.5%. Topic 43 (energy, emissions, renewable, solar) had relatively steady but low levels throughout the timeline, reflecting a background rather than a central priority. Topic 59 (police, officers, crime, policing) appeared sporadically with mild peaks around 2005 and 2011, and slight activity reemerged after 2020.

Figure 8

Temporal Distribution of Top 10 Topics in Presidential Executive Speeches (2003-2025)

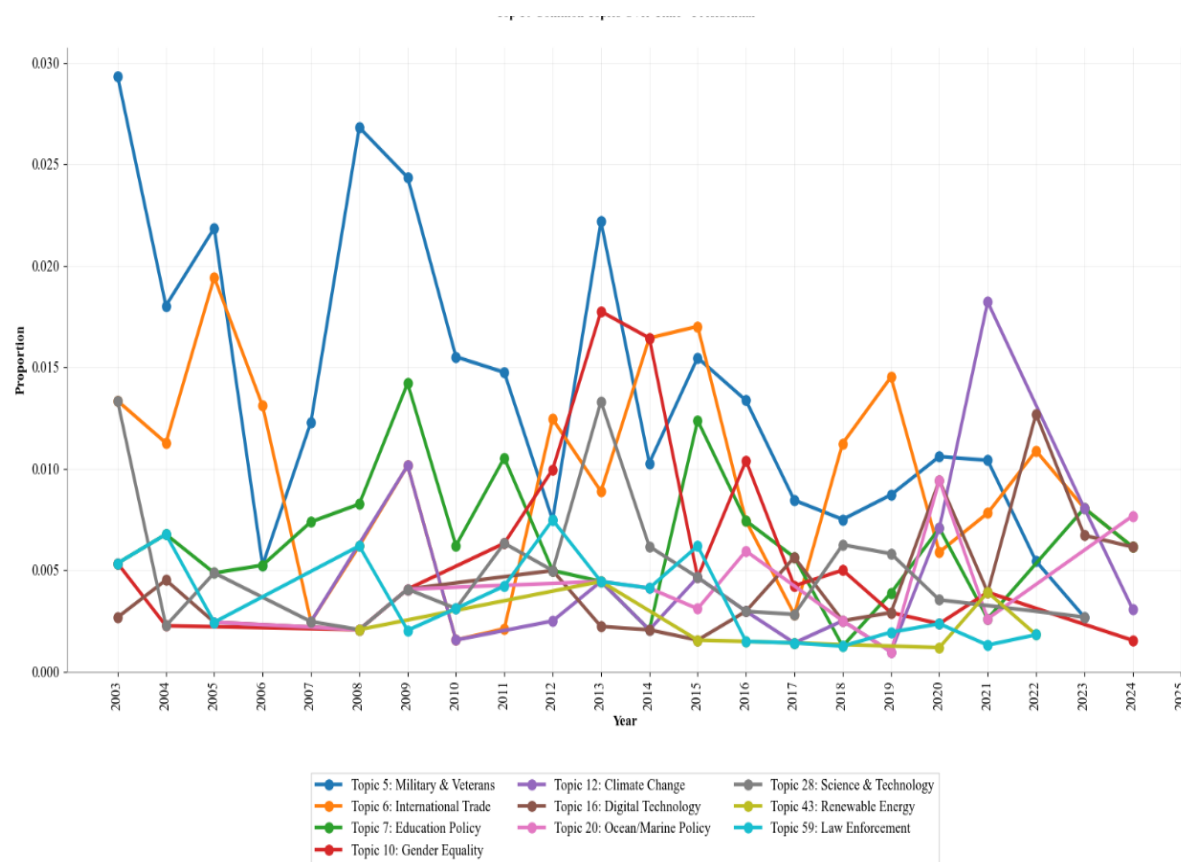
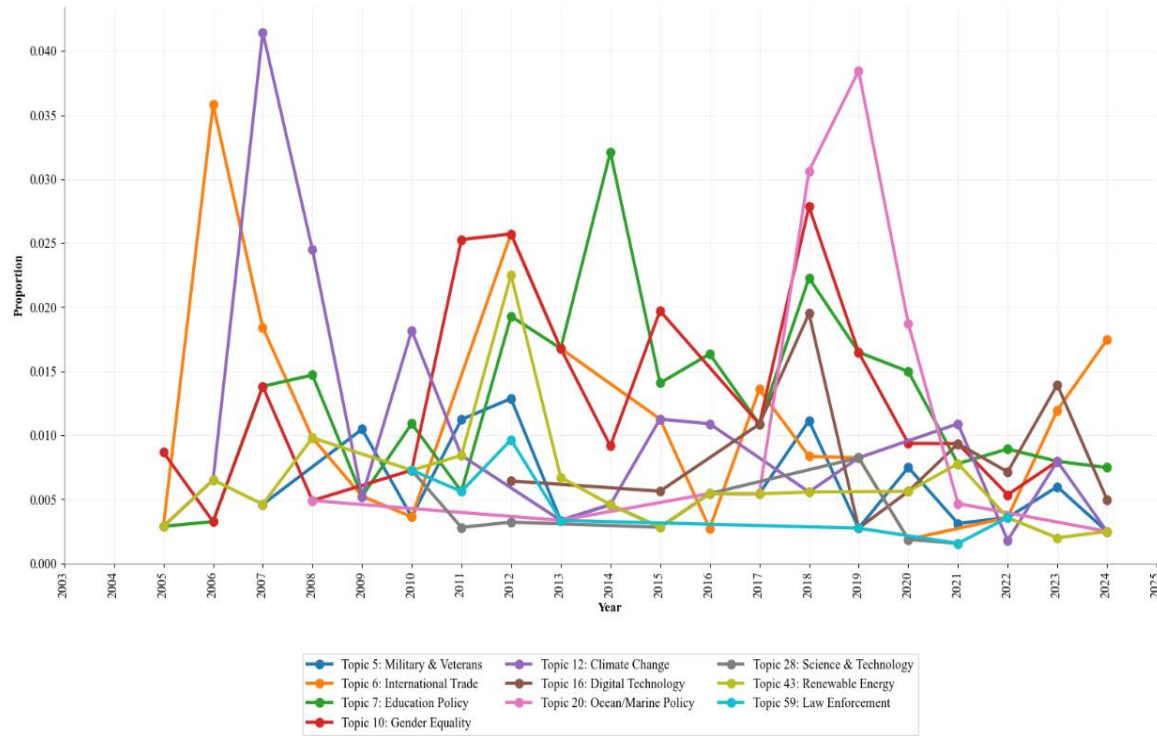


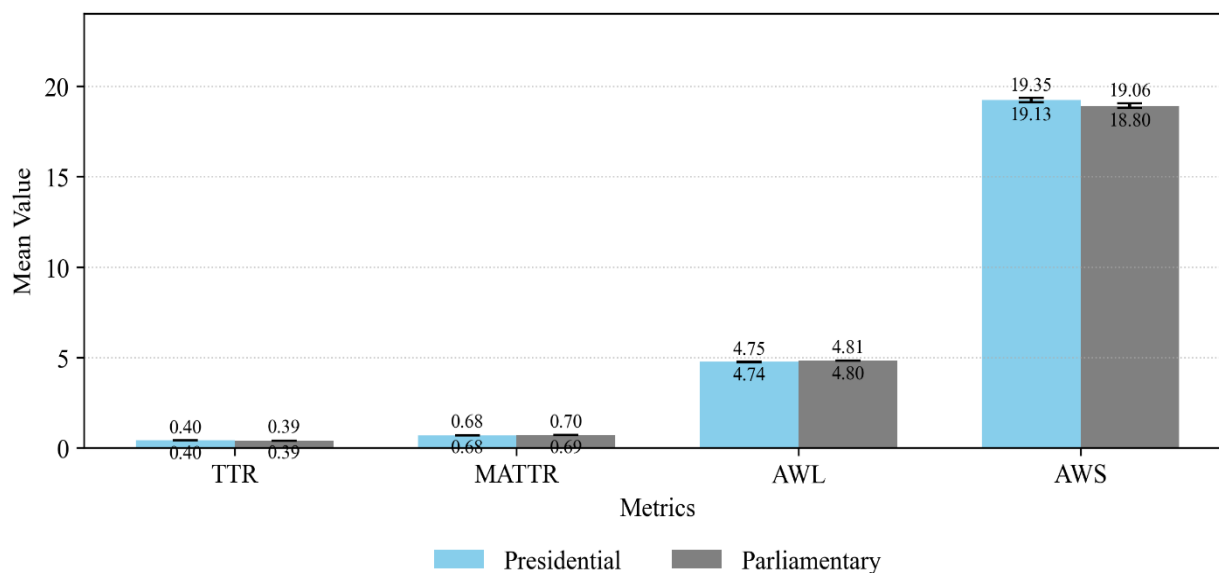
Figure 9

Temporal Distribution of Top 10 Topics in Parliamentary Executive Speeches (2003-2025)



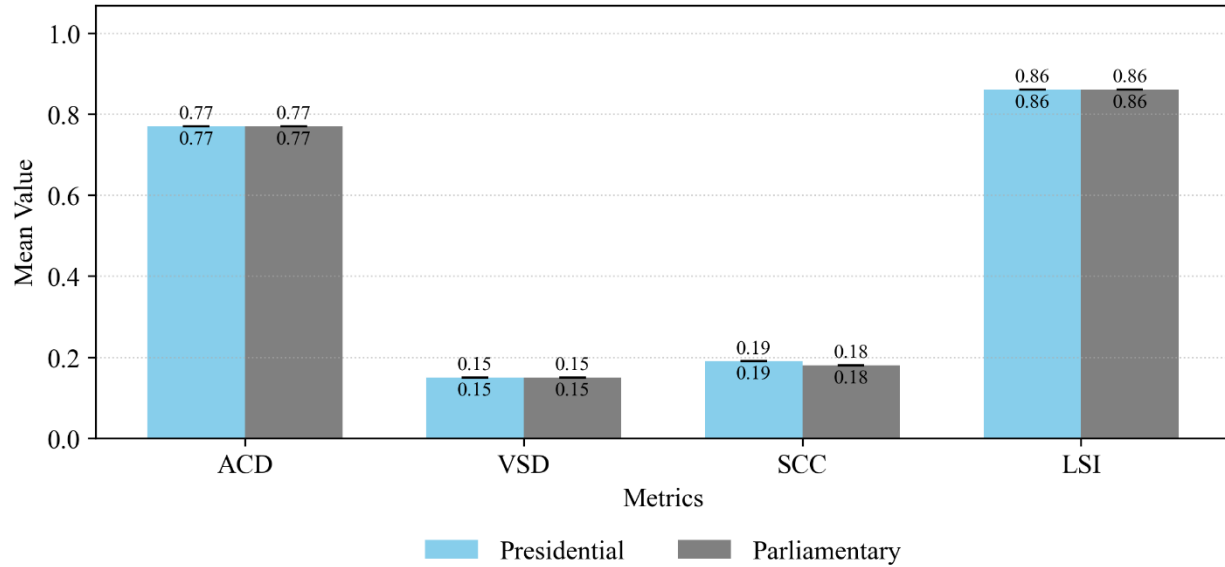
4.5 Linguistic Complexity

This study analyzed linguistic complexity in executive speeches using both spaCy- and BERT-based methods to compare presidential and parliamentary systems. The analysis covered four broad linguistic categories: lexical, semantic, syntactic, and contextual complexity with all metrics normalized for speech length. The group means, standard deviations, confidence intervals and Cohen's d was calculated as described for earlier sections.

Figure 10*Lexical Diversity by political system*

Note: TTR=Type Token Ratio, MATTR=Moving Average Type-Token Ratio, AWL=Average Word Length, AWS=Average Words per Sentence, Error Bars=95% CI.

Lexical diversity was analyzed using the type-token ratio (TTR), moving-average TTR (MATTR) with a 100-word sliding window, average sentence length, and average word length, with all metrics normalized for speech length. For lexical complexity, I observe that presidential speeches exhibited a slightly higher type-token ratio (TTR) ($M = 0.40$, $SD = 0.11$) than parliamentary speeches ($M = 0.39$, $SD = 0.09$, $d = 0.08$), but a lower moving-average TTR (MATTR) ($M = 0.68$, $SD = 0.05$) compared to prime ministers ($M = 0.70$, $SD = 0.04$, $d = -0.25$). Presidential speeches also featured slightly shorter average word length ($M = 4.74$, $SD = 0.37$) than parliamentary speeches ($M = 4.81$, $SD = 0.29$, $d = -0.21$), but longer average sentence length in terms of words ($M = 19.24$, $SD = 6.22$) than prime ministers ($M = 18.93$, $SD = 5.99$, $d = 0.05$).

Figure 11*Semantic Diversity by political system*

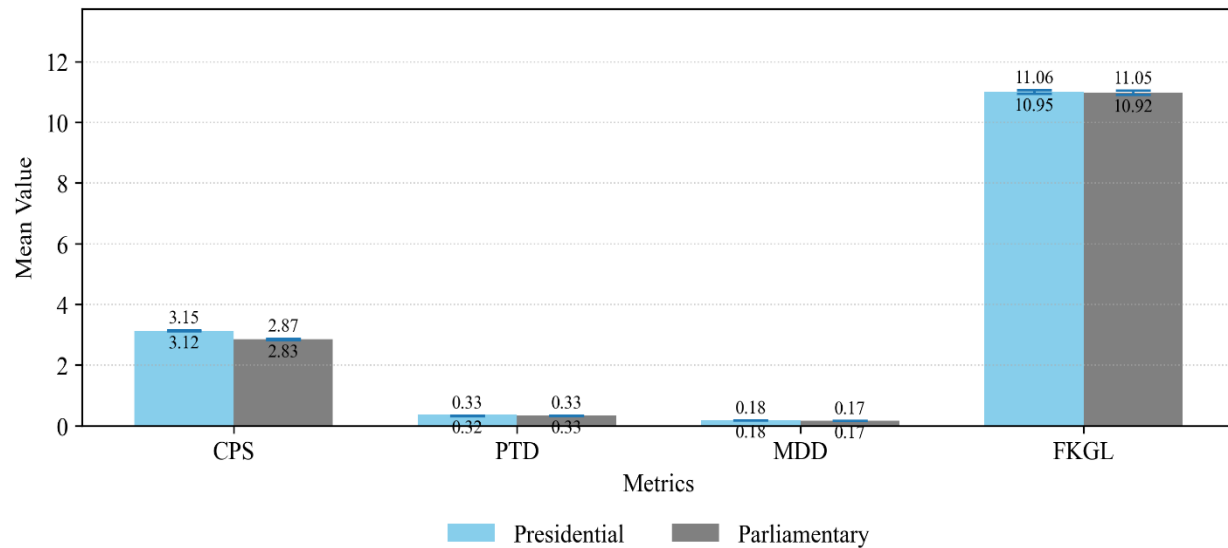
Note. ACD = Average Cosine Distance, VSD = Vector Space Density, SCC = Semantic Clustering Coefficient, LSI = Lexical Sophistication Index, CPS = Clauses per Sentence, PTD = Parse Tree Depth, MDD = Mean Dependency Distance, FKGL = Flesch-Kincaid Grade Level, Error Bars=95% CI.

Semantic sophistication was analyzed using spaCy vector representations of content tokens. Cosine distance and vector space density captured semantic dispersion and density, while k-means clustering and silhouette scores computed semantic clustering coefficients. A lexical sophistication index combines rare word usage and vector norm scores, reflecting both lexical frequency and semantic richness (Zhao et al., 2021; Crossley et al., 2011). My analysis on semantic complexity showed similar average cosine distance with no significant difference ($d = 0.02$). However, presidential speeches exhibited higher vector space density ($M = 0.15$, $SD = 0.004$) than parliamentary ones ($M = 0.15$, $SD = 0.05$, $d = 0.23$), and higher semantic clustering coefficients ($M = 0.19$, $SD = 0.06$) than parliamentary speeches ($M = 0.18$, $SD = 0.05$, $d = 0.25$).

Both systems showed comparable lexical sophistication index (LSI) values, indicating no difference in the use of rare or high-frequency words ($d = 0.00$).

Figure 12

Syntactic Diversity by political system



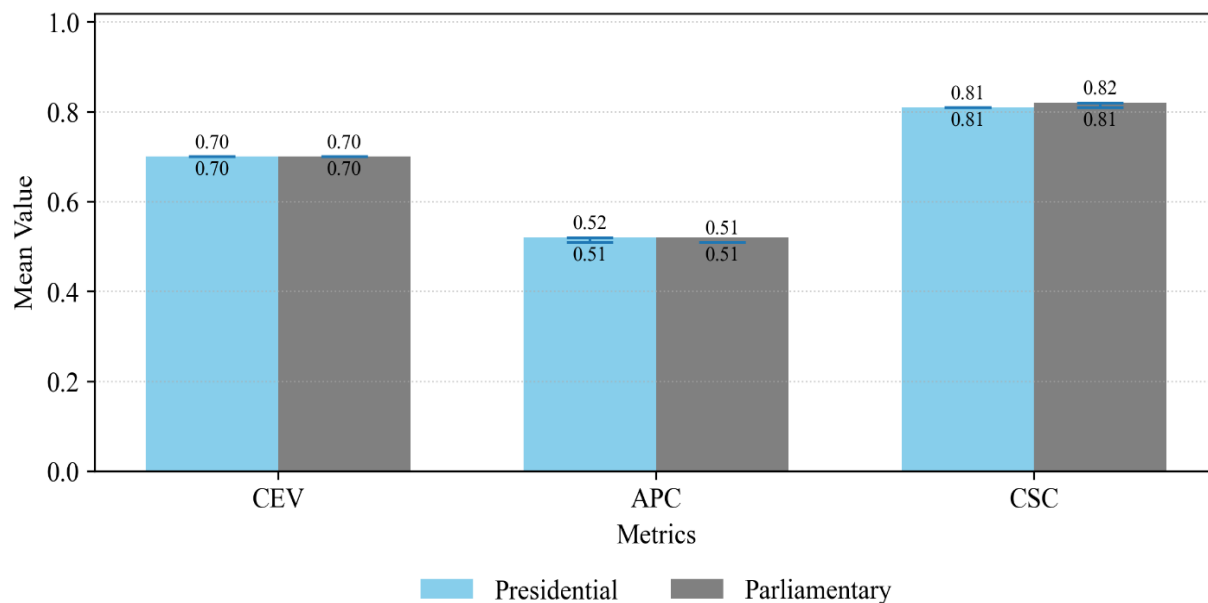
Note: CPS = Clauses per Sentence, PTD = Parse Tree Depth, MDD = Mean Dependency Distance, FKGL = Flesch-Kincaid Grade Level, Error Bars=95% CI.

Syntactic complexity was measured using spaCy dependency parsing. Clause density was calculated based on selected clause-type dependency labels (e.g., ROOT, ccomp, xcomp). The analysis normalized the parse tree depth and dependency distance by sentence length. I observed that presidential speeches included more clauses per sentence ($M = 3.13$, $SD = 0.90$) than parliamentary speeches ($M = 2.86$, $SD = 0.89$, $d = 0.31$). However, they had slightly lower parse tree depth ($M = 0.36$, $SD = 0.03$) compared to parliamentary speeches ($M = 0.33$, $SD = 0.03$, $d = -0.22$). Presidential speeches showed greater mean dependency distance ($M = 0.18$, $SD = 0.04$) than parliamentary ones ($M = 0.17$, $SD = 0.03$, $d = 0.40$).

As a more traditional measure of syntactic complexity, I calculated the Flesch-Kincaid Grade Level using the standard formula using a rule-based method to estimate syllables, and the analysis derived sentence length and word counts from spaCy-parsed tokens, normalizing all values for speech length. Both groups had similar Flesch-Kincaid Grade Level scores ($M = 11.00$; $d = 0.01$), indicating comparable readability levels.

Figure 13

Contextual Diversity by political system



Note: CEV = Contextual Embedding Variation, APC = Attention Pattern Complexity, CSC = Cross-Sentence Coherence, Sig=Significant, Error Bars=95% CI.

The analysis assessed contextual complexity using BERT-based metrics with full-token processing. To address BERT's 512-token input limit, it splits each speech into overlapping chunks with a stride of 128 tokens. For each chunk, the analysis computed contextual embedding variation by measuring the cosine distance between the first and last hidden layers for selected

tokens. It calculated attention pattern complexity by averaging entropy scores from attention heads across sampled layers. To estimate cross-sentence coherence, it computed cosine similarity between [CLS] vectors of adjacent chunks. The analysis implemented these metrics using bert-base-uncased, extracting hidden states and attention outputs, and aggregated results across all chunks for each speech.

Both systems showed nearly identical contextual embedding variation ($M=0.70$, $d=<-0.01$) and attention pattern complexity ($M = 0.52$, $d=0.01$). Finally, cross-sentence coherence was marginally higher in parliamentary speeches ($M = 0.82$, $SD = 0.06$,) than in presidential speeches ($M = 0.81$, $SD = 0.04$, $d = -0.12$).

Table 8

Descriptive Statistics and Effect Sizes for Complexity Analysis by Political Systems.

	Presidential			Parliamentary			Sig	<i>d</i>
	Mean	SD	95% CI	Mean	SD	95% CI		
TTR	0.40	0.11	0.40-0.40	0.39	0.09	0.39-0.39	Yes	0.08
MATTR	0.68	0.05	0.68-0.68	0.70	0.04	0.69-0.70	Yes	-0.25
AWL	4.74	0.37	4.74-4.75	4.81	0.29	4.81-4.82	Yes	-0.21
AWS	19.24	6.22	19.13-19.35	18.93	5.99	18.80-19.07	Yes	0.05
ACD	0.77	0.03	0.77-0.77	0.77	0.05	0.77-0.77	No	0.02
VSD	0.15	<0.01	0.15-0.15	0.15	0.05	0.15-0.15	Yes	0.23
SCC	0.19	0.06	0.19-0.19	0.18	0.05	0.18-0.18	Yes	0.25
LSI	0.86	0.04	0.86-0.86	0.86	0.06	0.86-0.86	No	< 0.01
CPS	3.13	0.90	3.12-3.15	2.86	0.89	2.83-2.87	Yes	0.31
PTD	0.36	0.03	0.32-0.33	0.33	0.03	0.33-0.33	Yes	-0.22
MDD	0.18	0.04	0.18-0.18	0.17	0.03	0.17-0.17	Yes	0.40
FKGL	11.00	3.19	10.95-11.06	10.99	2.81	10.92-11.05	No	0.01
CEV	0.70	0.01	0.70–0.70	0.70	0.02	0.70–0.70	No	< -0.01
APC	0.52	0.01	0.51–0.52	0.52	0.01	0.51–0.51	No	0.01
CSC	0.81	0.04	0.81-0.81	0.82	0.06	0.81–0.82	Yes	-0.12

Note TTR = Type-Token Ratio, MATTR = Moving-Average Type-Token Ratio, AWL = Average Word Length, AWS = Average Words per Sentence, ACD = Average Cosine Distance,

VSD = Vector Space Density, SCC = Semantic Clustering Coefficient, LSI = Lexical Sophistication Index, CPS = Clauses per Sentence, PTD = Parse Tree Depth, MDD = Mean Dependency Distance, FKGL = Flesch-Kincaid Grade Level, CEV = Contextual Embedding Variation, APC = Attention Pattern Complexity, CSC = Cross-Sentence Coherence, Sig=Significant, d=Cohens' d, Values rounding to .00 are shown as < .01.

4.6 Electoral Prediction

This study developed and evaluated multiple machine learning models to predict election outcomes (Win/Loss) based on 41 linguistic features derived from prior analyses of executive speeches on pronouns, complexity, repetition, narrativity, and topicality. The models were trained on a combined dataset of presidential and parliamentary systems to determine whether the language used in speeches could serve as a predictive signal for electoral success. The Results variable served as the binary target (1 = Win, 0 = Loss). I classified electoral outcomes as a Win or a Loss by mapping publicly available election results to the corresponding executive's tenure. For presidential systems, Win meant that the president's party retained or gained a majority in the Senate (USA, Brazil, Argentina), the DPR (Indonesia), or the National Assembly (South Korea, Seychelles). For parliamentary systems, a Win meant that the ruling party/coalition retained a majority in the lower house (e.g., India, Japan, UK, Australia, Canada) or the national legislature (e.g., Israel, Mauritius, Norway).

I evaluated three models: Support Vector Machines (SVM), Random Forest, and XGBoost. All models used stratified train-test splits (80/20); SMOTE for class balancing. After splitting, I applied SMOTE (Synthetic Minority Oversampling Technique) only to the training data to balance the imbalance, thereby preventing data leakage into the test set. SMOTE generated synthetic samples for the minority class (Results = 0) using default k-nearest neighbors. I then performed 3-fold cross-validation on the balanced training set for hyperparameter tuning. I assessed model performance using accuracy, precision, recall, F1-score,

and confusion matrices, and interpreted feature importance using both permutation importance and SHAP values to identify the most influential predictors.

Table 9

Model Evaluation Metrics

Model	Total	Train	Test	SMOTE	SMOTE Resulting Class Balance	CV	Feature Importance
SVM	19,398	15,518	3,880	Yes - Train	7,759 (Class 0) / 7,759 (Class 1)	3-fold	Permutation, SHAP
Random Forest	19,398	15,518	3,880	Yes - Train	7,759 / 7,759	3-fold	Permutation, SHAP
XGBoost	19,398	15,518	3,880	Yes - Train	7,759 / 7,759	3-fold	Permutation, SHAP

Support Vector Machine (SVM)

The best-performing model across all experiments was an SVM classifier using an RBF kernel, SMOTE balancing, and standardized input. GridSearchCV determined the optimal hyperparameters: $C = 10$, $\gamma = \text{"scale"}$, and $\text{kernel} = \text{"rbf"}$. On the combined dataset ($N = 3880$), the model achieved an accuracy of 73.3%, with balanced F1-scores across both classes ($F1 = 0.61$ for class 0; $F1 = 0.80$ for class 1). This model demonstrated the best recall for Loss (66%), which was a key weakness in other models. Permutation importance revealed the top predictors, including clauses per sentence, trigram TTR, first singular pronoun usage, and vector space density, as the most influential.

Random Forest

A Random Forest classifier with SMOTE, $n_estimators = 200$, and $max_depth = 20$ was the second-best performer. This model achieved the highest overall accuracy (75.8%), but its recall for the minority class (class 0) was substantially lower (28%), resulting in an F1-score of 0.41 for that group. This imbalance was noted despite efforts to tune and balance the model,

suggesting that tree-based ensemble methods may be prone to overfitting the majority class in this context. SHAP analysis indicated that syntactic complexity and narrativity measures (e.g., clauses per sentence, Narrativity Staging) drove most of the predictions.

XGBoost

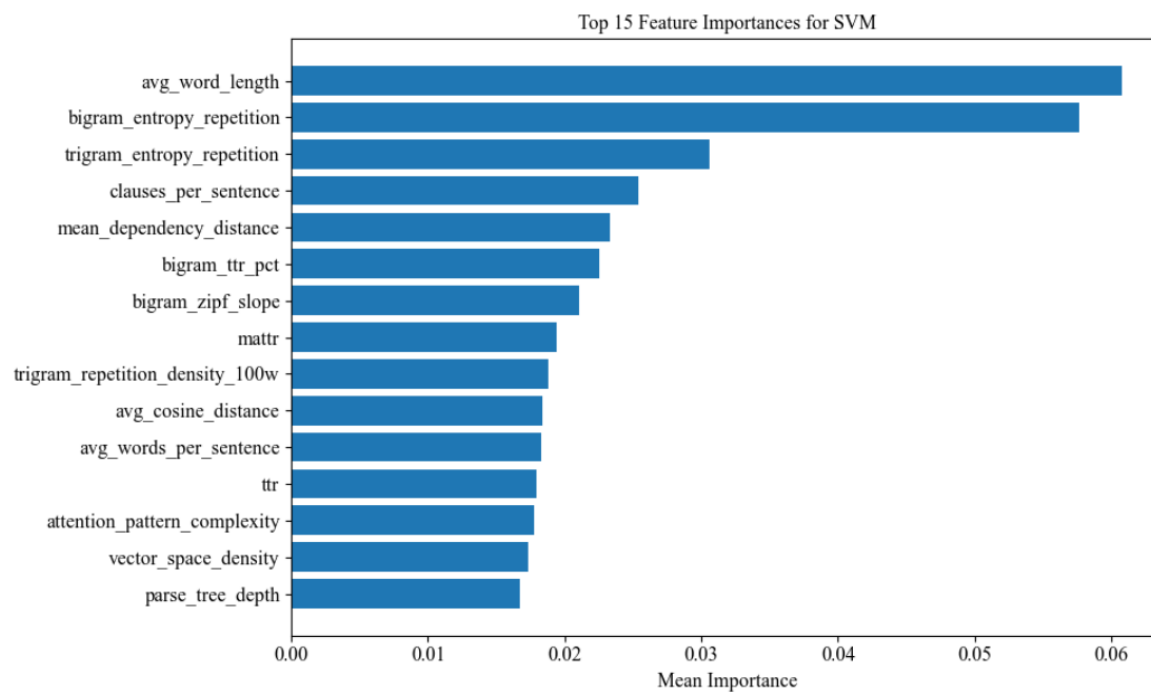
A tuned XGBoost classifier with default depth and learning rate, utilizing `scale_pos_weight` and an objective of "binary: logistic", ranked third overall. This model achieved 72.9% accuracy, with balanced performance across both classes (F1 score = 0.57 for class 0 and F1 score = 0.80 for class 1). Although its class 0 recall (60%) was slightly lower than SVM's, its overall balance and interpretability via built-in feature importance and SHAP values make it a strong alternative model, especially for understanding which linguistic features drive predictions.

The performance table (Table 9) shows that while Random Forest achieved the highest overall accuracy (0.758), it performed poorly in detecting Class 0 cases, with only a 0.28 recall, suggesting that it heavily favored the majority class despite achieving high precision and recall for Class 1. In contrast, the SVM model provided the most balanced performance across both classes, with a recall of 0.66 for Class 0 and 0.77 for Class 1 and balanced F1-scores for both classes (0.61 for Class 0 and 0.80 for Class 1), making it a better choice for detecting both outcomes. Although XGBoost's performance was similar, its Class 0 F1-score was slightly lower than that of SVM. Based on the best Recall for Class 0 and the most balanced F1 scores, I selected the SVM model as the final predictive model.

Table 10*Performance Metrics of Predictive Models*

		Class 0			Class 1		
Model	Accuracy	Precision	Recall	F1-Score	Precision	Recall	F1-Score
SVM	0.734	0.56	0.66	0.61	0.84	0.77	0.80
Random Forest	0.758	0.75	0.28	0.41	0.76	0.96	0.85
XGBoost	0.729	0.54	0.60	0.57	0.82	0.79	0.80

The SVM models feature importance analysis revealed that linguistic characteristics influence electoral outcomes across four key dimensions. Lexical sophistication emerged as the strongest predictor, with `avg_word_length` ranking highest, followed by bigram and trigram entropy repetition measures that capture phrase predictability. Syntactic complexity features, such as the number of clauses per sentence and the mean dependency distance, reveal how sentence structure influences electoral success. Furthermore, lexical diversity metrics, such as `bigram_ttr_pct` and `mattr`, underscore the importance of vocabulary richness for electoral performance. Finally, semantic features such as attention pattern complexity and vector space density influenced outcomes beyond surface-level choices. These findings demonstrate that effective political speech must strike a balance between clear communication and strategic language.

Figure 14*Top 15 Feature Importance for SVM*

5

Discussion

This chapter interprets the key findings presented in the previous chapter, linking them to research questions, communication theories, and existing literature to extend, challenge, or reinforce current understandings of executive political communication. This chapter also examines the broader implications of these findings by showing how executive leaders project leadership, shape political messaging strategies, and navigate institutional frameworks that constrain or enable specific rhetorical choices.

5.1 Pronoun Usage

My analysis of pronoun usage in executive speeches from presidential and parliamentary systems revealed systematic differences that support the rhetorical presidency theory (Tulis, 1987). These findings suggest that institutional design has a fundamental impact on executive rhetoric. Presidential speeches consistently used more first-person singular pronouns, emphasizing individual leadership, which aligns with prior findings from Gunsch et al. (2000) and Slatcher et al. (2007) that individual pronouns reinforce perceptions of personal authority and authenticity in executive speech. Presidents also used second-person pronouns more

frequently than prime ministers, indicating a greater sense of audience engagement, which supports Pennebaker's (2011) argument that pronoun usage reflects speaker–audience relationships. This finding aligns with the rhetorical presidency theory's claim that presidents must build direct public connections beyond institutional channels.

Beyond individual pronouns, a distinguishing feature of presidential rhetoric was the greater use of pronoun–verb constructions, which highlighted leadership and decisiveness. Presidents used significantly more individual past references than prime ministers, emphasizing completed personal actions. They also used more individual present and individual future in their communications, underlining active leadership and future intentions. These patterns suggest that presidential systems, shaped by the separation of powers, encourage more action-driven and personally accountable language, reinforcing Alavidze's (2016) findings that suggest pronoun usage reveals speakers' attitudes and intentions, as well as the rhetorical presidency theory, which views direct public action as central to presidential leadership. Presidents also used more references to the collective past and the collective future, indicating a stronger tendency to highlight both shared accomplishments and future collective goals. Interestingly, collective present usage was similar across both systems, suggesting that both systems employ a common language when describing ongoing collaborative efforts.

In addition to using more pronoun–verb constructions, presidents also demonstrated a greater tendency to alternate between individual and collective pronoun usage in their speeches. Both individual-to-collective and collective-to-individual switches were significantly higher in presidential speeches, suggesting that the separation of powers enables executives to navigate between personal authority and collective responsibility more effectively than prime ministers.

This rhetorical flexibility also highlights how presidents claim individual credit while calling for collective actions, all within the same speech.

However, collective pronoun usage did not differ significantly across systems. Both presidents and prime ministers used collective pronouns at similar rates, which appears to challenge Steffens and Haslam's (2013) assumptions that parliamentary systems rely more heavily on collective rhetoric. Similar rates of collective pronoun usage across systems suggest that appeals to collective responsibility are global rather than system specific. These findings validate the extension of rhetorical presidency theory beyond the U.S. context. While institutional structures shape personal pronoun patterns, the similarities in collective framing reveal common executive communication strategies across governmental systems.

5.2 Storytelling

Analysis of narrativity in executive speeches reveals that institutional differences between presidential and parliamentary systems systematically shape rhetorical strategies. Presidential speeches, when compared to parliamentary speeches, exhibit higher overall narrativity and transitions across the stages of staging, plotting, and cognitive tension, suggesting the presence of more dramatic and emotionally charged narrativity reflecting the direct communicative needs of presidents, who must engage audiences directly without institutional intermediaries.

Presidential speeches exhibited higher levels of staging, aligning with Fisher's (1984) argument that stories are most persuasive when they establish relevance early and follow a coherent structure. It also supports Hammer's (2010) finding that early storytelling helped forge immediate audience connection. These sharper presidential openings in early stages highlight the urgency of establishing immediate narrative engagement when communicating directly to

broader audiences. Plot development also varied between presidential and parliamentary systems. Presidential speeches demonstrated greater plot progression, while parliamentary speeches maintained a steady pace. This escalation aligns with Franch's (2018) observation of narrative intensification in wartime U.S. presidential speeches, where presidents invoked duty and sought to unify national sentiment. Jones and McBeth (2010) similarly noted that storytelling helps frame policy as urgent and worthy of public attention. Intensified plotting in presidential rhetoric likely reflects efforts to seize attention and assert authority without relying on institutional intermediaries. In contrast, the more measured plotting in parliamentary speeches suggest institutional constraints that favor collective deliberation. Furthermore, cognitive tension was higher in presidential speeches, and resolution was also higher, reflecting an overall tendency toward greater emotional buildup and faster resolution. These findings support Lerner et al.'s (2015) argument that emotional trajectories shape political judgment, particularly in direct communication contexts, and align with Polletta's (1998) observation that collective narratives often maintain steadier emotional tones, as seen in parliamentary speech data. These narrative patterns suggest that presidents are better problem solvers. Prime ministers, by contrast, rely on shared, deliberative institutional language, delaying resolution.

While apparent differences exist, the presence of narrativity in both presidential and parliamentary speeches suggests that both systems employ storytelling to capture audience attention and ensure message coherence, illustrating how institutional contexts influence the way leaders tell stories.

5.3 Repetition

An analysis of repetition patterns revealed systematic differences between presidential and parliamentary systems. Presidential speeches displayed greater lexical diversity through

higher bigram and trigram type-token ratios, but also featured steeper Zipf slopes with higher repetition rates, indicating that presidents draw from a broad vocabulary to reinforce key phrases, suggesting that presidents rely on a broader vocabulary overall but strategically reinforce key messages through a limited set of repeated phrases. Lower bigram entropy and the repeated use of common word pairs indicate a rhetorical style aimed at reinforcing leadership themes and ensuring consistency in messages. These patterns align with Tannen's (2007) and Benoit's (2001) views that repetition in presidential rhetoric serves persuasive, symbolic, and agenda-setting purposes.

In contrast, parliamentary speeches exhibited lower repetition rates, flatter Zipf slopes, and higher bigram entropy, suggesting a more evenly distributed and flexible rhetorical style. These patterns suggest that prime ministers may use a more varied, dynamic language consistent with Heritage and Greatbatch's (1986) framing of parliamentary discourse as consensus-oriented and multi-voiced. Parliamentary speeches suggest a more even lexical distribution, consistent with consensus-oriented rhetoric where repetition helps align multiple viewpoints rather than assert rhetorical dominance. Together, these findings show that repetition is not just a rhetorical choice but tailored to the demands of each political system.

Presidential speeches showed higher frequencies of both common bigram bundles like "of the," "it is," and "we have," and trigram bundles such as "president of the" and "of the republic," indicating a structured, repetitive, and leadership-focused rhetorical style. In contrast, parliamentary speeches used these bundles less often and favored more conversational or policy-driven phrases like "would like to" and "we need to," reflecting a more flexible and collaborative tone. These patterns are consistent with Heritage and Greatbatch's (1986) findings that repetition in parliamentary settings supports collective deliberation rather than individual persuasion.

Together, these differences highlight how institutional design shapes the communicative roles and rhetorical choices of executive leaders.

Repetition patterns in executive speeches reflect the strategic influence of institutional design. Presidents tend to use more frequent and structured phrasing to assert authority and maintain message control. In contrast, prime ministers adopt a more varied and flexible style suited to a collaborative and deliberative environment. These differences show that repetition is shaped not just by rhetorical intent, but by the communicative demands of each political system.

5.4 Agenda Setting

Thematic patterns observed in the top topics of executive speeches reflect how the institutional context shapes agenda-setting priorities, aligning with Agenda-Setting Theory (McCombs & Shaw, 1972). Presidential speeches demonstrated a clear emphasis on military and veterans' affairs, international cooperation and trade, science and technology, digital innovation, and law enforcement, which project leadership, national identity, and geopolitical relevance. This focus aligns with the findings of Wood (2007) and Eshbaugh-Soha et al. (2005), who argue that the direct electoral mandate and separation of powers enable presidents to shape public priorities with fewer constraints. Moreover, this breadth of issue aligns with Habel's (2012) finding that U.S. presidential speeches appeal to a wide range of constituencies, unconstrained by party platforms.

In contrast, parliamentary speeches prioritized education policy, gender equality and women's issues, renewable energy, ocean and marine policy, and public health, reflecting a more domestic, policy-oriented agenda aligned with the demands of coalition politics and legislative consensus. These patterns align with Binzer Hobolt and Klemmensen (2008) and Habel (2012),

who found that prime ministers operate within tighter institutional boundaries and party obligations, shaping agendas that resonate with legislative priorities and societal equity. Together, these results demonstrate that executive leaders use rhetoric not just to communicate, but to shape strategically and prioritize policies in ways that reflect their institutional design.

Analyzing the top 20 topics by weight revealed striking institutional differences: only seven topics overlapped between systems, while thirteen were unique to each. Presidential speeches prioritized themes such as agriculture and rural development, fertilizer and farming policy, fishing and fisheries management, and cultural arts—issues that projected symbolic identity and broad appeal. This aligns with Agenda-Setting Theory (McCombs & Shaw, 1972) and supports the findings of Habel (2012) and Wood (2007), which suggest that presidents use their institutional autonomy to highlight diverse and high-salience issues. In contrast, parliamentary speeches emphasized logistics and infrastructure, sports and cultural events, public health infrastructure, anti-corruption mechanisms, family and childcare policy, nuclear science and technology, healthcare services, and judicial system governance, reflecting a more grounded, policy-focused agenda. These patterns confirm that institutional constraints shape not just what issues are raised, but which issues executives can prioritize, reinforcing Binzer Hobolt and Klemmensen's (2008) observation that parliamentary leaders operate within tighter ideological and legislative bounds.

The temporal trends in the top ten topics in executive speeches from 2003 to 2025 reveal distinct patterns between the two systems. In the parliamentary system, the distribution was highly volatile. Climate and environmental topics showed the most dramatic early spikes, reaching approximately 4.2% in 2007 before declining and fluctuating moderately over the remainder of the period. International cooperation and trade topics peaked around 2006 (3.6%)

and displayed cyclical increases, particularly in 2012 and again in 2024. Education topics exhibited steady growth, reaching their highest point in 2014 (3.2%) and maintaining a consistent emphasis in recent years. Women and gender equality topics appeared with variable intensity, peaking in 2011 (2.5%) and again in 2018 (2.8%). Most notably, ocean and marine sustainability topics surged in prominence around 2018–2019, peaking at nearly 4% before declining to moderate levels by 2025.

The presidential system, although exhibiting volatility, displayed distinct temporal patterns. Military and veteran topics dominated the early years, beginning strongly in 2003 (2.9%) and peaking again in 2008 (2.7%) before gradually declining over time. International cooperation and trade topics showed moderate fluctuations with noticeable peaks around 2005 (1.9%) and 2014–2015 (1.7%). Education topics maintained a relatively stable presence, with a notable rise in 2009 (1.4%). Women and gender equality topics exhibited the most dramatic increase around 2013–2014 (1.8%) before declining significantly. Climate and environmental topics remained low throughout much of the period until a sharp rise occurred in 2021, reaching 1.8%. The apparent differences in topic focus and timing reflect the influence of institutional design. Presidents tend to highlight broad, high-profile issues, while prime ministers emphasize domestic policy shaped by legislative and party priorities.

5.5 Linguistic Complexity

My analysis of linguistic complexity in executive speeches from presidential and parliamentary systems revealed patterns consistent with Uses and Gratifications Theory (Katz et al., 1973), which emphasizes communication centered around the audience. Presidential speeches featured simpler vocabulary, shorter word lengths, and more repetitive phrasing, suggesting a deliberate effort to meet public demands for clarity, ease of understanding, and

emotional connection, aligning with Uses and Gratifications Theory's proposition that speakers adapt their messages to satisfy audience needs as demonstrated by Spirling (2016), who found that British parliamentary leaders varied linguistic complexity depending on whether they were addressing public or elite audiences. Parliamentary speeches demonstrated higher lexical diversity and complexity tailored to institutional audiences conditioned to policy nuance and deliberation. These patterns suggest that linguistic complexity is a strategic choice shaped by institutional setup.

Semantic complexity in executive speeches from presidential and parliamentary systems reveals that both systems exhibited similar average cosine distance and lexical sophistication indicating comparable semantic spread and vocabulary rarity. However, presidential speeches showed higher vector space density and stronger semantic clustering suggesting that presidential rhetoric is more thematically concentrated, likely intended to improve message accessibility and retention among mass audiences, consistent with Uses and Gratifications Theory emphasis on communicators shaping content to meet audience needs. This supports the findings of Urbaniak (2022), who demonstrated that U.S. presidents strategically employ linguistic complexity to project competence and shape public perception. In contrast, parliamentary speeches exhibited lower semantic density, indicating a more diffused communication style that aligns with the deliberative and multi-perspective demands of parliamentary communication. This suggests that semantic complexity is not uniform but strategically calibrated to institutional roles and audience expectations.

Syntactic complexity in executive speeches from presidential and parliamentary systems revealed that presidential speeches featured more clauses per sentence and exhibited greater mean dependency distance, indicating a preference for longer, more elaborated sentence

structures used to emphasize key points. However, they showed slightly lower parse tree depth, suggesting that while presidents favor clause-rich sentences, they tend to avoid deeply nested sentences, which are likely to compromise clarity for general audiences. Parliamentary speeches, by comparison, included fewer clauses per sentence but greater structural depth, pointing to more hierarchical and embedded constructions. This pattern aligns with findings by Bischof and Senninger (2018), who observed that parliamentary actors simplify or adapt their language in response to institutional processes, such as consensus-building. Despite these structural variations, both systems demonstrated similar Flesch-Kincaid Grade Level scores, suggesting surface-level readability remains consistent. These findings suggest that syntactic complexity, like other linguistic clues, is influenced by the communicative roles and institutional settings that executive leaders encounter.

Analyzing more complex patterns of contextual complexity in executive speeches it was observed that both systems showed similar contextual embedding variation, but parliamentary speeches displayed slightly higher attention pattern complexity and cross-sentence coherence. This suggests a greater reliance on connecting ideas across sentences, reflecting their deliberative structure. These patterns align with Schoonvelde et al. (2019), who found that institutional and ideological contexts influence the complexity of political language. In contrast, presidential speeches appeared more segmented, focusing on clarity and immediate impact.

Similar levels of readability and word sophistication across systems suggest that making speeches easy to understand is a common goal for executive leaders. However, apparent differences in word variety, meaning structure, sentence patterns, and how ideas are connected show that institutional roles influence how leaders shape their messages. These findings suggest that linguistic complexity reflects strategic choices where presidents use precise, focused

language to reach broad audiences, while prime ministers employ language suited to parliamentary debate and collaboration.

5.6 Electoral Prediction

This study demonstrates that linguistic patterns in executive speeches can predict electoral outcomes using a unified model across both presidential and parliamentary systems. The Support Vector Machine achieved balanced performance by identifying key predictive features: average word length, bigram and trigram entropy, and type-token ratios, which highlight the importance of lexical richness and repetition patterns. These findings align with Ali et al. (2022), who emphasized the value of deep textual structures in political prediction. Syntactic complexity features such as clauses per sentence, mean dependency distance, and parse tree depth also proved predictive, supporting Polletta's (1998) and Lerner et al.'s (2015) observations that narrative coherence and structure shape how audiences process political messages. The prominence of semantic cohesion metrics, such as vector space density, further suggests that political texts encode institutional signals even without explicit sentiment markers. Ultimately, this study demonstrates that a leader's ability to structure ideas and manage rhetorical complexity in their speeches plays a crucial role in determining electoral success.

5.7 Summary

Our analysis demonstrated that institutional designs fundamentally shape how presidents and prime ministers communicate with the public. Presidents consistently use more personal language, tell stories with dramatic ups and downs, repeat key phrases strategically while still using varied vocabulary, focus on national themes and international issues, and use simpler words arranged in longer but less complex sentences to reach broad audiences. While Prime Ministers take a more collaborative approach, referencing other people and institutions more

often, telling steadier stories without sharp emotional peaks, repetition spread evenly across their speeches, emphasizing domestic policies and building consensus, and using more sophisticated vocabulary with complex sentence structures suited to policy debates.

However, some similarities challenge our understanding of how different political systems work. Both presidents and prime ministers use collective language at nearly identical rates, structure the middle portions of their stories, similarly, maintain comparable reading difficulty levels. Presidents manage to be both more varied and more repetitive at the same time.

Importantly, these linguistic patterns carry predictive power, and our findings show that the way executive leaders communicate can help predict whether they win or lose elections, regardless of the political system. The findings suggest that while institutional setups influence how leaders communicate, the basic requirements of executive leadership, irrespective of the system, are to create communication strategies that convey and convince audiences.

5.8 Limitations

While this study offers a comprehensive and data-driven comparison of executive speeches between presidential and parliamentary systems, several limitations exist. First, the use of machine learning to translate non-English speeches introduces inconsistencies. Although ML translation capabilities have improved, they may still fail to preserve some rhetorical and linguistic features.

Second, although institutional, regional, and linguistic balance guided the creation of the corpus, the dataset remains skewed toward democracies with accessible digital records. Additionally, countries vary widely in the temporal span of available speeches, which can introduce historical confounds, as rhetorical norms may change over time due to shifts in the

political climate, media environments, or global events, thereby complicating cross-system comparisons.

Third, executive leadership globally is a male-dominated field. As a result, this corpus includes a relatively small number of female leaders, which limits the study's ability to generalize across genders. Finally, while the analysis focuses on institutional differences, other factors, such as election cycles, national crises, economic downturns, or individual leadership styles, can influence executive language.

5.9 Recommendations

This comparative analysis of executive speeches between presidential and parliamentary systems proves that each system has its unique linguistic patterns. The way leaders speak, project, and connect with their audience varies across different systems, providing insightful guidance to speechwriters, administrators, campaign managers, foreign policy specialists, and academics.

Communication teams and speechwriters working with presidential executives should focus more on using individual pronouns in combination with action sequences, as well as switching between individual and collective voices, to build a shared purpose. The need for direct appeal in presidential speeches requires simpler language, with clause-rich sentences and dramatic storytelling, featuring sharp openings that build up escalating tension. Strategic repetition of phrases that reinforce leadership themes and agenda-setting, prioritizing national security and international relations amidst the current challenges of the time.

Parliamentary communication teams face different demands that require references to institutions and external actors rather than individual authority. The deliberative nature requires complex vocabulary and multi-clause sentences to communicate policy positions and facilitate

consensus rather than dramatic impact. Lower repetition rates, combined with conversational phrases like "would like to," signal collaborative openness and a responsive governance agenda that prioritizes adapting domestic priorities.

Campaign managers require system-specific competencies, as presidential techniques may not be applicable in parliamentary contexts. Presidential candidates should focus on optimizing word length, sentence complexity, and repetition patterns, while parliamentary candidates should emphasize different linguistic structural elements.

Training programs should recognize systematic differences rather than universal approaches. Executive grooming should match each system, with presidential leaders trained in strategic leadership and parliamentary leaders in coalition management. Political professionals working across systems should interpret these differences as institutional traits rather than personal preferences.

Academics can extend this research by investigating how these patterns vary in a digital social world, examining temporal changes in communication, exploring the roles of gender, age, and economic context in system-specific executive communication, and how communication patterns evolve as leaders gain experience. Additionally, studies should be conducted on hybrid institutional systems, such as France's semi-presidential model or Switzerland's directorial system, to understand how communication patterns differ.

Audience reception is another area that could benefit from research on system-specific communication styles, including how cross-system communication affects international versus domestic audiences and how media filters or amplifies institutional messaging patterns.

Future studies could investigate how institutional patterns vary across various platforms, including traditional speeches, social media interactions, live debates, and unscripted encounters.

Acknowledging recent trends in Artificial Intelligence, further studies could investigate how LLMs and generative AI tools affect institutional communication patterns and whether they preserve institutional linguistic style.

The documented differences provide evidence for developing institutional-specific communication strategies rather than universal approaches. Presidential systems require individual-focused, accountable language that emphasizes action, while parliamentary systems demand institutional references and deliberative complexity. Understanding that communication differences reflect institutional setup rather than personal preferences improves cross-cultural political engagement, diplomatic effectiveness, and professional development. Practitioners who recognize these systematic differences can develop strategies that align with institutional requirements, enhancing the quality of executive communication, fostering citizen engagement, and public trust in executive offices.

5.10 Contributions

This research makes several interconnected contributions to the analysis of political communication. First, we created custom pipelines to extract and build the first comprehensive comparative dataset of executive political communication, comprising over 20,000 speeches with 27 million words across 16 nations and 13 languages, from both presidential and parliamentary systems. This multilingual corpus enables systematic comparisons across nations, cultures, and institutions, establishing a benchmark for evaluating political texts using data science approaches.

Methodologically, we make two contributions. First, we developed a comprehensive, quantifiable linguistic feature extraction framework leveraging advanced NLP techniques. This approach yields 41 quantifiable lexical, syntactic, and semantic metrics based on pronoun

analysis, storytelling patterns, agenda setting, repetition structures, and linguistic complexity measures, capturing the multifaceted nature of executive political rhetoric. Second, we applied these linguistically informed features to a Support Vector Machine that successfully predicts electoral outcomes across various governmental systems, while revealing the most significant linguistic patterns that influence electoral success. This dual approach bridges data science with established theories in political communication, demonstrating that executive rhetoric is measurably predictive rather than stylistic or symbolic.

Practically, this work provides an electoral forecasting tool based solely on publicly accessible speeches. The universal, interpretable linguistic features deliver actionable insights for campaign strategists and communication professionals while enabling journalists and researchers to study rhetoric systematically. This data science approach enhances democratic transparency, accountability, and trust in public offices across diverse institutional, linguistic, and cultural contexts.

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Appendix

Appendix A

Pronoun Categories and Forms Used in the Study

Category	Forms
first_singular	i,i'd,i'd've,i'll,i'm,i've,me,mine,my,myself
first_plural	let's,lets,our,ours,ourselves,us,we,we'd,we'll,we're,we've,weve
second_person	you,you'd,you'll,you're,you've,youd,youll,your,youre,yours,youve
third_singular	he,he'd,he's,her,hers,herself,hes,him,himself,his,hissel*,oneself,she,she'd,she'll,she's,shes
third_plural	their*,them,themselves,they,they'd,they'll,they've,theyd,theyll,theyve

Appendix B

Verb Categories and Forms Use in the Study

Category	Forms
VB	verb, base form
VBD	verb, past tense
VBG	verb, gerund or present participle
VCN	verb, past participle
VBP	verb, non-3rd person singular present
VBZ	verb, 3rd person singular present

Appendix C

BERTopic Script

```
from bertopic import BERTopic from sklearn.feature_extraction.text import CountVectorizer
from tqdm import tqdm import json import pandas as pd from datetime import datetime import
matplotlib.pyplot as plt import re import spacy from umap import UMAP import numpy as np
import random import os
```

SET RANDOM SEEDS FOR REPRODUCIBILITY

```
RANDOM_SEED = 42 np.random.seed(RANDOM_SEED) random.seed(RANDOM_SEED)
os.environ['PYTHONHASHSEED'] = str(RANDOM_SEED)
```

```
nlp = spacy.load("en_core_web_sm") nlp.max_length = 2_000_000
```

```
with open("Presidential_Speeches_with_dates.json", "r", encoding="utf-8") as f: pres_data =
json.load(f) with open("Parliamentary_Speeches_with_dates.json", "r", encoding="utf-8") as f:
parl_data = json.load(f)
```

Text cleaning with noise and NER filtering

```
def clean_text(text): noise_patterns = [ # Original patterns
r'\b(applause|laughter|cheers|clapping|thank you|ladies and gentlemen|your excellency)\b',
r'\b(welcome|pleased to|honor to|begin by|let me|conclude|in closing|god bless)\b',
r'\b(happy|merry|new year|festival|celebrate|easter|eid|diwali|ramadan|republic day)\b',
r'\b(because|that's why|you know|we have to|i think|uh|um)\b',
r'\b(nonenglish|brazil|bharat|ji|indonesia|african)\b', r'(press statement by president joko
widodo|greetings from president joko widodo|remarks by president joko widodo|speech by
president joko widodo|president joko widodo s statement|transcript of president joko widodo
s|address by president joko widodo|republic of kenya press statement|republic of kenya
remarks)',
r'\b(britain|uk|eu|canada|canadians|australia|australian|seychelles|seychellois|netanyahu|ruto|wil
liam|commanderinchief)\b', r'\b(hon|ble|mr speaker|madam speaker|speaker|national
assembly|assembly|parliament|parliamentary|hon member|honourable|minister|deputy
minister)\b', r'\b(going|i am|i will|i have|we are|we will|we have|this is|that is|it is)\b',
r'\b(order|point of order|question|questions|motion|bill|act|clause|section|amendment)\b',
r'\b(nonenglish|joko widodo|republic joko|widodo|greet dear|like greet)\b',
r'\b(brazilian|burma|aung|la les des|et en der du|tata|000d|tni nonenglish)\b', r'\b(africa
000|commanderinchief|theyre america weve|stoltenberg|bondevik)\b', r'\b(solberg|kjell
magne|hon ble|mr speaker|speaker|assembly)\b',
r'\b(going|said|think|know|time|way|make|take|get|give|work|help)\b',
r'\b(order|question|motion|bill|act|clause|member|minister|deputy)\b',
r'\b(thank|thanks|welcome|pleased|honor|begin|conclude|closing)\b',
r'\b(ram|lord|sisters|saints|guardian|muhammadiyah|siti|maria|ikn)\b',

# Additional patterns to filter out topics that ranked higher than our Top 10
# Personal Names/Officials (high-ranking noisy topics)
r'\b(trump|obama|nato|relationship|barack|david)\b',
```

r\b(widodo|joko|rice|journalist|republic|yes|sake)\b',
 r\b(solberg|erna|war|preparedness|office|outdoor|oil|competence)\b',
 r\b(stoltenberg|jens|party|unemployment|kroner|july)\b',
 r\b(gandhi|mahatma|gandhiji|satyagraha|communal)\b',
 r\b(modi|baat|mann|ki|countrymen)\b',
 r\b(bondevik|kjell|welfare|kr|storting)\b',
 r\b(jugnauth|pravind|hrs|kumar|mauritius)\b',

Country/Region-Specific (high-ranking noisy topics)

r\b(australians|indigenous|labor|sydney|commonwealth|alliance|tonight|immigration)\b',
 r\b(sisters|brothers|friends|projects|railway|railways|dki|lrt|governor|transportation)\b',
 r\b(respected|tni|rp|nation|republic|language|rodrigues|languages)\b',
 r\b(temple|agung|bali|hindus|pura)\b',
 r\b(peninsula|unification|north|korea|peace|advisory)\b',
 r\b(asean|mekong|excellencies|region|indopacific)\b',
 r\b(mercosur|paraguay|integration|uruguay|kirchner)\b',
 r\b(argentines|buenos|aires|truth|works)\b',
 r\b(rio|janeiro|cabral|eduardo)\b',
 r\b(ilbo|residents|gangwondo|local|gwangju)\b',

Institution/Agency-Specific

r\b(kpk|prosecution|investigation|cbi|anticorruption)\b',
 r\b(bnpb|damage|reconstruction|evacuation)\b',
 r\b(iit|iits|university|students|drdo|manchester)\b',
 r\b(secretarygeneral|excellencies)\b',

Event/Summit-Focused (high-ranking)

r\b(g20|g7|summit|financial|crisis|global)\b',
 r\b(sports|athletes|olympics|games|winter|players)\b',
 r\b(disaster|disasters|earthquake|emergency|spread)\b',

Conversational/Informal (high-ranking)

r\b(folks|weve|theyre|youre|audience|jobs|im)\b',

Specific noisy combinations that ranked high

r\b(god|guedes|federal|army|wonderful)\b',
 r\b(party|remain|unity|certain|housing)\b',
 r\b(dam|regency|water)\b',
 r\b(toll|road|airport|sumatra|inaugurate)\b',
 r\b(quarantine|covid19|confirmed|careful|ppkm)\b',

```

r'\b(certificate|certificates|land)\b',
r'\b(independence|movement|provisional|march|activists|1942|freedom)\b',
r'\b(hunger|poverty|food|2004)\b',
r'\b(buddhism|buddha|teachings|monk|monks|temple|fdi)\b',
]
for pattern in noise_patterns:
    text = re.sub(pattern, "", text, flags=re.IGNORECASE)

# Remove named entities: NORP, FAC, GPE, LOC
doc = nlp(text)
spans_to_remove = [ent.text for ent in doc.ents if ent.label_ in {'NORP', 'FAC', 'GPE', 'LOC'}]
for span in spans_to_remove:
    text = text.replace(span, "")

return text.strip()

pres_docs = [clean_text(speech["content"]) for speech in tqdm(pres_data, desc="Cleaning Presidential")]
parl_docs = [clean_text(speech["content"]) for speech in tqdm(parl_data, desc="Cleaning Parliamentary")]

def safe_extract_year(date_str):
    try:
        return datetime.strptime(date_str, "%Y-%m-%d").year
    except:
        return None

pres_years = [safe_extract_year(s.get("date", "")) for s in pres_data]
parl_years = [safe_extract_year(s.get("date", "")) for s in parl_data]

docs = pres_docs + parl_docs
classes = ["Presidential"] * len(pres_docs) + ["Parliamentary"] * len(parl_docs)
years = pres_years + parl_years

Custom stop words

custom_stop_words = [
'hon', 'ble', 'mr', 'mrs', 'ms', 'speaker', 'madam', 'sir', 'thank', 'thanks',
'going', 'come', 'came', 'go', 'went', 'said', 'say', 'says', 'telling', 'told', 'think', 'thought', 'know',
'knew', 'see', 'saw', 'look', 'looked', 'want', 'wanted', 'need', 'needed', 'time', 'times', 'way', 'ways',
'make', 'made', 'take', 'taken', 'get', 'got', 'give', 'given', 'work', 'worked', 'working', 'help',
'helped', 'order', 'question', 'questions', 'motion', 'bill', 'act', 'clause', 'section', 'amendment',
'debate', 'discussion', 'vote', 'voting', 'member', 'members', 'honourable', 'minister', 'deputy',
'prime', 'president', 'vice', 'excellency' ]

```

Vectorizer configuration

```
vectorizer_model =
CountVectorizer( stop_words=list(set(list(CountVectorizer(stop_words='english').get_stop_wo
rds()) + custom_stop_words)), min_df=10, max_df=0.95 )
```

UMAP model with random state for reproducibility

```
umap_model = UMAP(n_neighbors=15, n_components=5, min_dist=0.0, metric='cosine',
random_state=RANDOM_SEED)
```

Fit BERTopic

```
topic_model = BERTopic( verbose=True, calculate_probabilities=True,
vectorizer_model=vectorizer_model, umap_model=umap_model ) topics, probs =
topic_model.fit_transform(docs)
```

Get topics per class - ALL TOPICS

```
topics_per_class = topic_model.topics_per_class(docs, classes=classes)

print(f"\nALL TOPICS DISCOVERED:") print(f"Total topics found:
{len(topics_per_class['Topic'].unique())} (including outliers)")
```

Save ALL Topics Per Class with Weight

```
topics_pc_df = topics_per_class.copy() topics_pc_df["Weight%"] =
topics_pc_df.groupby("Class")["Frequency"].transform(lambda x: x / x.sum() * 100)
topics_pc_df.to_csv("topic_weight_by_system.csv", index=False) print("Saved
topic_weight_by_system.csv")
```

Save Per-Speech Topic Assignment and Probability - KEEP ALL SPEECHES

```
speech_topic_df = pd.DataFrame({ "speech_id": list(range(1, len(docs) + 1)), "system":
classes, "dominant_topic": topics, "dominant_topic_prob": [float(p) if isinstance(p, (float, int))
else (max(p) if p is not None and len(p) > 0 else 0.0) for p in probs], "year": years })
```

Keep ALL speeches - only year filtering for complete probability data


```
speech_topic_df = speech_topic_df[(speech_topic_df["year"] >= 2000) &
(speech_topic_df["year"] <= 2025)]
speech_topic_df.to_csv("speech_topic_mapping_with_probs.csv", index=False) print("Saved
speech_topic_mapping_with_probs.csv")
```

Topic Distributions Over Time - ALL TOPICS

```
topic_over_time_data = []

for system in ["Presidential", "Parliamentary"]: system_df =
speech_topic_df[speech_topic_df.system == system]

if not system_df.empty: # Only proceed if we have data
    grouped = system_df.groupby(["year",
"dominant_topic"]).size().reset_index(name="count")
    totals = system_df.groupby("year").size().reset_index(name="total")
    merged = pd.merge(grouped, totals, on="year")
    merged["proportion"] = merged["count"] / merged["total"]
    merged["system"] = system
    topic_over_time_data.append(merged)

if topic_over_time_data: # Only create if we have data final_topic_trends =
pd.concat(topic_over_time_data)
final_topic_trends.to_csv("topic_distributions_over_time.csv", index=False) print("Saved
topic_distributions_over_time.csv")

print("Processing complete.")

from google.colab import files files.download('speech_topic_mapping_with_probs.csv')
files.download('topic_weight_by_system.csv')
files.download('topic_distributions_over_time.csv')
```

Appendix D

Sample Web Extraction Script

```
import requests from bs4 import BeautifulSoup import json import time import re

class NorwegianSpeechScraper: def init(self): self.base_url =
"https://www.regjeringen.no/no/dokumentarkiv/id115322/" self.headers = { 'User-Agent':
```

```
'Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36', 'Accept':
'text/html,application/xhtml+xml,application/xml;q=0.9,;q=0.8', 'Accept-Language': 'nb-
NO,nb;q=0.9,no-NO;q=0.8,en-US;q=0.7,en;q=0.6', } self.session = requests.Session() self.utid
=
```

```
"742672,235039,246375,612139,279845,611883,234245,612019,231991,612064,611826,612
151,235960,2464455,612201,2465324,2465405,2465409,2465242,2465412,612243,2465320"
```

```
def detect_language(self, title, content):
```

```
    """Detect if the speech is in Norwegian or English"""
```

```
    if title.startswith('Prime Minister'):    return 'English'
```

```
    elif title.startswith('Statsminister'):
```

```
        return 'Norwegian'
```

```
    # Fallback to content analysis if title is ambiguous
```

```
    norwegian_words = ['og', 'til', 'fra', 'med', 'har', 'vil']
```

```
    english_words = ['and', 'to', 'from', 'with', 'has', 'will']
```

```
    nor_count = sum(1 for word in norwegian_words if f' {word} ' in content.lower())
```

```
    eng_count = sum(1 for word in english_words if f' {word} ' in content.lower())
```

```
    return 'Norwegian' if nor_count > eng_count else 'English'
```

```
def extract_date(self, content):
```

```
    """Extract date from speech content"""
```

```
    # First try to find date in format "Dato: DD.MM.YYYY"
```

```
    date_match = re.search(r'Dato:\s*(\d{2}\.\d{2}\.\d{4})', content)
```

```
    if date_match:
```

```
        return date_match.group(1)
```

```
    # Try Norwegian date format
```

```
    nor_months = {
```

```
        'januar': '01', 'februar': '02', 'mars': '03', 'april': '04',
```

```
        'mai': '05', 'juni': '06', 'juli': '07', 'august': '08',
```

```
        'september': '09', 'oktober': '10', 'november': '11', 'desember': '12'
```

```
    }
```

```
    # Try English date format
```

```
    eng_months = {
```

```
        'January': '01', 'February': '02', 'March': '03', 'April': '04',
```

```
        'May': '05', 'June': '06', 'July': '07', 'August': '08',
```

```

    'September': '09', 'October': '10', 'November': '11', 'December': '12'
}

# Try Norwegian format
for month in nor_months:
    pattern = rf'\b(\d{{1,2}})\.s*{month}s*(\d{{4}})\b'
    match = re.search(pattern, content, re.IGNORECASE)
    if match:
        day = match.group(1).zfill(2)
        year = match.group(2)
        return f'{day}. {nor_months[month]}. {year}'

# Try English format
for month in eng_months:
    pattern = rf'\b(\d{{1,2}})s*{month}s*(\d{{4}})\b'
    match = re.search(pattern, content, re.IGNORECASE)
    if match:
        day = match.group(1).zfill(2)
        year = match.group(2)
        return f'{day}. {eng_months[month]}. {year}'

return None

def get_speech_content(self, url):
    """Fetch and extract the content of a speech"""
    try:
        response = self.session.get(url, headers=self.headers)
        soup = BeautifulSoup(response.text, 'html.parser')

        content_div = soup.find('div', class_='[article-content, article-body]')
        if content_div:
            # Remove script and style elements
            for element in content_div.find_all(['script', 'style']):
                element.decompose()
            return content_div.get_text(strip=True)
        return None
    except Exception as e:
        print(f'Error fetching speech content: {str(e)}')
        return None

```

```

def get_page_speeches(self, params, page_num):
    """Get speeches from a specific page"""
    try:
        final_params = {
            **params,
            'page': str(page_num)
        }

        response = self.session.get(self.base_url, params=final_params, headers=self.headers)
        print(f"\nFetching page {page_num}")

        soup = BeautifulSoup(response.text, 'html.parser')
        speeches = []

        # Check for "no results" message
        no_results = soup.find(string="Søket ditt gav 0 treff.")
        if no_results:
            print("No more results found")
            return None

        # Look for titles directly
        titles = soup.find_all(['h2', 'h3'], class_='title')
        articles = [t.parent for t in titles if t.parent]
        print(f"Found {len(articles)} articles on page {page_num}")

        for article in articles:
            try:
                title_elem = article.find(['h2', 'h3'])
                link = article.find('a', href=True)

                if not title_elem or not link:
                    continue

                title = title_elem.get_text(strip=True)
                if title == "Om nettstedet":
                    continue

                speech_url = f"https://www.regjeringen.no{link['href']}"
                content = self.get_speech_content(speech_url)

```

```

        speech = {
            'title': title,
            'url': speech_url,
            'content': content,
            'page_number': page_num,
            'language': self.detect_language(title, content or ""),
            'date': self.extract_date(content or "")
        }

        speeches.append(speech)
        print(f'Processed speech: {title} ({speech["language"]}) - Date: {speech["date"]}')
        time.sleep(1)

    except Exception as e:
        print(f'Error processing article: {str(e)}')
        continue

    return speeches

except Exception as e:
    print(f'Error fetching page {page_num}: {str(e)}')
    return None

def get_speeches_for_year(self, year):
    """Get all speeches for a specific year"""
    all_speeches = []
    page = 1

    params = {
        'documenttype': 'aktuelt/talerogartikler',
        'term': "",
        'utid': self.utid,
        'from': f'01.01.{year}',
        'to': f'31.12.{year}'
    }

    while True:
        page_speeches = self.get_page_speeches(params, page)

        # If None returned, we've hit the end

```

```

    if page_speeches is None:
        break

    # If no speeches found on this page, stop
    if not page_speeches:
        break

    all_speeches.extend(page_speeches)
    page += 1
    time.sleep(2) # Polite delay between pages

return all_speeches

def save_speeches_by_language(self, speeches, year):
    """Save speeches separated by language"""
    # Separate speeches by language
    english_speeches = [s for s in speeches if s['language'] == 'English']
    norwegian_speeches = [s for s in speeches if s['language'] == 'Norwegian']

    # Save English speeches
    with open(f'speeches_{year}_english.json', 'w', encoding='utf-8') as f:
        json.dump(english_speeches, f, ensure_ascii=False, indent=2)
        print(f'Saved {len(english_speeches)} English speeches for {year}')

    # Save Norwegian speeches
    with open(f'speeches_{year}_norwegian.json', 'w', encoding='utf-8') as f:
        json.dump(norwegian_speeches, f, ensure_ascii=False, indent=2)
        print(f'Saved {len(norwegian_speeches)} Norwegian speeches for {year}')

    # Save combined file for reference
    with open(f'speeches_{year}_all.json', 'w', encoding='utf-8') as f:
        json.dump(speeches, f, ensure_ascii=False, indent=2)
        print(f'Saved {len(speeches)} total speeches for {year}')

def main():
    scraper = NorwegianSpeechScraper()
    all_english_speeches = []
    all_norwegian_speeches = []

    # Process years from 2005 to 2021
    for year in range(2005, 2022):

```

```

print(f"\nProcessing year {year}...")
year_speeches = scraper.get_speeches_for_year(year)

# Save speeches separated by language
scraper.save_speeches_by_language(year_speeches, year)

# Add to overall collections
all_english_speeches.extend([s for s in year_speeches if s['language'] == 'English'])
all_norwegian_speeches.extend([s for s in year_speeches if s['language'] == 'Norwegian'])

time.sleep(3) # Polite delay between years

# Save complete collections
with open('all_speeches_english.json', 'w', encoding='utf-8') as f:
    json.dump(all_english_speeches, f, ensure_ascii=False, indent=2)
    print(f"\nSaved total of {len(all_english_speeches)} English speeches")

with open('all_speeches_norwegian.json', 'w', encoding='utf-8') as f:
    json.dump(all_norwegian_speeches, f, ensure_ascii=False, indent=2)
    print(f"\nSaved total of {len(all_norwegian_speeches)} Norwegian speeches")

if name == "main": main()

```

Appendix E

Parliamentary system - Pronoun usage by country

Country	Metric	Mean	SD	95% CI
Australia	1PL	3.02	0.07	2.89-3.16
Canada	1PL	3.9	0.06	3.78-4.01
Israel	1PL	2.36	0.12	2.13-2.60
India	1PL	2.44	0.02	2.39-2.49
Japan	1PL	1.64	0.03	1.57-1.71
Mauritius	1PL	2.02	0.16	1.72-2.35
Norway	1PL	2.67	0.04	2.60-2.75
United Kingdom	1PL	2.57	0.05	2.48-2.67
Australia	1SG	1.22	0.05	1.13-1.32
Canada	1SG	1.06	0.03	0.99-1.12

Israel	1SG	1.35	0.08	1.20-1.52
India	1SG	1.44	0.01	1.41-1.46
Japan	1SG	2.16	0.05	2.07-2.26
Mauritius	1SG	0.89	0.06	0.77-1.00
Norway	1SG	0.74	0.02	0.70-0.77
United Kingdom	1SG	1.21	0.03	1.15-1.27
Australia	2P	1.02	0.06	0.90-1.14
Canada	2P	1.48	0.07	1.36-1.61
Israel	2P	0.78	0.07	0.64-0.94
India	2P	0.81	0.01	0.78-0.83
Japan	2P	0.56	0.02	0.53-0.60
Mauritius	2P	0.71	0.1	0.53-0.92
Norway	2P	0.63	0.02	0.58-0.68
United Kingdom	2P	0.77	0.03	0.71-0.82
Australia	3PL	0.79	0.03	0.73-0.86
Canada	3PL	1.09	0.04	1.01-1.17
Israel	3PL	0.78	0.05	0.69-0.88
India	3PL	0.55	0.01	0.54-0.57
Japan	3PL	0.3	0.01	0.27-0.33
Mauritius	3PL	0.27	0.04	0.20-0.36
Norway	3PL	0.53	0.01	0.51-0.56
United Kingdom	3PL	0.83	0.02	0.79-0.86
Australia	3SG	0.41	0.03	0.34-0.47
Canada	3SG	0.33	0.05	0.25-0.43
Israel	3SG	0.72	0.07	0.58-0.87
India	3SG	0.44	0.01	0.41-0.46
Japan	3SG	0.21	0.02	0.17-0.26
Mauritius	3SG	0.13	0.02	0.09-0.17
Norway	3SG	0.27	0.02	0.23-0.32
United Kingdom	3SG	0.39	0.02	0.35-0.43
Australia	C to I	10.33	0.31	9.74-10.95
Canada	C to I	9.45	0.23	9.00-9.91
Israel	C to I	10.31	0.56	9.23-11.42
India	C to I	10.35	0.11	10.14-10.57
Japan	C to I	12.78	0.32	12.16-13.40
Mauritius	C to I	8.48	0.6	7.32-9.68
Norway	C to I	5.57	0.12	5.35-5.80

United Kingdom	C to I	12.35	0.36	11.67-13.07
Australia	CFU	0.09	0.01	0.07-0.10
Canada	CFU	0.23	0.01	0.21-0.26
Israel	CFU	0.17	0.02	0.14-0.21
India	CFU	0.07	0	0.07-0.08
Japan	CFU	0.3	0.01	0.28-0.33
Mauritius	CFU	0.05	0.01	0.03-0.08
Norway	CFU	0.15	0.01	0.14-0.16
United Kingdom	CFU	0.14	0.01	0.13-0.15
Australia	CPA	0.15	0.01	0.13-0.17
Canada	CPA	0.27	0.01	0.25-0.30
Israel	CPA	0.22	0.02	0.18-0.27
India	CPA	0.12	0	0.11-0.12
Japan	CPA	0.16	0.01	0.14-0.18
Mauritius	CPA	0.06	0.01	0.03-0.08
Norway	CPA	0.13	0.01	0.12-0.14
United Kingdom	CPA	0.11	0	0.10-0.12
Australia	CPR	1.07	0.03	1.02-1.13
Canada	CPR	1.49	0.03	1.42-1.55
Israel	CPR	0.7	0.05	0.62-0.80
India	CPR	0.7	0.01	0.68-0.71
Japan	CPR	0.47	0.02	0.44-0.50
Mauritius	CPR	0.65	0.05	0.54-0.75
Norway	CPR	1.08	0.02	1.05-1.12
United Kingdom	CPR	0.99	0.02	0.94-1.03
Australia	I to C	10.53	0.31	9.92-11.16
Canada	I to C	10.13	0.23	9.68-10.59
Israel	I to C	10.8	0.58	9.67-11.96
India	I to C	10.69	0.12	10.46-10.92
Japan	I to C	13.37	0.33	12.73-14.03
Mauritius	I to C	8.84	0.6	7.68-10.03
Norway	I to C	5.71	0.12	5.48-5.95
United Kingdom	I to C	12.9	0.38	12.20-13.65
Australia	IFU	0.02	0	0.02-0.03
Canada	IFU	0.02	0	0.02-0.03
Israel	IFU	0.04	0.01	0.03-0.06
India	IFU	0.02	0	0.02-0.03

Japan	IFU	0.07	0.01	0.05-0.08
Mauritius	IFU	0.01	0	0.00-0.02
Norway	IFU	0.01	0	0.01-0.02
United Kingdom	IFU	0.02	0	0.02-0.03
Australia	IPA	0.15	0.01	0.14-0.17
Canada	IPA	0.16	0.01	0.14-0.18
Israel	IPA	0.24	0.02	0.20-0.28
India	IPA	0.15	0	0.14-0.16
Japan	IPA	0.37	0.02	0.34-0.41
Mauritius	IPA	0.01	0.01	0.00-0.02
Norway	IPA	0.08	0	0.07-0.08
United Kingdom	IPA	0.14	0.01	0.12-0.15
Australia	IPR	0.63	0.02	0.58-0.68
Canada	IPR	0.52	0.02	0.49-0.55
Israel	IPR	0.59	0.04	0.52-0.67
India	IPR	0.71	0.01	0.69-0.72
Japan	IPR	0.85	0.02	0.80-0.90
Mauritius	IPR	0.44	0.03	0.38-0.50
Norway	IPR	0.35	0.01	0.33-0.37
United Kingdom	IPR	0.69	0.02	0.65-0.73

Note. 1SG=First Person Singular, 1PL=First Person Plural, 2P=Second Person, 3SG=Third Person Singular, 3PL=Third Person Plural, I to C=Individual to Collective, C to I=Collective to Individual, IPA=Individual Past, IPR=Individual Present, IFU=Individual Future, CPA=Collective Past, CPR=Collective Present, CFU=Collective Future.

Appendix F

Presidential system - Pronoun usage by country

Country	Metric	Mean	SD	95% CI
Argentina	1PL	3.13	0.06	3.01-3.25
Brazil	1PL	2.8	0.02	2.77-2.84
Egypt	1PL	1.58	0.1	1.39-1.77
Indonesia	1PL	1.96	0.02	1.91-2.00
Kenya	1PL	1.89	0.07	1.76-2.02
Korea	1PL	2.06	0.02	2.02-2.10
Seychelles	1PL	4.42	0.13	4.16-4.67
United States	1PL	3.26	0.03	3.19-3.33

Argentina	1SG	1.78	0.05	1.68-1.87
Brazil	1SG	2.33	0.02	2.30-2.37
Egypt	1SG	1.32	0.09	1.15-1.50
Indonesia	1SG	2.09	0.02	2.05-2.13
Kenya	1SG	0.67	0.03	0.61-0.73
Korea	1SG	1.85	0.02	1.81-1.89
Seychelles	1SG	1.4	0.07	1.27-1.55
United States	1SG	2.27	0.03	2.21-2.33
Argentina	2P	1.06	0.05	0.96-1.17
Brazil	2P	0.86	0.01	0.84-0.88
Egypt	2P	0.81	0.06	0.70-0.92
Indonesia	2P	0.66	0.01	0.63-0.68
Kenya	2P	0.49	0.04	0.42-0.56
Korea	2P	1.02	0.02	0.98-1.06
Seychelles	2P	0.77	0.06	0.65-0.90
United States	2P	2.12	0.03	2.06-2.19
Argentina	3PL	0.86	0.03	0.80-0.92
Brazil	3PL	0.59	0.01	0.58-0.60
Egypt	3PL	0.6	0.05	0.50-0.71
Indonesia	3PL	0.28	0.01	0.27-0.30
Kenya	3PL	0.3	0.02	0.27-0.33
Korea	3PL	0.49	0.01	0.48-0.51
Seychelles	3PL	0.75	0.04	0.67-0.83
United States	3PL	1.19	0.02	1.15-1.23
Argentina	3SG	0.48	0.03	0.43-0.53
Brazil	3SG	0.4	0.01	0.38-0.41
Egypt	3SG	0.5	0.07	0.38-0.66
Indonesia	3SG	0.14	0.01	0.13-0.16
Kenya	3SG	0.45	0.02	0.41-0.50
Korea	3SG	0.16	0.01	0.14-0.18
Seychelles	3SG	0.18	0.02	0.14-0.21
United States	3SG	0.7	0.02	0.66-0.74
Argentina	C to I	20.23	0.49	19.27-21.19
Brazil	C to I	15.4	0.1	15.20-15.60
Egypt	C to I	19.78	1.38	17.08-22.49
Indonesia	C to I	10.2	0.12	9.97-10.44
Kenya	C to I	6.87	0.32	6.25-7.49
Korea	C to I	11.84	0.13	11.59-12.09
Seychelles	C to I	10.44	0.39	9.70-11.20

United States	C to I	11.89	0.14	11.61-12.18
Argentina	CFU	0.07	0.01	0.06-0.08
Brazil	CFU	0.15	0	0.15-0.16
Egypt	CFU	0.03	0.01	0.02-0.04
Indonesia	CFU	0.12	0	0.11-0.13
Kenya	CFU	0.02	0	0.01-0.02
Korea	CFU	0.18	0	0.17-0.19
Seychelles	CFU	0.11	0.01	0.10-0.13
United States	CFU	0.12	0	0.11-0.13
Argentina	CPA	0.43	0.02	0.39-0.46
Brazil	CPA	0.29	0	0.28-0.30
Egypt	CPA	0.13	0.02	0.09-0.18
Indonesia	CPA	0.08	0	0.07-0.09
Kenya	CPA	0.09	0.01	0.07-0.10
Korea	CPA	0.11	0	0.10-0.11
Seychelles	CPA	0.15	0.01	0.12-0.18
United States	CPA	0.25	0.01	0.24-0.27
Argentina	CPR	1.32	0.03	1.26-1.39
Brazil	CPR	1.04	0.01	1.03-1.06
Egypt	CPR	0.35	0.03	0.28-0.41
Indonesia	CPR	0.73	0.01	0.71-0.75
Kenya	CPR	0.5	0.02	0.45-0.54
Korea	CPR	0.44	0.01	0.43-0.45
Seychelles	CPR	1.22	0.04	1.13-1.31
United States	CPR	1.37	0.02	1.34-1.41
Argentina	I to C	20.97	0.49	20.00-21.93
Brazil	I to C	16.12	0.11	15.92-16.33
Egypt	I to C	24.48	1.54	21.50-27.54
Indonesia	I to C	9.78	0.12	9.54-10.01
Kenya	I to C	7.02	0.34	6.36-7.70
Korea	I to C	11.97	0.13	11.71-12.22
Seychelles	I to C	10.41	0.4	9.64-11.18
United States	I to C	12.28	0.15	11.99-12.58
Argentina	IFU	0.03	0	0.02-0.03
Brazil	IFU	0.06	0	0.06-0.07
Egypt	IFU	0	0	0.00-0.01
Indonesia	IFU	0.06	0	0.06-0.07
Kenya	IFU	0	0	0.00-0.01
Korea	IFU	0.05	0	0.05-0.06

Seychelles	IFU	0.03	0	0.02-0.04
United States	IFU	0.07	0	0.06-0.07
Argentina	IPA	0.29	0.01	0.27-0.32
Brazil	IPA	0.35	0.01	0.34-0.36
Egypt	IPA	0.11	0.02	0.07-0.15
Indonesia	IPA	0.27	0.01	0.25-0.28
Kenya	IPA	0.07	0.01	0.06-0.09
Korea	IPA	0.16	0.01	0.15-0.17
Seychelles	IPA	0.1	0.01	0.07-0.12
United States	IPA	0.34	0.01	0.32-0.35
Argentina	IPR	0.85	0.03	0.80-0.90
Brazil	IPR	0.91	0.01	0.90-0.93
Egypt	IPR	0.4	0.04	0.33-0.48
Indonesia	IPR	1.11	0.01	1.08-1.13
Kenya	IPR	0.41	0.02	0.37-0.45
Korea	IPR	0.91	0.01	0.89-0.93
Seychelles	IPR	0.65	0.03	0.58-0.71
United States	IPR	1.14	0.02	1.11-1.18

Note. 1SG=First Person Singular, 1PL=First Person Plural, 2P=Second Person, 3SG=Third Person Singular, 3PL=Third Person Plural, I to C=Individual to Collective, C to I=Collective to Individual, IPA=Individual Past, IPR=Individual Present, IFU=Individual Future, CPA=Collective Past, CPR=Collective Present, CFU=Collective Future.

Appendix G

Parliamentary system – Narrativity by country

Country	Metric	Mean	SD	95% CI
Australia	CogTen	5.61	2.39	0.92-10.30
Canada	CogTen	9.44	2.24	4.98-13.73
India	CogTen	14.94	0.71	13.57-16.34
Israel	CogTen	20.03	3.34	13.45-26.56
Japan	CogTen	18.79	1.76	15.27-22.23
Mauritius	CogTen	12.37	5.3	1.64-22.63
Norway	CogTen	19.18	1.17	16.88-21.51
United Kingdom	CogTen	8.16	1.37	5.51-10.81
Australia	Overall	10.78	1.55	7.81-13.84
Canada	Overall	16.31	1.49	13.34-19.20
India	Overall	20.31	0.46	19.41-21.21

Israel	Overall	24.4	2.25	19.90-28.76
Japan	Overall	20.72	1.18	18.39-22.98
Mauritius	Overall	18.48	3.12	12.26-24.49
Norway	Overall	24.71	0.75	23.23-26.16
United Kingdom	Overall	17.55	0.92	15.74-19.34
Australia	Plotting	-0.08	2.02	-4.03-3.90
Canada	Plotting	1.85	1.92	-1.95-5.62
India	Plotting	11.28	0.61	10.08-12.48
Israel	Plotting	32.92	2.85	27.32-38.48
Japan	Plotting	23.65	1.61	20.48-26.77
Mauritius	Plotting	31.04	4.93	21.25-40.52
Norway	Plotting	38.95	0.96	37.11-40.86
United Kingdom	Plotting	8.76	1.17	6.44-11.07
Australia	Staging	26.82	2.39	22.14-31.51
Canada	Staging	37.64	2.07	33.61-41.71
India	Staging	34.69	0.68	33.35-36.04
Israel	Staging	20.21	3.18	14.00-26.40
Japan	Staging	19.71	1.82	16.14-23.27
Mauritius	Staging	12.04	5.2	1.88-22.21
Norway	Staging	15.97	1.16	13.67-18.25
United Kingdom	Staging	35.72	1.26	33.22-38.19

Note. Overall=Narrativity Overall, Staging=Narrativity Staging, Plotting=Narrativity Cogten=Narrative Cognitive Tension.

Appendix H

Presidential system - Narrativity by country

Country	Metric	Mean	SD	95% CI
Argentina	Cogten	18.82	2.1	14.69-22.96
Argentina	Overall	33.72	1.35	31.06-36.32
Argentina	Plotting	28.57	1.77	25.11-32.03
Argentina	Staging	53.76	1.59	50.65-56.84
Brazil	Cogten	24.3	0.67	22.98-25.64
Brazil	Overall	30.92	0.44	30.07-31.81
Brazil	Plotting	36.04	0.6	34.85-37.21
Brazil	Staging	32.43	0.66	31.13-33.74

Egypt	Cogten	21.43	4.07	13.46-29.25
Egypt	Overall	14.67	2.81	9.28-20.26
Egypt	Plotting	-2.33	3.69	-9.50-5.17
Egypt	Staging	24.9	4.44	16.27-33.68
Indonesia	Cogten	18.47	0.81	16.88-20.05
Indonesia	Overall	29.21	0.56	28.11-30.30
Indonesia	Plotting	30.94	0.73	29.52-32.37
Indonesia	Staging	38.21	0.76	36.73-39.67
Kenya	Cogten	7.27	2.87	1.62-12.88
Kenya	Overall	23.16	1.64	19.98-26.37
Kenya	Plotting	16.98	2.29	12.47-21.50
Kenya	Staging	45.23	2.4	40.52-49.87
Korea	Cogten	13.22	0.75	11.76-14.69
Korea	Overall	17.71	0.49	16.74-18.68
Korea	Plotting	5.07	0.65	3.79-6.33
Korea	Staging	34.84	0.71	33.44-36.22
Seychelles	Cogten	7.97	2.69	2.75-13.19
Seychelles	Overall	17.45	1.8	13.88-20.97
Seychelles	Plotting	16.79	2.52	11.93-21.70
Seychelles	Staging	27.24	2.61	22.23-32.25
United States	Cogten	12.99	1.05	10.96-15.05
United States	Overall	19.29	0.71	17.89-20.65
United States	Plotting	9.23	0.93	7.40-11.09
United States	Staging	35.64	0.98	33.73-37.54

Note. Overall=Narrativity Overall, Staging=Narrativity Staging, Plotting=Narrativity
Cogten=Narrative Cognitive Tension.

Appendix I

Parliamentary system - Repetition by country

Country	Metric	Mean	SD	95% CI
Australia	Bigram Entropy	5.48	0.08	5.32-5.64
Canada	Bigram Entropy	4.66	0.06	4.55-4.77
India	Bigram Entropy	4.98	0.03	4.93-5.04
Israel	Bigram Entropy	4.88	0.12	4.65-5.11
Japan	Bigram Entropy	4.68	0.07	4.53-4.83

Mauritius	Bigram Entropy	3.96	0.18	3.58-4.29
Norway	Bigram Entropy	4.4	0.04	4.32-4.48
United Kingdom	Bigram Entropy	5.02	0.06	4.91-5.14
Australia	Bigram Repetition	14.73	0.37	14.00-15.48
Canada	Bigram Repetition	10.01	0.21	9.61-10.43
India	Bigram Repetition	14.01	0.11	13.80-14.22
Israel	Bigram Repetition	15.82	0.58	14.73-16.98
Japan	Bigram Repetition	17.2	0.36	16.49-17.90
Mauritius	Bigram Repetition	9.89	0.54	8.80-10.93
Norway	Bigram Repetition	11.6	0.16	11.29-11.90
United Kingdom	Bigram Repetition	14.59	0.24	14.13-15.05
Australia	Bigram TTR	3.32	0.07	3.19-3.46
Canada	Bigram TTR	2.6	0.04	2.51-2.69
India	Bigram TTR	3.1	0.02	3.06-3.14
Israel	Bigram TTR	3.49	0.1	3.30-3.68
Japan	Bigram TTR	4.11	0.06	3.98-4.23
Mauritius	Bigram TTR	2.25	0.11	2.03-2.45
Norway	Bigram TTR	2.98	0.03	2.92-3.04
United Kingdom	Bigram TTR	3.29	0.04	3.21-3.37
Australia	Trigram Entropy	3.18	0.1	2.99-3.38
Canada	Trigram Entropy	1.92	0.07	1.78-2.07
India	Trigram Entropy	2.78	0.03	2.72-2.84
Israel	Trigram Entropy	3.04	0.13	2.78-3.30
Japan	Trigram Entropy	3.22	0.08	3.07-3.37
Mauritius	Trigram Entropy	2.07	0.18	1.71-2.42
Norway	Trigram Entropy	2.27	0.04	2.19-2.36
United Kingdom	Trigram Entropy	2.93	0.06	2.81-3.05
Australia	Trigram Repetition	2.68	0.11	2.46-2.90
Canada	Trigram Repetition	1.49	0.06	1.37-1.61
India	Trigram Repetition	2.54	0.04	2.48-2.62
Israel	Trigram Repetition	3.9	0.35	3.28-4.65
Japan	Trigram Repetition	5.6	0.17	5.27-5.94
Mauritius	Trigram Repetition	2.21	0.23	1.78-2.67
Norway	Trigram Repetition	2.1	0.05	2.00-2.20
United Kingdom	Trigram Repetition	2.86	0.08	2.72-3.02
Australia	Trigram TTR	0.73	0.03	0.68-0.79
Canada	Trigram TTR	0.43	0.02	0.39-0.46

India	Trigram TTR	0.68	0.01	0.67-0.70
Israel	Trigram TTR	1.05	0.07	0.91-1.20
Japan	Trigram TTR	1.59	0.04	1.50-1.67
Mauritius	Trigram TTR	0.6	0.06	0.49-0.71
Norway	Trigram TTR	0.74	0.02	0.70-0.77
United Kingdom	Trigram TTR	0.78	0.02	0.74-0.81
Australia	Zipf Slope (Bigrams)	-0.28	0	-0.29--0.27
Canada	Zipf Slope (Bigrams)	-0.24	0	-0.24--0.23
India	Zipf Slope (Bigrams)	-0.27	0	-0.27--0.27
Israel	Zipf Slope (Bigrams)	-0.3	0.01	-0.31--0.29
Japan	Zipf Slope (Bigrams)	-0.32	0	-0.33--0.31
Mauritius	Zipf Slope (Bigrams)	-0.21	0.01	-0.23--0.2
Norway	Zipf Slope (Bigrams)	-0.27	0	-0.27--0.26
United Kingdom	Zipf Slope (Bigrams)	-0.28	0	-0.28--0.28
Australia	Zipf Slope (Trigrams)	-0.1	0	-0.11--0.1
Canada	Zipf Slope (Trigrams)	-0.08	0	-0.08--0.08
India	Zipf Slope (Trigrams)	-0.1	0	-0.1--0.1
Israel	Zipf Slope (Trigrams)	-0.14	0	-0.15--0.13
Japan	Zipf Slope (Trigrams)	-0.17	0	-0.17--0.16
Mauritius	Zipf Slope (Trigrams)	-0.08	0	-0.09--0.07
Norway	Zipf Slope (Trigrams)	-0.12	0	-0.12--0.11
United Kingdom	Zipf Slope (Trigrams)	-0.11	0	-0.11--0.11

Note. TTR = Type Token Ratio.

Appendix J

Presidential system – Repetition by country

Country	Metric	Mean	SD	95% CI
Argentina	Bigram Entropy	5.34	0.08	5.18-5.51
Brazil	Bigram Entropy	5.58	0.03	5.53-5.63
Egypt	Bigram Entropy	3.96	0.08	3.80-4.11
Indonesia	Bigram Entropy	3.73	0.03	3.67-3.80
Kenya	Bigram Entropy	5.19	0.06	5.07-5.31
Korea	Bigram Entropy	3.84	0.03	3.78-3.90
Seychelles	Bigram Entropy	5.4	0.1	5.21-5.59
United States	Bigram Entropy	5.43	0.05	5.34-5.53

Argentina	Bigram Repetition	17.39	0.37	16.67-18.10
Brazil	Bigram Repetition	17.62	0.12	17.39-17.85
Egypt	Bigram Repetition	10.92	0.54	9.98-12.09
Indonesia	Bigram Repetition	14.27	0.16	13.95-14.59
Kenya	Bigram Repetition	18.35	0.36	17.63-19.05
Korea	Bigram Repetition	12.11	0.11	11.90-12.33
Seychelles	Bigram Repetition	16.67	0.48	15.73-17.61
United States	Bigram Repetition	16.82	0.23	16.36-17.27
Argentina	Bigram TTR	3.85	0.06	3.73-3.97
Brazil	Bigram TTR	3.88	0.02	3.84-3.93
Egypt	Bigram TTR	2.65	0.11	2.45-2.89
Indonesia	Bigram TTR	3.27	0.03	3.22-3.32
Kenya	Bigram TTR	3.63	0.08	3.48-3.79
Korea	Bigram TTR	2.89	0.02	2.85-2.93
Seychelles	Bigram TTR	3.54	0.08	3.38-3.69
United States	Bigram TTR	3.72	0.04	3.64-3.79
Argentina	Trigram Entropy	3.25	0.1	3.06-3.44
Brazil	Trigram Entropy	3.62	0.03	3.56-3.68
Egypt	Trigram Entropy	1.85	0.12	1.61-2.07
Indonesia	Trigram Entropy	2.18	0.03	2.12-2.23
Kenya	Trigram Entropy	4.49	0.08	4.33-4.65
Korea	Trigram Entropy	2.13	0.03	2.07-2.18
Seychelles	Trigram Entropy	3.36	0.12	3.12-3.60
United States	Trigram Entropy	3.39	0.06	3.28-3.50
Argentina	Trigram Repetition	3.3	0.12	3.07-3.54
Brazil	Trigram Repetition	3.85	0.04	3.77-3.92
Egypt	Trigram Repetition	2.5	0.38	1.91-3.37
Indonesia	Trigram Repetition	5.07	0.12	4.85-5.30
Kenya	Trigram Repetition	11.56	0.34	10.90-12.22
Korea	Trigram Repetition	3.14	0.05	3.05-3.23
Seychelles	Trigram Repetition	3.4	0.15	3.11-3.68
United States	Trigram Repetition	3.58	0.07	3.44-3.72
Argentina	Trigram TTR	0.9	0.03	0.84-0.96
Brazil	Trigram TTR	1.02	0.01	1.00-1.04
Egypt	Trigram TTR	0.71	0.09	0.56-0.91
Indonesia	Trigram TTR	1.13	0.02	1.09-1.16
Kenya	Trigram TTR	2.42	0.1	2.23-2.60

Korea	Trigram TTR	0.85	0.01	0.83-0.87
Seychelles	Trigram TTR	0.9	0.03	0.83-0.97
United States	Trigram TTR	0.95	0.02	0.91-0.98
Argentina	Zipf Slope (Bigrams)	-0.31	0	-0.32--0.3
Brazil	Zipf Slope (Bigrams)	-0.31	0	-0.31--0.31
Egypt	Zipf Slope (Bigrams)	-0.25	0.01	-0.26--0.24
Indonesia	Zipf Slope (Bigrams)	-0.28	0	-0.29--0.28
Kenya	Zipf Slope (Bigrams)	-0.3	0	-0.31--0.29
Korea	Zipf Slope (Bigrams)	-0.26	0	-0.26--0.25
Seychelles	Zipf Slope (Bigrams)	-0.3	0	-0.31--0.29
United States	Zipf Slope (Bigrams)	-0.31	0	-0.31--0.3
Argentina	Zipf Slope (Trigrams)	-0.12	0	-0.13--0.12
Brazil	Zipf Slope (Trigrams)	-0.13	0	-0.13--0.13
Egypt	Zipf Slope (Trigrams)	-0.11	0.01	-0.12--0.1
Indonesia	Zipf Slope (Trigrams)	-0.14	0	-0.14--0.14
Kenya	Zipf Slope (Trigrams)	-0.19	0	-0.2--0.18
Korea	Zipf Slope (Trigrams)	-0.11	0	-0.11--0.11
Seychelles	Zipf Slope (Trigrams)	-0.12	0	-0.12--0.11
United States	Zipf Slope (Trigrams)	-0.12	0	-0.13--0.12

Note. TTR = Type Token Ratio.

Appendix K

Parliamentary system – Linguistic Complexity by country

Country	Metric	Mean	SD	95% CI
Australia	ACD	0.77	0	0.77-0.77
Canada	ACD	0.76	0	0.76-0.76
India	ACD	0.77	0	0.77-0.78
Israel	ACD	0.79	0	0.78-0.79
Japan	ACD	0.77	0	0.77-0.77
Mauritius	ACD	0.79	0	0.79-0.80
Norway	ACD	0.77	0	0.77-0.77
United Kingdom	ACD	0.76	0	0.76-0.76
Australia	APC	0.52	0	0.52-0.52
Canada	APC	0.52	0	0.52-0.52
India	APC	0.52	0	0.51-0.52

Israel	APC	0.51	0	0.51-0.51
Japan	APC	0.51	0	0.51-0.51
Mauritius	APC	0.51	0	0.51-0.52
Norway	APC	0.51	0	0.51-0.51
United Kingdom	APC	0.51	0	0.51-0.52
Australia	AWL	4.72	0.01	4.69-4.75
Canada	AWL	4.64	0.01	4.62-4.66
India	AWL	4.82	0	4.81-4.83
Israel	AWL	4.74	0.02	4.69-4.79
Japan	AWL	4.94	0.01	4.92-4.95
Mauritius	AWL	5.52	0.04	5.45-5.59
Norway	AWL	4.87	0.01	4.86-4.89
United Kingdom	AWL	4.69	0.01	4.67-4.71
Australia	CEV	0.7	0	0.70-0.70
Canada	CEV	0.71	0	0.71-0.71
India	CEV	0.7	0	0.70-0.70
Israel	CEV	0.7	0	0.70-0.70
Japan	CEV	0.7	0	0.70-0.70
Mauritius	CEV	0.7	0	0.69-0.70
Norway	CEV	0.69	0	0.69-0.69
United Kingdom	CEV	0.7	0	0.70-0.70
Australia	CPS	2.92	0.03	2.86-2.98
Canada	CPS	2.89	0.02	2.85-2.92
India	CPS	2.62	0.01	2.60-2.63
Israel	CPS	3.19	0.06	3.08-3.30
Japan	CPS	3.41	0.02	3.36-3.46
Mauritius	CPS	2.98	0.06	2.87-3.10
Norway	CPS	2.37	0.01	2.34-2.39
United Kingdom	CPS	3.84	0.05	3.75-3.94
Australia	CSC	0.82	0	0.82-0.82
Canada	CSC	0.83	0	0.83-0.84
India	CSC	0.81	0	0.81-0.81
Israel	CSC	0.8	0.01	0.79-0.81
Japan	CSC	0.82	0	0.82-0.82
Mauritius	CSC	0.8	0	0.80-0.81
Norway	CSC	0.81	0	0.81-0.81
United Kingdom	CSC	0.83	0	0.83-0.83

Australia	FKG	10.3	0.11	10.09-10.53
Canada	FKG	8.93	0.07	8.79-9.06
India	FKG	10.83	0.04	10.76-10.90
Israel	FKG	12.15	0.31	11.58-12.81
Japan	FKG	13.49	0.08	13.32-13.65
Mauritius	FKG	15.18	0.26	14.68-15.71
Norway	FKG	9.61	0.05	9.51-9.71
United Kingdom	FKG	12.11	0.14	11.84-12.40
Australia	LSI	0.85	0	0.85-0.86
Canada	LSI	0.87	0	0.86-0.87
India	LSI	0.86	0	0.86-0.86
Israel	LSI	0.86	0	0.86-0.87
Japan	LSI	0.85	0	0.84-0.85
Mauritius	LSI	0.89	0	0.89-0.90
Norway	LSI	0.86	0	0.86-0.86
United Kingdom	LSI	0.86	0	0.86-0.86
Australia	MATTR	0.69	0	0.69-0.69
Canada	MATTR	0.71	0	0.71-0.72
India	MATTR	0.69	0	0.69-0.69
Israel	MATTR	0.67	0	0.66-0.67
Japan	MATTR	0.69	0	0.69-0.69
Mauritius	MATTR	0.72	0	0.72-0.73
Norway	MATTR	0.69	0	0.69-0.69
United Kingdom	MATTR	0.71	0	0.70-0.71
Australia	MDD	0.17	0	0.17-0.17
Canada	MDD	0.18	0	0.18-0.18
India	MDD	0.17	0	0.17-0.17
Israel	MDD	0.16	0	0.16-0.17
Japan	MDD	0.15	0	0.15-0.15
Mauritius	MDD	0.15	0	0.15-0.16
Norway	MDD	0.19	0	0.19-0.19
United Kingdom	MDD	0.15	0	0.15-0.15
Australia	PTD	0.33	0	0.33-0.34
Canada	PTD	0.34	0	0.34-0.34
India	PTD	0.33	0	0.33-0.34
Israel	PTD	0.32	0	0.32-0.33
Japan	PTD	0.31	0	0.31-0.32

Mauritius	PTD	0.32	0	0.32-0.33
Norway	PTD	0.35	0	0.35-0.35
United Kingdom	PTD	0.31	0	0.31-0.31
Australia	SCC	0.16	0	0.16-0.16
Canada	SCC	0.15	0	0.15-0.16
India	SCC	0.17	0	0.17-0.17
Israel	SCC	0.18	0	0.18-0.19
Japan	SCC	0.21	0	0.20-0.21
Mauritius	SCC	0.16	0	0.16-0.17
Norway	SCC	0.19	0	0.19-0.20
United Kingdom	SCC	0.18	0	0.17-0.18
Australia	TTR	0.35	0	0.34-0.36
Canada	TTR	0.39	0	0.39-0.40
India	TTR	0.38	0	0.38-0.39
Israel	TTR	0.39	0.01	0.37-0.40
Japan	TTR	0.4	0	0.39-0.41
Mauritius	TTR	0.49	0.01	0.47-0.50
Norway	TTR	0.41	0	0.41-0.41
United Kingdom	TTR	0.38	0	0.37-0.38
Australia	VSD	0.15	0	0.15-0.15
Canada	VSD	0.15	0	0.15-0.15
India	VSD	0.15	0	0.15-0.15
Israel	VSD	0.15	0	0.15-0.15
Japan	VSD	0.15	0	0.15-0.15
Mauritius	VSD	0.15	0	0.14-0.15
Norway	VSD	0.15	0	0.15-0.15
United Kingdom	VSD	0.15	0	0.15-0.15

Note TTR = Type-Token Ratio, MATTR = Moving-Average Type-Token Ratio, AWL = Average Word Length, AWS = Average Words per Sentence, ACD = Average Cosine Distance, VSD = Vector Space Density, SCC = Semantic Clustering Coefficient, LSI = Lexical Sophistication Index, CPS = Clauses per Sentence, PTD = Parse Tree Depth, MDD = Mean Dependency Distance, FKGL = Flesch-Kincaid Grade Level, CEV = Contextual Embedding Variation, APC = Attention Pattern Complexity, CSC = Cross-Sentence Coherence.

Appendix L

Presidential system – Linguistic Complexity by country

Country	Metric	Mean	SD	95% CI
Argentina	ACD	0.77	0	0.77-0.77
Brazil	ACD	0.78	0	0.78-0.78
Egypt	ACD	0.76	0	0.76-0.77
Indonesia	ACD	0.78	0	0.78-0.78
Kenya	ACD	0.79	0	0.79-0.79
Korea	ACD	0.76	0	0.76-0.76
Seychelles	ACD	0.77	0	0.76-0.77
United States	ACD	0.75	0	0.75-0.75
Argentina	APC	0.52	0	0.52-0.52
Brazil	APC	0.51	0	0.51-0.51
Egypt	APC	0.51	0	0.51-0.51
Indonesia	APC	0.51	0	0.51-0.51
Kenya	APC	0.51	0	0.51-0.51
Korea	APC	0.51	0	0.51-0.51
Seychelles	APC	0.51	0	0.51-0.51
United States	APC	0.52	0	0.52-0.52
Argentina	AWL	4.49	0.01	4.46-4.52
Brazil	AWL	4.62	0.01	4.61-4.63
Egypt	AWL	5.13	0.03	5.07-5.19
Indonesia	AWL	4.78	0.01	4.77-4.79
Kenya	AWL	5.71	0.01	5.68-5.74
Korea	AWL	4.95	0	4.94-4.96
Seychelles	AWL	4.86	0.02	4.83-4.89
United States	AWL	4.39	0.01	4.37-4.40
Argentina	CEV	0.7	0	0.70-0.70
Brazil	CEV	0.7	0	0.70-0.70
Egypt	CEV	0.7	0	0.70-0.70
Indonesia	CEV	0.7	0	0.70-0.70
Kenya	CEV	0.7	0	0.70-0.70
Korea	CEV	0.7	0	0.70-0.70
Seychelles	CEV	0.7	0	0.70-0.70
United States	CEV	0.71	0	0.71-0.71
Argentina	CPS	5.16	0.06	5.04-5.28

Brazil	CPS	3.49	0.01	3.47-3.51
Egypt	CPS	7.01	0.17	6.69-7.35
Indonesia	CPS	2.63	0.01	2.61-2.65
Kenya	CPS	3.09	0.09	2.93-3.29
Korea	CPS	3.1	0.01	3.08-3.12
Seychelles	CPS	2.82	0.04	2.74-2.90
United States	CPS	2.71	0.02	2.68-2.74
Argentina	CSC	0.8	0	0.79-0.80
Brazil	CSC	0.8	0	0.80-0.80
Egypt	CSC	0.84	0	0.83-0.84
Indonesia	CSC	0.79	0	0.79-0.79
Kenya	CSC	0.77	0	0.77-0.78
Korea	CSC	0.83	0	0.83-0.84
Seychelles	CSC	0.81	0	0.81-0.82
United States	CSC	0.83	0	0.83-0.83
Argentina	FKG	14.35	0.15	14.07-14.63
Brazil	FKG	11.28	0.04	11.21-11.36
Egypt	FKG	24.09	0.48	23.14-25.05
Indonesia	FKG	10.05	0.04	9.97-10.13
Kenya	FKG	16.15	0.32	15.60-16.82
Korea	FKG	12.2	0.04	12.13-12.28
Seychelles	FKG	10.93	0.15	10.63-11.24
United States	FKG	7.46	0.06	7.34-7.58
Argentina	LSI	0.85	0	0.85-0.85
Brazil	LSI	0.85	0	0.85-0.85
Egypt	LSI	0.88	0	0.88-0.89
Indonesia	LSI	0.85	0	0.85-0.85
Kenya	LSI	0.87	0	0.87-0.88
Korea	LSI	0.88	0	0.88-0.88
Seychelles	LSI	0.85	0	0.85-0.86
United States	LSI	0.84	0	0.84-0.85
Argentina	MATTR	0.66	0	0.65-0.66
Brazil	MATTR	0.67	0	0.67-0.67
Egypt	MATTR	0.68	0	0.67-0.68
Indonesia	MATTR	0.67	0	0.67-0.68
Kenya	MATTR	0.74	0	0.73-0.74
Korea	MATTR	0.71	0	0.71-0.71

Seychelles	MATTR	0.68	0	0.67-0.68
United States	MATTR	0.68	0	0.68-0.68
Argentina	MDD	0.14	0	0.14-0.14
Brazil	MDD	0.17	0	0.17-0.17
Egypt	MDD	0.1	0	0.10-0.11
Indonesia	MDD	0.2	0	0.20-0.21
Kenya	MDD	0.19	0	0.19-0.20
Korea	MDD	0.16	0	0.16-0.16
Seychelles	MDD	0.18	0	0.17-0.18
United States	MDD	0.21	0	0.21-0.22
Argentina	PTD	0.3	0	0.29-0.30
Brazil	PTD	0.32	0	0.32-0.32
Egypt	PTD	0.27	0	0.27-0.28
Indonesia	PTD	0.33	0	0.33-0.33
Kenya	PTD	0.31	0	0.31-0.32
Korea	PTD	0.32	0	0.31-0.32
Seychelles	PTD	0.35	0	0.35-0.35
United States	PTD	0.35	0	0.35-0.35
Argentina	SCC	0.16	0	0.16-0.16
Brazil	SCC	0.16	0	0.15-0.16
Egypt	SCC	0.17	0	0.16-0.18
Indonesia	SCC	0.23	0	0.23-0.23
Kenya	SCC	0.16	0	0.16-0.17
Korea	SCC	0.21	0	0.20-0.21
Seychelles	SCC	0.17	0	0.16-0.17
United States	SCC	0.18	0	0.18-0.19
Argentina	TTR	0.34	0	0.33-0.35
Brazil	TTR	0.34	0	0.34-0.35
Egypt	TTR	0.44	0	0.43-0.45
Indonesia	TTR	0.44	0	0.43-0.44
Kenya	TTR	0.43	0	0.43-0.44
Korea	TTR	0.45	0	0.45-0.45
Seychelles	TTR	0.36	0.01	0.35-0.37
United States	TTR	0.34	0	0.33-0.34
Argentina	VSD	0.15	0	0.15-0.15
Brazil	VSD	0.15	0	0.15-0.15
Egypt	VSD	0.15	0	0.15-0.15

Indonesia	VSD	0.15	0	0.15-0.15
Kenya	VSD	0.15	0	0.15-0.15
Korea	VSD	0.15	0	0.15-0.15
Seychelles	VSD	0.15	0	0.15-0.15
United States	VSD	0.15	0	0.15-0.15

Note TTR = Type-Token Ratio, MATTR = Moving-Average Type-Token Ratio, AWL = Average Word Length, AWS = Average Words per Sentence, ACD = Average Cosine Distance, VSD = Vector Space Density, SCC = Semantic Clustering Coefficient, LSI = Lexical Sophistication Index, CPS = Clauses per Sentence, PTD = Parse Tree Depth, MDD = Mean Dependency Distance, FKGL = Flesch-Kincaid Grade Level, CEV = Contextual Embedding Variation, APC = Attention Pattern Complexity, CSC = Cross-Sentence Coherence.

Appendix M

Repetition Analysis – Non-Translated text

	Presidential Language			Parliamentary Language			Sig	<i>d</i>
	Mean	95% CI	SD	Mean	95% CI	SD		
Bigram TTR	2.22	2.19 - 2.24	0.014	2.77	2.72-2.81	0.024	Yes	-0.32
Trigram TTR	0.59	0.57-0.60	0.005	0.72	0.69-0.74	0.012	Yes	-0.18
Bigram Entropy	4.69	4.65-4.72	0.017	4.36	4.30-4.41	0.029	Yes	0.16
Trigram Entropy	3.10	3.06-3.13	0.017	2.39	2.34-2.45	0.028	Yes	0.35
Bigram Repetition	16.36	16.19-16.53	0.087	13.06	12.75-13.36	0.156	Yes	0.31
Trigram Repetition	5.41	5.33-5.50	0.045	2.80	2.69-2.91	0.055	Yes	0.55

Note. TTR = Type Token Ratio, Sig = *d*=Cohen's *d*.